Supplementary material: Complex energy learning from finite magnetic-rod observations

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The proofs concern the full 64-coordinate constant-mass mechanical model and its three-coordinate conservative restriction defined in the article. A separate full-coordinate neural certificate retains the exact source operators and all 64 transverse coordinates of the same saved weights; its cubic-corrected twist is distinct from the target-family twist. The selected network has two complex split-tanh heads and 918 real parameters. Its checkpoint SHA-256 prefix is `f51bcea12c2f`. All numerical candidate conclusions below refer to that checkpoint when identified as calibrated. Earlier time-step and fitting diagnostics are explicitly distinguished. Certificates for other architectures and momentum-dependent historical models are not inherited.

The force/coefficient interface uses exact decimal scales $\ell = 1/1000$ m and $E_s = 3/500000000000$ J. Network feature scales are their exact saved binary values and need not equal these decimal numbers. A numerical recorded-state IVP, a compiled simulation trajectory and a physical measurement are different objects; their distinct error requirements remain stated in each result.

S1 Complex canonical coordinates and symplectic discretization of the saved full-coordinate energy

S1.1 The fixed mechanical and network object

Let $n = 64$, let $M = M_0 + M_r \succ 0$ be the exact sum of the saved source matrices, and let $K_0 \succ 0$ be the saved elastic matrix. All saved floating-point entries and network parameters are interpreted as exact real numbers. With $b = e_{64}$, the conservative Hamiltonian is

$$H_N(x,p) = \tfrac{1}{2}p^T M^{-1} p + U_N(x), \qquad U_N(x) = E_0(x) + \kappa[1-\cos(b^T x)] + F_N(x), \quad E_0(x) = \tfrac{1}{2}x^T K_0 x. \tag{S1}$$

The energy correction is precisely the saved configuration graph

$$\begin{aligned} F_N(x) &= c_E E_0(x) + E_{\mathrm{net}}\left[N(s(x)) - N(0) - \sum_{j=1}^{3}\gamma_j s_j(x)\right], \\ s(x) &= \left((Bx)_1/Q_1, (Bx)_2/Q_2, (Bx)_3/Q_3, E_0(x)/E_{\mathrm{net}}\right), \qquad \gamma_j = \partial_j N(0). \end{aligned} \tag{S2}$$

Here $B = (V^T M V)^{-1} V^T M$, where V is the full-column-rank preparation basis supplied by the source record. Its saved rational entries, the source matrices, and all operations defining B are interpreted exactly; a separately rounded runtime projection buffer is not substituted. Here N is the real readout of the saved $4 \to 12 \to 12 \to 1$ complex split-tanh energy head. Together with its dissipation head and the two real skips the model has 918 real parameters; the present conservative Hamiltonian does not depend on the dissipation head. The checkpoint is `f51bcea12c2f2640c0ea33c3493a691d8ab240578018a6d44a9f68904a1a60a9`. There is no learned

momentum dependence and no configuration-dependent mass. The conservative subsystem sets the dissipation and magnetic input force to zero while retaining the magnetic potential in (S1).

For $q = Bx$ and $E = E_0(x)$ write the right-hand side of (S2) as $F(q, E)$. Differentiation of both features gives

$$\nabla_x F_N = B^T F_q + F_E K_0 x, \qquad \nabla U_N = (1 + F_E)K_0 x + \kappa \sin(b^T x)b + B^T F_q. \tag{S3}$$

In particular,

$$(F_q)_j = \frac{E_{\text{net}}}{Q_j}[\partial_j N(s) - \gamma_j], \qquad F_E = c_E + \partial_4 N(s). \tag{S4}$$

The fourth feature is not removed by the affine correction. The Hamilton equations are consequently

$$\dot{x} = M^{-1}p, \qquad \dot{p} = -\nabla U_N(x), \qquad (H_N)_{pp} = M^{-1}, \quad (H_N)_{xp} = 0. \tag{S5}$$

These formulas hold on the full 128-dimensional transverse phase space. They use neither invariance of the three-coordinate trial space nor the scalar target-family prior.

S1.2 Dimensionless canonical variables and the complex equation

The configuration entries contain both displacements and rotations. To avoid adding a physical configuration to a physical momentum, choose a fixed invertible configuration scale L_s, an energy scale $\mathcal{E} > 0$ and an action scale $\mathcal{A} > 0$. Define dimensionless variables by

$$x = L_s Q, \qquad p = \mathcal{A}L_s^{-T}P, \qquad t = (\mathcal{A}/\mathcal{E})\tau, \qquad h(Q, P) = \mathcal{E}^{-1}H_N(L_s Q, \mathcal{A}L_s^{-T}P). \tag{S6}$$

The physical one-form and two-form pull back to

$$p^T dx = \mathcal{A}P^T dQ, \qquad \sum_j dx_j \wedge dp_j = \mathcal{A}\sum_j dQ_j \wedge dP_j. \tag{S7}$$

Thus the common action factor is removed together with the energy/time normalization; the configuration/momentum transform alone is conformally symplectic with constant factor $\mathcal{A}$. Direct use of the chain rule, rather than a change of units alone, gives

$$\begin{aligned} Q' = \partial_P h = A_s P, \qquad & P' = -\partial_Q h = -g(Q), \\ h(Q, P) = \tfrac{1}{2}P^T A_s P + \mathcal{U}(Q), \qquad & A_s = \frac{\mathcal{A}^2}{\mathcal{E}}L_s^{-1}M^{-1}L_s^{-T} \succ 0, \\ \mathcal{U}(Q) = U_N(L_s Q)/\mathcal{E}, \qquad & g(Q) = L_s^T \nabla U_N(L_s Q)/\mathcal{E}. \end{aligned} \tag{S8}$$

A prime means differentiation with respect to τ; the matrix A_s is not the action scale $\mathcal{A}$.

Now define complex canonical coordinates and their inverse by

$$z = (Q + iP)/\sqrt{2}, \quad \bar{z} = (Q - iP)/\sqrt{2}, \qquad Q = (z + \bar{z})/\sqrt{2}, \quad P = (z - \bar{z})/(i\sqrt{2}), \tag{S9}$$

and put $\mathfrak{h}(z, \bar{z}) = h(Q(z, \bar{z}), P(z, \bar{z}))$. The Wirtinger chain rules are

$$\partial_z \mathfrak{h} = (h_Q - ih_P)/\sqrt{2}, \qquad \partial_{\bar{z}} \mathfrak{h} = (h_Q + ih_P)/\sqrt{2}. \tag{S10}$$

Consequently the same real system is

$$z' = -i\partial_{\bar{z}} \mathfrak{h}, \qquad \bar{z}' = i\partial_z \mathfrak{h}, \qquad \omega = \sum_j dQ_j \wedge dP_j = i\sum_j dz_j \wedge d\bar{z}_j. \tag{S11}$$

For example, for real differentiable observables the corresponding Poisson bracket is

$$\{f,k\} = -i\sum_j \left[(\partial_{z_j} f)(\partial_{\bar{z}_j} k) - (\partial_{\bar{z}_j} f)(\partial_{z_j} k)\right], \qquad \{z_j, \bar{z}_k\} = -i\delta_{jk}. \tag{S12}$$

This is a complex coordinate representation of a real scalar Hamiltonian. It does not require that $\mathfrak{h}$ be holomorphic in z alone, or identify a hidden complex neuron with a canonical state.

The analytic extension used for this saved network is equally precise. Each complex affine layer is replaced by its real block matrix $\left(\begin{smallmatrix} W_R & -W_I \\ W_I & W_R \end{smallmatrix}\right)$, and the split activation acts componentwise by real tanh. Complexifying these real variables gives a holomorphic continuation wherever its tanh denominators are nonzero. If Z, W are independent variables, the canonical extension is

$$\mathfrak{h}^{\mathbb{C}}(Z,W) = h^{\mathbb{C}}\left((Z+W)/\sqrt{2}, (Z-W)/(i\sqrt{2})\right). \tag{S13}$$

The physical slice has $W = \bar{Z}$; no conjugation is performed during holomorphic differentiation in the independent variables. The existing source-bound analytic-domain certificate concerns the same realified potential. No literal-complex-tanh replacement is made here.

S1.3 Deriving the Störmer–Verlet difference scheme

Because (S8) is separable, its kinetic and potential subsystems have exact flows

$$D_s(Q,P) = (Q + sA_sP, P), \qquad K_s(Q,P) = (Q, P - sg(Q)). \tag{S14}$$

Their symmetric composition $\Phi_\delta = K_{\delta/2} \circ D_\delta \circ K_{\delta/2}$ gives

$$\begin{aligned} P_{n+1/2} &= P_n - \tfrac{\delta}{2} g(Q_n), \\ Q_{n+1} &= Q_n + \delta A_s P_{n+1/2}, \\ P_{n+1} &= P_{n+1/2} - \tfrac{\delta}{2} g(Q_{n+1}). \end{aligned} \tag{S15}$$

This is the classical Störmer–Verlet construction (Hairer 2010, Lecture 2, Theorem 2 and Section 5, Example 3). It is explicit apart from application of the fixed inverse mass; it does not solve a nonlinear implicit network equation. In physical time with step $\Delta t = (\mathcal{A}/\mathcal{E})\delta$, its identical form is

$$\begin{aligned} p_{n+1/2} &= p_n - \tfrac{\Delta t}{2} \nabla U_N(x_n), \\ x_{n+1} &= x_n + \Delta t M^{-1} p_{n+1/2}, \\ p_{n+1} &= p_{n+1/2} - \tfrac{\Delta t}{2} \nabla U_N(x_{n+1}). \end{aligned} \tag{S16}$$

Both endpoint forces must include (S3). Eliminating momenta gives the associated position difference equation

$$M(x_{n+1} - 2x_n + x_{n-1}) = -(\Delta t)^2 \nabla U_N(x_n). \tag{S17}$$

For the complex version of each substep, recover Q, P from (S9) and apply

$$z \xmapsto{K_s} z - isg(Q)/\sqrt{2}, \qquad z \xmapsto{D_s} z + sA_sP/\sqrt{2}, \tag{S18}$$

with the conjugate updates for $\bar{z}$. Intermediate values of Q, P are updated after each substep. This is the same difference map in complex coordinates, rather than a new discretization obtained by inserting complex states into a real-output network routine.

Exact symplecticity and reversibility. Put $J = \left(\begin{smallmatrix} 0 & I_n \\ -I_n & 0 \end{smallmatrix}\right)$. Since A_s and $G(Q) = Dg(Q)$ are symmetric,

$$DK_s = \begin{pmatrix} I & 0 \\ -sG(Q) & I \end{pmatrix}, \qquad DD_s = \begin{pmatrix} I & sA_s \\ 0 & I \end{pmatrix}, \qquad (DK_s)^T J DK_s = (DD_s)^T J DD_s = J. \tag{S19}$$

The chain rule therefore gives $(D\Phi_\delta)^T J D\Phi_\delta = J$. The physical map preserves $\sum dx_j \wedge dp_j$ by (S7); the complex map preserves $i\sum dz_j \wedge d\bar{z}_j$. Also $K_s^{-1} = K_{-s}$ and $D_s^{-1} = D_{-s}$, whence $\Phi_{-\delta} = \Phi_\delta^{-1}$. These identities concern exact mathematical evaluation of the map. They do not assert exact conservation of the nonlinear energy.

The physical linear stability threshold. Let $H_2 = U_N''(0) \succ 0$ and choose a mass-normalized basis S with $S^T M S = I$ and $S^T H_2 S = \mathrm{diag}(\omega_1^2, \ldots, \omega_n^2)$. The transformation $x = S\xi$, $p = MS\pi$ is canonical, since $S^{-T} = MS$. Each physical-time mode of the linearized Verlet map is

$$\begin{pmatrix} \xi_{j,n+1} \\ \pi_{j,n+1} \end{pmatrix} = \begin{pmatrix} 1 - r_j^2/2 & \Delta t \\ -\Delta t \omega_j^2 (1 - r_j^2/4) & 1 - r_j^2/2 \end{pmatrix} \begin{pmatrix} \xi_{j,n} \\ \pi_{j,n} \end{pmatrix}, \qquad r_j = \Delta t \omega_j. \tag{S20}$$

Its determinant is one and its trace is $2 - r_j^2$. For $0 < r_j < 2$ it preserves the positive quadratic form $\pi_j^2 + \omega_j^2(1 - r_j^2/4)\xi_j^2$, so its powers are bounded. At $r_j = 2$ the matrix is $\left(\begin{smallmatrix} -1 & \Delta t \\ 0 & -1 \end{smallmatrix}\right)$, whose powers are generally unbounded. For $r_j > 2$ one eigenvalue has modulus greater than one. Thus, for a positive step, the full linearized map is power bounded exactly when

$$0 < \Delta t < 2/\omega_{\max}. \tag{S21}$$

The existing exact-source candidate frequency enclosures give

$$\omega_{\max} \le \frac{2441081236299847251}{17592186044416}\ \mathrm{rad\,s^{-1}}. \tag{S22}$$

Their exact rational arithmetic therefore supplies the sufficient linear condition

$$0 < \Delta t < \frac{35184372088832}{2441081236299847251}\ \mathrm{s} \simeq 1.44134376052818 \times 10^{-5}\ \mathrm{s}. \tag{S23}$$

This reuses a received spectrum rather than running an integration experiment. It is a condition for the equilibrium linearization of the conservative saved-source model, not a nonlinear safety guarantee, a control hold duration, an error-tolerance choice, or a discrete KAM certificate. The finite-time floating-point integrations in article Section 3.2 are a separate numerical check; this spectral threshold uses no integration output.

S1.4 A full local and global second-order estimate

The following estimates make explicit the domain condition that is otherwise hidden in order notation. Work in the dimensionless coordinates and choose a convex region containing the configurations of the exact solutions and all compared kick/drift stages. Assume $\mathcal{U} \in C^3$ there and, in a fixed compatible vector norm,

$$\|A_s\| \le a, \quad \|g\| \le G_0, \quad \|Dg\| \le G_1, \quad \|D^2 g\| \le G_2, \quad \|A_s P(\tau)\| \le V \tag{S24}$$

The bounds on $g, Dg, D^2 g$ hold uniformly throughout the stated convex configuration region; only the bound on $A_s P(\tau)$ is restricted to the relevant exact arcs. These are assumptions for this estimate, not numerical constants supplied by a finite training loss. Let $0 < \delta \le \delta_0$ and use the phase norm $\|(Q,P)\|_\times = \max(\|Q\|, \|P\|)$.

Starting at the same (Q,P), write $Q_e(s), P_e(s)$ for the exact solution. Two integrations of the equations give

$$Q_e(\delta) = Q + \delta A_s P - \int_0^\delta (\delta - s) A_s g(Q_e(s))\, ds. \tag{S25}$$

The Verlet endpoint $Q_v = Q + \delta A_s P - \delta^2 A_s g(Q)/2$ then satisfies

$$\|Q_v - Q_e(\delta)\| \le a G_1 V \int_0^\delta (\delta - s) s\, ds = C_Q \delta^3, \qquad C_Q = a G_1 V/6. \tag{S26}$$

For $f(s) = g(Q_e(s))$,

$$f''(s) = D^2 g(Q_e(s))[A_s P_e(s), A_s P_e(s)] - Dg(Q_e(s)) A_s g(Q_e(s)), \qquad \|f''\| \le W := G_2 V^2 + G_1 a G_0. \tag{S27}$$

The integral remainder of the trapezoidal rule is bounded by $W\delta^3/12$. Replacing its exact endpoint by Q_v adds at most $\delta G_1\|Q_v - Q_e(\delta)\|/2$. Consequently

$$\|P_v - P_e(\delta)\| \le C_P\delta^3, \qquad C_P = \frac{G_2V^2 + G_1aG_0 + \delta_0 aG_1^2V}{12}, \qquad \|\Phi_\delta - \varphi_\delta\|_\times \le C\delta^3, \quad C = \max(C_Q, C_P). \tag{S28}$$

This proves local order three without assigning an unspecified constant to the network.

On compared stage segments, the two shear Lipschitz constants give

$$\mathrm{Lip}(\Phi_\delta) \le (1 + \delta G_1/2)^2(1 + \delta a) \le e^{\Lambda\delta}, \qquad \Lambda = G_1 + a > 0. \tag{S29}$$

For $T = N\delta$ and matching initial conditions, iteration of $e_{n+1} \le e^{\Lambda\delta}e_n + C\delta^3$ yields

$$e_N \le C\delta^3\frac{e^{\Lambda T} - 1}{e^{\Lambda\delta} - 1} \le C\delta^2\frac{e^{\Lambda T} - 1}{\Lambda}. \tag{S30}$$

Thus the method has global order two on a fixed time interval while the stated exact and numerical stages remain in the prescribed region. A buffer-domain bootstrap may establish this containment once the right-hand side is smaller than the buffer width; containment is not assumed to follow from symplecticity alone.

If an implementation makes a per-step error at most η_n relative to the mathematical Verlet map, the same argument instead gives

$$e_N \le e^{\Lambda T}e_0 + C\delta^2\frac{e^{\Lambda T} - 1}{\Lambda} + \sum_{n=0}^{N-1} e^{\Lambda(N-1-n)\delta}\eta_n. \tag{S31}$$

Network evaluation and inverse-mass solve errors belong to η_n; they are not covered by the exact shear identity. No numerical value of these implementation errors, a selected integration step, or the domain-dependent error constants is supplied here.

S1.5 Distinguishing integration, learning and KAM statements

For a target with the same normalized kinetic energy let $g_*(Q)$ be its conservative force, and suppose on the common solution region that $\|g - g_*\| \le \varepsilon$ and that one of the two complete Hamiltonian vector fields has Lipschitz constant L_*. The integral difference equation gives, for identical initial states,

$$\|\varphi_\tau^N - \varphi_\tau^*\|_\times \le \varepsilon\frac{e^{L_*\tau} - 1}{L_*}, \tag{S32}$$

with τ on the right when $L_* = 0$. Combining this estimate with (S30) separates a δ^2 integration term from a learning term that does not vanish when the step is reduced. Finite node losses do not by themselves imply the required uniform force bound. The existing coefficient allowances, source priors and candidate analytic certificates must be used with their own stated domains and quantifiers to obtain any such bound.

The continuous local elliptic KAM certificate concerns the flow of (S1). Symplecticity and second-order convergence of Φ_δ do not turn it into a KAM certificate for the discrete map: map nonresonances may also involve $\delta k \cdot \omega - 2\pi m$, and a discrete KAM application needs its own hypotheses and bounds. No finite numerical torus radius or all-time trajectory error is deduced from (S30).

Finally, let $D_0 = \alpha M_0 \succ 0$ with $\alpha > 0$, let $a_N(x) \ge 0$ be the saved configuration-dependent squared-readout dissipation factor, and let $g_{\mathrm{mag}}(x) = \mu\cos(b^Tx)b$ be the configuration input port. The full port model has the additional force $-a_N(x)D_0M^{-1}p + g_{\mathrm{mag}}(x)u$. When u is independent of momentum its phase divergence contains $-a_N(x)\,\mathrm{tr}(D_0M^{-1})$, which is negative wherever $a_N > 0$; such a flow cannot preserve the canonical phase volume there. The conservative shear proof is therefore not a symplecticity claim for the damped, forced or feedback-controlled model. Its established power and sampled-work safety bounds supply the separate control argument. Nor does changing canonical coordinates prove that the complex network representation outperforms a real network.

S2 Source-conditioned target regularity and the legacy-model boundary

The native source files distinguish the generated benchmark from an unknown physical device. The actual configured and solved models used for all eleven development configurations retain an Euler–Bernoulli beam with isotropic linear elasticity, spatially constant Young's modulus, fixed cross-section and geometric nonlinearity disabled. The only added weak contribution is rotational inertia. The specified point load is the magnetic moment already represented by the known cosine potential. These are properties of the saved source models, independently of the fitted neural weights.

For the ideal fixed-mesh semidiscrete model specified by these settings, the existing full-FE energy derivation gives

$$U_E(x) = \tfrac{1}{2}x^T K_E x + \kappa(1 - \cos(b^T x)), \qquad K_E = (E/E_{\mathrm{nom}})K_0.$$

The source modulus is $E_{\mathrm{nom}} = 5 \times 10^6$ Pa and the generated target modulus is $E = 5.3 \times 10^6$ Pa. Consequently the material correction relative to the nominal simulator is exactly

$$W_*(x) = \beta E_0(x), \qquad \beta = \frac{3}{50}, \qquad E_0(x) = \tfrac{1}{2}x^T K_0 x.$$

Here $E_0(x)$ denotes the nominal elastic energy. This source-conditioned identity applies to the ideal model specified by the saved settings; it is not inferred from the finite trajectory observations and does not certify exact arithmetic inside a compiled solver.

On the same restriction $x = V\ell z$ and normalization E_s, put $w_*(z) = W_*(V\ell z)/E_s$. Then

$$w_*(z) = \frac{\beta\ell^2}{2E_s} z^T K_{03} z, \quad b_1(w_*) \le \frac{|\beta|\ell^2 R}{E_s} \max_i \sum_j |(K_{03})_{ij}|, \quad b_2(w_*) \le \frac{|\beta|\ell^2}{E_s} \max_{i,j} |(K_{03})_{ij}|,$$

and $b_3(w_*) = b_4(w_*) = b_5(w_*) = 0$. The source-conditioned material fifth-order remainder therefore vanishes. The known magnetic term has not been set to zero: it is retained exactly and cancels only in the material learning error. For the specified three-coordinate conservative restriction the quadratic Hessian is

$$K_* = (53/50)K_{03} + \kappa d d^T.$$

The exact source ratio $3/50$ lies inside the previously received scalar compatibility interval, so the existing uniform positive spectrum and finite-order-divisor certificate covers this ideal simulator member. This is a source-conditioned simulator conclusion, not a completed learning-to-KAM transfer for an unknown device.

Historical geometrically nonlinear models have different kinematics, coordinate charts and discretizations. They supply no interchangeable derivative prior for this fixed-mesh target without an independent coordinate transport and remainder estimate.

Thus the source files justify the displayed regularity bounds for the ideal generated benchmark. They do not identify an unknown material's analytic tail or quadratic spectrum, resolve the legacy nonlinear inertia mapping, establish complex-network superiority, or prove physical-target KAM. For an unknown physical target these remain independent constitutive and measurement premises.

S3 Learning a three-dimensional conservative correction

The learning variable can be the configuration rather than the complete canonical state when the kinetic energy is known and the unknown term is a configuration potential. This reduction concerns the domain on which an error function must be learned. It does not reduce the number of degrees of freedom of the Hamiltonian system.

S3.1 The restricted physical model and its learning dimension

Let A and K_3 be fixed symmetric positive definite three-by-three mass and elastic matrices, respectively. Let $d \in \mathbb{R}^3$ and $\kappa > 0$ be the fixed magnetic port and bias coefficient. Consider the real conservative Hamiltonians

$$\begin{aligned} H_\theta(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}A^{-1}p + U_0(q) + W_\theta(q), \\ H_*(q,p) &= \tfrac{1}{2}p^{\mathsf{T}}A^{-1}p + U_0(q) + W_*(q), \\ U_0(q) &= \tfrac{1}{2}q^{\mathsf{T}}K_3 q + \kappa[1-\cos(d^{\mathsf{T}}q)]. \end{aligned} \tag{S33}$$

Here $q, p \in \mathbb{R}^3$, the mass is not fitted, and damping and time-dependent input are excluded from this conservative subsystem. The target W_* and a candidate W_θ are functions of q alone.

The energy-feature candidate has this structure on its trial subspace. More precisely, let M, K_0 be the full mass and elastic matrices and let V be a fixed full-column-rank trial matrix. Set $A = V^{\mathsf{T}}MV$, $K_3 = V^{\mathsf{T}}K_0V$, and $B = A^{-1}V^{\mathsf{T}}M$, so that $BV = I_3$. Its full potential correction is $F_\theta(Bx, x^{\mathsf{T}}K_0x/2)$, and its restriction is

$$W_\theta(q) = F_\theta(q, E_0(q)), \qquad E_0(q) = \tfrac{1}{2}q^{\mathsf{T}}K_3q. \tag{S34}$$

The four feature entries $(q, E_0(q))$ have only three independent real variables on this surface. For example, the centered scalar readout is

$$\begin{aligned} s(q) &= (q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0(q)/E_*), \\ W_\theta(q) &= c_{\mathrm{E}}E_0(q) + E_*\left[N_\theta(s(q)) - N_\theta(0) - \sum_{j=1}^{3}\frac{q_j}{Q_j}\partial_j N_\theta(0)\right]. \end{aligned} \tag{S35}$$

The positive Q_j, E_* are the frozen exported feature scales; N_θ is the real scalar readout of the real or realified complex network. Equation (S35) implies $W_\theta(0) = 0$ and $\nabla W_\theta(0) = 0$ because $\nabla E_0(0) = 0$. It does not impose zero quadratic, cubic, or quartic corrections. The coefficient c_{E} occurs linearly.

For arbitrary full configurations, $E_0(x)$ is not determined by Bx. In that case

$$V^{\mathsf{T}}\nabla_x F_\theta(Bx, E_0(x)) = F_{\theta,q} + F_{\theta,E}V^{\mathsf{T}}K_0x,$$

whereas the restricted force is $\nabla_q W_\theta = F_{\theta,q} + F_{\theta,E}K_3q$ at $(q, E_0(q))$. Thus unrestricted full-field observations require the separate preparation, transfer, and conservative-label error estimates. The reduction below does not assert autonomous closure of a projected full trajectory.

Fix $\ell > 0$, $E_s > 0$, and $R > 0$, and use the normalized configuration cube and potentials

$$\Omega = [-R, R]^3, \qquad q = \ell z, \qquad w_\theta(z) = W_\theta(\ell z)/E_s, \quad w_*(z) = W_*(\ell z)/E_s. \tag{S36}$$

In the physical interface $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$ J. The normalization E_s need not be identified with the exact exported binary value E_* in (S35). Write $f_\theta = \nabla_z w_\theta$, $f_* = \nabla_z w_*$ and $g_\theta = f_\theta - f_*$. A normalized conservative-force label means a label for f_*, including its sign and the factor ℓ/E_s. For a vector field g, write $\|g\|_{\infty,\Omega} = \max_j \sup_{z\in\Omega}|g_j(z)|$.

Proposition S1. Exact reduction of the error domain. *Let Π be any probability law for (Z, P) on $\Omega \times \mathcal{P}$, with configuration marginal ν. For $1 \le p < \infty$, the physical canonical vector-field error obeys*

$$\begin{aligned} X_{H_\theta} - X_{H_*} &= (0, -(E_s/\ell)g_\theta(z)), \\ \mathbb{E}_\Pi\|X_{H_\theta}(\ell Z, P) - X_{H_*}(\ell Z, P)\|_p^p &= (E_s/\ell)^p\mathbb{E}_\nu\|g_\theta(Z)\|_p^p. \end{aligned} \tag{S37}$$

Moreover, a uniform potential-error bound on Ω is the same bound, multiplied by E_s, on every momentum fiber above that cube. No positive density in the momentum variables is needed for these identities.

Proof. The common kinetic and reference potential terms cancel in $H_\theta - H_*$. Its physical configuration gradient is $(E_s/\ell)g_\theta$, and its momentum gradient is zero. Applying the canonical matrix $J = \left(\begin{smallmatrix} 0 & I_3 \\ -I_3 & 0 \end{smallmatrix}\right)$ gives the first identity. The integrand is independent of P, so integration with respect to the joint law gives its integral against the configuration marginal. The potential statement follows from the same cancellation and the definition of $w_\theta - w_*$.

S3.2 A configuration-space version of the statistical energy bound

The statistical component used here is the uniform bounded-loss generalization inequality in Chen, Matsubara and Yaguchi, *KAM Theory Meets Statistical Learning Theory: Hamiltonian Neural Networks with Non-zero Training Loss*, AAAI 2022, Lemma 2. Their Poincare–Sobolev step is on the complete phase domain. Proposition S1 permits that step to be performed on Ω for the present model.

Let Θ be a hypothesis class fixed before sampling. Suppose $Z_1, \dots, Z_N$ are iid with law ν whose Lebesgue density satisfies $\rho \geq m > 0$ on Ω. For a fixed finite $p \geq 1$, suppose

$$0 \leq \|g_\theta(z)\|_p^p \leq \Lambda_p \quad (z \in \Omega,\ \theta \in \Theta).$$

Let $\overline{\mathfrak{R}}_{p,N}$ be a valid upper bound for the empirical Rademacher complexity of this true loss class, with the normalization of the cited lemma. If observed labels Y_i satisfy $\|Y_i - f_*(Z_i)\|_p \leq \eta$ for every i, set

$$\begin{aligned} \widehat{L}_{p,N}(\theta) &= N^{-1} \sum_i \|f_\theta(Z_i) - Y_i\|_p^p, \\ A_{p,N}(\theta, \delta) &= \left(\widehat{L}_{p,N}(\theta)^{1/p} + \eta\right)^p + 2\overline{\mathfrak{R}}_{p,N} + 3\Lambda_p \sqrt{\frac{2\log(4/\delta)}{N}}. \end{aligned} \tag{S38}$$

With probability at least $1 - \delta$, simultaneously for all θ,

$$\mathbb{E}_\nu \|g_\theta\|_p^p \leq A_{p,N}(\theta, \delta). \tag{S39}$$

Indeed, apply the cited lemma to the iid pairs $(Z_i, f_*(Z_i))$. The triangle inequality for the finite-sample L^p norm bounds the unobserved true empirical loss by the first term of (S38). This argument allows arbitrary label errors satisfying the stated deterministic cap; it does not require those errors to be iid. The class bounds and the complexity bound must still be justified for the whole fixed class.

Theorem S2. Three-dimensional energy reception. *Assume $p > 3$, $w_*, w_\theta \in W^{1,p}(\Omega)$, and the preceding statistical hypotheses. Define*

$$e_\theta = w_\theta - w_* - \frac{1}{|\Omega|} \int_\Omega (w_\theta - w_*)\, dz.$$

On the event (S39), uniformly in θ,

$$\|e_\theta\|_{C^0(\Omega)} \leq C_{p,R} \left(\frac{A_{p,N}(\theta, \delta)}{m|\Omega|}\right)^{1/p}, \qquad C_{p,R} = 2R\, 3^{1-1/p} \frac{p}{p-3}. \tag{S40}$$

The physical error is E_s times this quantity on every momentum fiber. If $W_\theta(0) = W_*(0) = 0$, the absolute error in this fixed physical energy gauge is at most twice the right-hand side of (S40), multiplied by E_s.

Proof. First take a continuously differentiable scalar u. Convexity of the cube and the segment identity imply, for $z \in \Omega$,

$$|u(z) - \overline{u}| \leq \operatorname{diam}_{p'}(\Omega) \int_0^1 \frac{1}{|\Omega|} \int_\Omega \|\nabla u(z + t(y - z))\|_p \, dy\, dt, \qquad p' = \frac{p}{p-1}.$$

Holder's inequality for the inner probability integral and the change of variables $x = z + t(y - z)$ bound that inner integral by $t^{-3/p}(|\Omega|^{-1} \int_\Omega \|\nabla u\|_p^p)^{1/p}$. Here $z + t(\Omega - z) \subset \Omega$ and the Jacobian is t^3. The integral of $t^{-3/p}$ is $p/(p-3)$, and $\operatorname{diam}_{p'}(\Omega) = 2R3^{1-1/p}$. This proves the asserted deterministic inequality for smooth u. Smooth approximation in $W^{1,p}$ on a cube extends it to the continuous representative: apply the same inequality to differences of smooth approximants

after mean centering to obtain uniform convergence, and identify the limit with their L^p limit. Finally, the density and risk bounds give $\int_\Omega \|\nabla e_\theta\|_p^p \, dz \le A_{p,N}/m$. Proposition S1 gives the momentum-fiber statement. For the anchored gauge, $w_\theta(z) - w_*(z) = e_\theta(z) - e_\theta(0)$; the triangle inequality gives the additional factor two.

Thus $p = 4$ is sufficient for this first-order energy estimate; the corresponding six-dimensional argument requires $p > 6$. The pth root in (S40) cannot be omitted for a pth-power loss. Ordinary mean-square training has $p = 2$ and does not itself meet $p > 3$. If an independent uniform force bound $\|g_\theta\|_\infty \le B_0$ is available, then for $p \ge 2$, $\mathbb{E}\|g_\theta\|_p^p \le B_0^{p-2}\mathbb{E}\|g_\theta\|_2^2$; this is an additional boundedness argument, not equivalence of Sobolev embeddings in different dimensions.

S3.3 A mean-square route with a quantitative force regularity prior

The following deterministic result can use any valid population-risk bound, including (S39) with $p = 2$. It avoids replacing mean-square training by an unexecuted fourth-power objective, at the cost of an independent force Lipschitz bound.

Theorem S3. A boundary-valid three-dimensional cone estimate. *Let $g : \Omega \to \mathbb{R}^3$ be continuous and let every component be L-Lipschitz with respect to Euclidean distance. Suppose ν has density at least $m > 0$ and $\mathbb{E}_\nu\|g\|_2^2 \le a$. For every $0 < h \le R$,*

$$\|g\|_{\infty,\Omega} \le Lh + \sqrt{\frac{6a}{m\pi h^3}}. \tag{S41}$$

Consequently define the sufficient uniform-force allowance

$$\varepsilon_3(a; L, m, R) = \inf_{0<h\le R} \left(Lh + \sqrt{\frac{6a}{m\pi h^3}} \right). \tag{S42}$$

When $a = 0$ the allowance is zero. If $a, L > 0$, its minimizing radius is

$$h = \min\left\{ R, \left(\frac{3}{2L}\sqrt{\frac{6a}{m\pi}} \right)^{2/5} \right\}.$$

In the interior regime the allowance is proportional to $a^{1/5}$.

Proof. Fix $z \in \Omega$ and a component j, and put $b = |g_j(z)|$. For each coordinate choose the direction toward the more distant face of the cube. The corresponding octant of the radius-h ball centered at z is contained in Ω for $h \le R$, including when z is on a face, edge, or corner. Its volume is $\pi h^3/6$. Throughout this octant $|g_j| \ge (b - Lh)_+$, so

$$a \ge m\frac{\pi h^3}{6}(b - Lh)_+^2.$$

If $b \le Lh$, (S41) already holds; otherwise solve the last inequality for b. Take the supremum in z, j and then the infimum in h. If $a = 0$, continuity and a positive density imply $g = 0$ everywhere. For $a, L > 0$, differentiation of the displayed one-variable objective gives the stated minimizer. If $L = 0$, the objective is minimized at $h = R$; a sharper bound in this constant-field case is $\|g\|_{\infty,\Omega} \le \sqrt{a}$.

For a d-dimensional cube the same proof uses the inward orthant volume $c_d h^d$, where $c_d = |B_1^d|/2^d$, and gives $\inf_h\{Lh + \sqrt{a/(mc_d h^d)}\}$. Its interior risk exponent is $1/(d+2)$: $1/5$ for three configuration variables and $1/8$ for six phase variables. This improves the interpolation exponent. It does not remove the need for iid configuration sampling, a density lower bound, class-wide bounds, or a small observed training loss.

If $g_\theta = \nabla(w_\theta - w_*)$ has the allowance ε, integration on line segments gives

$$\sup_\Omega |w_\theta - w_* - w_\theta(0) + w_*(0)| \le 3R\varepsilon. \tag{S43}$$

For the mean-zero energy gauge, the sufficient bound is instead $6R\varepsilon$. Both follow by integrating the componentwise gradient bound along a segment in the convex cube, and, for the latter, averaging the resulting difference between two points.

S3.4 Reception of the Hessian and the cubic and quartic jets

Small force error does not alone control the derivatives needed by the normal form. The next result lists the additional orders separately. For a multi-index α, $|\alpha|$ denotes its total degree, $\alpha! = \prod_i \alpha_i!$, and D^α denotes the corresponding ordinary real partial derivative.

Theorem S4. Low-order derivative reception. *Suppose $g = \nabla(w_\theta - w_*)$ and $\|g\|_{\infty,\Omega} \le \varepsilon$. For $k \in \{1,2,3\}$, assume $g \in C^{k+1}(\Omega)$ and that every component derivative of total order $k+1$ has absolute value at most B_{k+1} on Ω. For $|\alpha| = k$, every component satisfies*

$$|D^\alpha g_j(0)| \le J_k(\varepsilon, B_{k+1}, R) := \inf_{0<h\le R/k} \left(\frac{2^k \varepsilon}{h^k} + \frac{k B_{k+1} h}{2} \right). \tag{S44}$$

When $\varepsilon, B_{k+1} > 0$, take

$$h = \min\{R/k, (2^{k+1}\varepsilon/B_{k+1})^{1/(k+1)}\}.$$

In the interior regime the bound equals $(k+1)B_{k+1}^{k/(k+1)}\varepsilon^{1/(k+1)}$.

Let $c_{\theta,\beta}$ and $c_{*,\beta}$ be ordinary Taylor coefficients of the normalized potentials, so their definition includes division by $\beta!$. For $d = 2,3,4$ and $|\beta| = d$,

$$|c_{\theta,\beta} - c_{*,\beta}| \le \frac{J_{d-1}(\varepsilon, B_d, R)}{\beta!}. \tag{S45}$$

In particular, the normalized Hessian operator error is at most $3J_1$.

Proof. For a coordinate unit vector e_i define $\Delta_{h,i}u(z) = u(z + he_i) - u(z)$ and $\Delta_h^\alpha = \prod_i \Delta_{h,i}^{\alpha_i}$. Choose $i_1, \dots, i_k$ with α_i occurrences of coordinate i. Repeated use of the fundamental theorem of calculus gives

$$h^{-k}\Delta_h^\alpha g_j(0) = \int_{[0,1]^k} D^\alpha g_j \left(h \sum_{r=1}^k t_r e_{i_r} \right) dt.$$

All evaluation points and their segments from zero lie in Ω when $h \le R/k$. The derivative bounds imply that the integrand differs from $D^\alpha g_j(0)$ by at most $B_{k+1} h \sum_r t_r$. Its integral is at most $kB_{k+1}h/2$. On the other hand, the binomial coefficients in Δ_h^α have absolute sum 2^k, so $|\Delta_h^\alpha g_j(0)| \le 2^k\varepsilon$. Combining the two estimates proves (S44). Direct minimization gives the stated radius and interior value. Zero bounds follow by taking limits.

For $|\beta| = d$, choose j with $\beta_j > 0$ and set $\alpha = \beta - e_j$. Then $D^\beta(w_\theta - w_*) = D^\alpha g_j$. Divide by $\beta!$ to obtain (S45). Every Hessian entry is at most J_1; its maximum row sum and hence its symmetric operator norm are at most $3J_1$.

The three required priors are bounds on $D^3(w_\theta - w_*)$, $D^4(w_\theta - w_*)$, and $D^5(w_\theta - w_*)$, respectively. A fifth-derivative bound alone is not substituted for the third- and fourth-derivative bounds. A certified sum of the candidate and target bounds suffices for each order. The force Lipschitz constant in Theorem S3 instead concerns $D^2(w_\theta - w_*)$ and is a separate, lower-order prior. For physical derivatives the degree-d allowance is multiplied by $E_s\ell^{-d}$; for coefficients of $W(\ell z)$ it is multiplied only by E_s.

Here is an explicit spectral consequence. Suppose the physical target Hessian $K_* = D_q^2(U_0 + W_*)(0)$ is positive definite. Set

$$\zeta = 3E_s\ell^{-2}J_1, \qquad \upsilon = \|A^{-1}\|_2\zeta. \tag{S46}$$

Then $\zeta < \lambda_{\min}(K_*)$ implies $K_\theta > 0$. If $\omega_{*,1} \le \omega_{*,2} \le \omega_{*,3}$ are the positive target frequencies and $\upsilon < \omega_{*,1}^2$, the corresponding ordered candidate frequencies satisfy

$$|\omega_{\theta,j} - \omega_{*,j}| \le \frac{\upsilon}{\sqrt{\omega_{*,1}^2 - \upsilon} + \omega_{*,1}} =: \Delta_\omega. \tag{S47}$$

Indeed, the mass-normalized symmetric Hessian perturbation has norm at most υ. The variational eigenvalue bounds therefore give $|\omega_{\theta,j}^2 - \omega_{*,j}^2| \le \upsilon$; factor the difference of squares to obtain (S47). Thus a positive target divisor margin

$$\Delta_* := \min_{k\in\mathbb{Z}^3,\, 1\le|k|_1\le 4} |k\cdot\omega_*|$$

remains positive for the candidate if $4\Delta_\omega < \Delta_*$. This finite-order condition supports the degree-four normal form; it is not an all-integer Diophantine assertion.

The six quadratic, ten cubic, and fifteen quartic coefficient intervals in (S45), with their physical scales, can be passed to a spectrum and Birkhoff normal-form verifier. The verifier must include the cubic homological correction to the quartic term and prove a nonzero twist determinant uniformly over the admitted intervals. A center-point determinant is insufficient for the whole family. If these conditions and the separate analytic hypotheses of the local elliptic KAM criterion hold for a selected candidate with an equilibrium at the origin, that criterion applies to this same candidate. The learning estimates supply accuracy and jet uncertainty; they do not replace any of those KAM hypotheses.

S3.5 What the dimension reduction establishes and what it leaves open

The real estimates above do not establish a quantitative pole-free complex neighborhood. A realified split-tanh network is continued in the complexified physical coordinates, with independent continued real and imaginary channels. Evaluating a real-part operation at complex arguments is not that continuation. A quantitative outer-domain bound, target regularity when needed, and the same selected weights are required for analytic interpolation or an analytic KAM certificate. Existing strip-interpolation and local normal-form results may then be used with their own stated hypotheses.

Two examples separate the necessary assumptions. First, on the fixed cube take $u_a(z) = \exp(-|z|_2^2/a^2)$ and subtract its mean. These are real-analytic functions. Their force mean-square integral is at most $3\pi^{3/2}a/2^{3/2}$, while their mean-zero sup norm does not tend to zero: the mean is at most $\pi^{3/2}a^3/|\Omega|$ and $u_a(0) = 1$. Thus three dimensions do not make $W^{1,2}$ embed in C^0, even within real-analytic functions without uniform regularity bounds. Second,

$$v_k(z) = k^{-2}[1 - \cos(kz_1)]$$

has uniformly vanishing force, a force Lipschitz constant at most one, and $\partial_{11}v_k(0) = 1$ for every k. Its fourth derivative at zero has magnitude k^2. Therefore neither a small force nor the cone estimate alone gives small Hessian and quartic errors. The separately stated derivative priors or a higher-order jet receiver are substantive.

The present statements implement the configuration-potential part of the proposed learning-to-KAM route. They do not assert that an optimizer attains a small population risk. In particular, repeated velocities at one configuration, two finite-difference step diagnostics at the same configuration, or correlated trajectory windows do not provide iid configuration coverage with a positive three-dimensional density. A force-label cap received at one prepared point is not a population-risk estimate. Actual selected weights must have their own training record and all the required class, label, regularity, analytic, and normal-form bounds. The arguments apply to real and realified complex potentials alike and do not prove a complex-network advantage or a safety statement for a driven dissipative full-state system.

S4 Real regularity bounds for the current energy-feature network

This statement concerns the selected 918-real-parameter architecture, with two complex $4 \to 12 \to 12 \to 1$ branches. Only the energy branch enters the following conservative bounds. The checkpoint identity is `f51bcea12c2f2640c0ea33c3493a691d8ab240578018a6d44a9f68904a1a60a9`. The result does not reuse the earlier 736-parameter phase-space class.

Let $w_N(z) = W_N(\ell z)/E_s$, with $\ell = 1/1000$ and $E_s = 3/500000000000$ J. The exported feature-energy scale E_* retains its exact saved binary value, separately from E_s. For a scalar function f, let $b_k(f)$ be an upper bound for the absolute value of every ordered mixed partial derivative of order k on the cube $|z|_\infty \le R$.

Proposition S5. A derivative majorant for the actual graph. *Let x_k bound every mixed derivative of order k of a scalar preactivation. Define the exponential partial Bell polynomials by*

$$B_{0,0} = 1, \qquad B_{k,j}(x) = \sum_{i=1}^{k-j+1} \binom{k-1}{i-1} x_i B_{k-i,j-1}(x),$$

with the other zero-index cases zero. Put $t_1 = 1, t_2 = 2, t_3 = 4, t_4 = 16, t_5 = 16$. Then

$$b_k(\tanh \circ f) \le \sum_{j=1}^{k} t_j B_{k,j}(x), \qquad 1 \le k \le 5.$$

For an affine layer, the corresponding derivative bounds are the absolute-weighted sums of its input bounds. These two operations, applied to the realification of the complex layers and to the quadratic energy feature, bound $D^2 w_N, D^3 w_N, D^4 w_N, D^5 w_N$ on the entire cube. The subtraction of the three first-feature derivatives at the origin changes only the first derivative and the constant term.

Proof. Write $t = \tanh x$, so $|t| \le 1$ for real x. The first five derivative polynomials factor as

$$1 - t^2, \quad -2t(1 - t^2), \quad -2(1 - t^2)(1 - 3t^2), \quad 8t(1 - t^2)(2 - 3t^2), \quad 8(1 - t^2)(2 - 15t^2 + 15t^4).$$

The stated bounds follow from $0 \le t^2 \le 1$; for the last expression, $2 - 15u + 15u^2 \in [-7/4, 2]$ when $u \in [0, 1]$. The multivariate chain rule is the sum over partitions of the k ordered differentiation positions. Grouping partitions by their block sizes gives the displayed Bell recurrence. Absolute values and the triangle inequality give the affine operation.

For $e(z) = \ell^2 z^{\mathsf{T}} K_3 z/2$, the feature bounds used are

$$b_1(e) \le R\ell^2 \max_i \sum_j |(K_3)_{ij}|, \qquad b_2(e) \le \ell^2 \max_{i,j} |(K_3)_{ij}|, \qquad b_k(e) = 0 \ (k > 2).$$

The first three features have first derivative at most ℓ/Q_j and no higher derivatives. Divide the fourth feature by the exact E_*, propagate through the two realified hidden layers and the scalar readout, multiply by E_*/E_s, and add $|c_{\mathrm{E}}| b_k(e)/E_s$. This is precisely the graph used for the selected potential. Its constant and linear subtractions do not affect orders two through five. All reported arithmetic uses rational numbers. Absolute stiffness entries are rounded upward to dyadics before the recurrence, and final bounds are rounded upward once more. Every operation in the majorant is monotone in these nonnegative inputs, so both roundings preserve the inequalities.

The resulting conservative upward decimal bounds for the same selected weights are

R	$b_2(w_N)$	$b_3(w_N)$	$b_4(w_N)$	$b_5(w_N)$
$1/2$	0.324211	0.0108748	0.0376192	0.154884
$1/64$	0.322470	0.00368385	0.0136982	0.0339880

These are bounds on normalized derivatives, not ordinary Taylor coefficients or measured errors. For an independently justified target bound $b_k(w_*)$, the sum $b_k(w_N) + b_k(w_*)$ bounds the error derivative of order k. Thus the candidate side of the separate third-, fourth-, and fifth-derivative assumptions in a force-to-jet reception theorem can be received without substituting one derivative order for another. No target bound follows from the candidate graph.

Two containing classes have also been bounded by the same recurrence: first, frozen hidden layers with realified readout ℓ^1 norm at most one and $|c_{\mathrm{E}}| \le 1$; second, the complete current graph with every trainable real parameter component at most two in absolute value and $|c_{\mathrm{E}}| \le 1$. The selected weights belong to both classes. These containing classes were defined after the preceding training; they do not retroactively provide a same-data selection event. Fresh independent validation can instead treat the selected output as fixed. Neither class construction supplies the unknown target's regularity, the sampling density, a population risk bound, a pole-free complex neighborhood for every member, or physical-target KAM.

S5 Finite-trajectory conservative-force reception

S5.1 Removing finite-window bias without assuming exact observations

The preceding finite label is an average along a trajectory. It is not an instantaneous force measurement. This distinction persists even when the momentum increment and the magnetic integral are computed exactly. Let $s_\sigma(t)$ be the dimensionless projected sum of the unknown conservative force and dissipative force on the physical trajectory with sign label $\sigma \in \{+,-\}$. More precisely, for the full native model let

$$s_\sigma(t) = \frac{\ell}{E_s} V^T \{\nabla(E_{\mathrm{el},0} + \mathcal{W}_*)(x_\sigma(t)) + D_*(x_\sigma(t))v_\sigma(t)\}.$$

Here $E_{\mathrm{el},0}(x) = x^T K_0 x/2$ is the nominal elastic potential and $\mathcal{W}_*$ is its unknown full-coordinate correction; its reduced restriction is $W_*(q) = \mathcal{W}_*(Vq)$. The symbol E_s remains the constant energy scale. Put

$$\begin{aligned} P_\sigma &= (\ell/E_s)V^T M v_\sigma, \\ m_\sigma &= (\ell/E_s)V^T b\{\kappa \sin(b^T x_\sigma) - \mu u_\sigma \cos(b^T x_\sigma)\}, \\ f_* &= (\ell/E_s)V^T \nabla \mathcal{W}_*(x_{\mathrm{prep}}). \end{aligned}$$

The sign convention is $\dot{P}_\sigma + s_\sigma + m_\sigma = 0$; the input u_σ is the applied transverse magnetic field. Thus an exact observed-momentum expression is

$$Y(T) = -\frac{1}{2}\sum_{\sigma=+,-}\left\{\frac{P_\sigma(T) - P_\sigma(0)}{T} + \frac{1}{T}\int_0^T m_\sigma(t)\,dt\right\} - f_0.$$

The magnetic quantities B_m, ϵ_m below refer to this scaled vector m_σ, and ϵ_p refers to P_σ. The prepared full-coordinate position may include saved rounding; the additional transfer to an ideal reduced design node is stated below. The known magnetic force is accounted for separately. Define $S = (s_+ + s_-)/2$ and $f_0 = \ell V^T K_0 x_{\mathrm{prep}}/E_s$. With exact momentum increments and exact magnetic integrals, the paired label is

$$Y(T) = \frac{1}{T}\int_0^T S(t)\,dt - f_0.$$

This identity uses the physical equations. Exact initial-position and opposite-velocity symmetry would remove the initial dissipative term; the following result instead retains an explicit initial allowance.

Proposition S6. A finite-window force receiver. *Suppose $S \in C^3([0,4T])$, $T > 0$, and $\sup_{[0,4T]} \|S'''\|_\infty \le B_3$. Put*

$$\mathcal{R}_T Y = \tfrac{8}{3}Y(T) - 2Y(2T) + \tfrac{1}{3}Y(4T).$$

If f_* is the desired conservative correction force at the prepared configuration and $\|S(0) - f_0 - f_*\|_\infty \le I_0$, then

$$\|\mathcal{R}_T Y - f_*\|_\infty \le I_0 + \tfrac{1}{3}B_3 T^3. \tag{FW1}$$

For a receiver based on saved endpoint momenta and composite trapezoids, assume additionally that: each scaled projected endpoint momentum has error at most ϵ_p; each sampled projected magnetic force has error at most ϵ_m; and the exact projected magnetic force on each signed trajectory has second derivative bounded by B_m. Use exact times, with endpoints $T, 2T, 4T$ and maximum panel width h. Then the observed receiver satisfies

$$\|\widehat{\mathcal{R}_T Y} - f_*\|_\infty \le I_0 + \frac{B_3 T^3}{3} + \frac{5B_m h^2}{12} + 5\epsilon_m + \frac{11\epsilon_p}{2T}. \tag{FW2}$$

All bounds concern the same physical model, signed trajectories and normalization. None of $I_0, B_3, B_m, \epsilon_p, \epsilon_m$ is inferred solely from a small observed weak residual or a coarse–fine difference.

Proof. The three weights sum to one. Their first and second moments at $(T, 2T, 4T)$ vanish. Thus constants are preserved and the linear and quadratic temporal terms are removed. Taylor's formula with integral remainder gives

$$\mathcal{R}_T Y - (S(0) - f_0) = \int_0^{4T} K_T(u) S'''(u)\, du,$$

where, writing $v = u/T$, $K_T(u) = T^2 k(v)$ and

$$k(v) = \begin{cases} v^2(12 - 7v)/24, & 0 \le v \le 1, \\ \{(4 - v)^3 - 12(2 - v)^3\}/72, & 1 \le v \le 2, \\ (4 - v)^3/72, & 2 \le v \le 4. \end{cases}$$

This kernel is nonnegative: on $[1, 2]$ use $4 - v \ge 3(2 - v) \ge 0$; the other pieces have the stated sign directly. Exact integration gives $\int_0^4 k(v)\, dv = 1/3$. This proves (FW1), including the initial allowance.

The absolute sum of the averaging weights is 5. A composite trapezoidal average has error at most $B_m h^2/12$, and the sampled-force error contributes at most ϵ_m. Averaging the two signs retains these common bounds. For either sign, the momentum combination has endpoint coefficients, in the order $0, T, 2T, 4T$,

$$\frac{1}{T}\left(-\tfrac{7}{4},\ \tfrac{8}{3},\ -1,\ \tfrac{1}{12}\right).$$

Their absolute sum is $11/(2T)$. These facts give (FW2).

If the two signs have different exact saved clocks, each receiver must be formed on its own clock. Exact Lagrange weights are obtained from the three actual durations; the corresponding kernel and observation constants must then be received for those durations. Changing the negative-velocity label does not change the direction of time.

Interpretation of the saved-data diagnostics. The predeclared diagnostic evaluated all windows of $1, 2, 4, 8, 12, 16, 24, 32, 64, 128, 256$ panels, all specified first-order pairs, and all specified second-order triples. No model was refitted and no best window was selected. At direction 3 on the finer record, the old 12-panel label differs from the declared material force by about 2.15835×10^{-4} in its largest component. The decomposition contains an elastic time-average drift of about 2.16072×10^{-4}, an even dissipative contribution of about 3.32821×10^{-7}, and a negative averaged weak-residual contribution of magnitude about 5.69840×10^{-7}. These are signed vector terms; their maxima must not be added as if their signs and maximizing coordinates agreed.

The fixed $1, 2, 4$-panel formula reduces the observed discrepancy from the declared material force to about 2.32485×10^{-8} at direction 2 and 1.76758×10^{-7} at direction 3 on the finer records. These are evaluator diagnostics. In contrast, the first-order $12, 24$-panel formula makes the direction-3 discrepancy about 4.28863×10^{-4}, illustrating why a temporal-bias model matters. The old 12-panel momentum slope has endpoint-noise factor $1/(6T)$; the new formula has factor $11/(2T)$, which is 33 times larger. Reduced bias therefore comes with increased sensitivity to endpoint errors. The saved-data comparison supplies neither the continuous observation allowances nor the derivative bounds in (FW2).

The refined receiver uses the recorded observations and known native operators. Constitutive evaluator coefficients enter only the subsequent comparison. This calculation does not identify a general unknown quartic, satisfy iid sampling assumptions, or establish a KAM transfer. Its role is to expose and remove a specific finite-window bias before further configuration measurements are interpreted.

A fourth window and the regularity required by the jet receiver. The C^5 potential regularity used in the fourth-order jet receiver below also motivates one further temporal cancellation, provided the dissipative law and input have the corresponding regularity along the trajectories. For $S \in C^m([0, a_m T])$ define $A(t) = \int_0^1 S(ut)\, du$, including $A(0) = S(0)$. Then $\|A^{(m)}\|_\infty \le \mathcal{B}_m/(m + 1)$ when $\|S^{(m)}\|_\infty \le \mathcal{B}_m$. Polynomial interpolation at the positive nodes $a_1 T, \dots, a_m T$, evaluated at zero, consequently has error at most

$$\frac{\mathcal{B}_m T^m \prod_i a_i}{(m + 1)!}.$$

Indeed the scalar interpolation remainder applies on the convex hull of zero and all interpolation nodes, component by component. No statistical independence or measured derivative is used in this formula.

For $(a_1, a_2, a_3, a_4) = (1, 2, 4, 8)$ the weights are

$$(64/21, -8/3, 2/3, -1/21),$$

and the continuous temporal allowance is $8B_4T^4/15$. Under the endpoint, magnetic and initial allowances of Proposition S6, the four-window version obeys

$$\|\widehat{\mathcal{R}_T^{(4)}Y} - f_*\|_\infty \le I_0 + \frac{8B_4T^4}{15} + \frac{15B_mh^2}{28} + \frac{45\epsilon_m}{7} + \frac{45\epsilon_p}{7T}.$$

Here B_4 bounds the fourth time derivative of the paired nonmagnetic force, while B_m retains its meaning as a magnetic second-derivative bound. The magnetic weight norm is $45/7$. The combined momentum coefficients at $0, T, 2T, 4T, 8T$ are

$$T^{-1}(-15/8, 64/21, -4/3, 1/6, -1/168),$$

whose absolute sum is also $45/(7T)$. Relative to the three-window formula, endpoint-noise amplification increases by the factor $90/77$. This modest additional cost does not by itself guarantee a better result: the fourth temporal derivative and all observation allowances must still be justified. No observed temporal fit supplies those bounds.

The predeclared four-window diagnostic did not improve the fixed three-window result on either new configuration at either time step. For example, at direction 3 on the fine record its maximum declared-family discrepancy is approximately 1.77537×10^{-7}, compared with 1.76758×10^{-7} for three windows in the same floating diagnostic. These are evaluations of saved finite labels, not continuous-error bounds. The interval reception in Proposition S7 uses its own exact native arithmetic. The negative comparison motivates a separate time-step diagnostic rather than further increasing the extrapolation order.

Proposition S7. Certified arithmetic for a fixed finite-window force label. *Use the same exact native full matrices M, K_0, V, tip vector b, and $d = V^Tb$. For each signed observation record $\sigma \in \{+, -\}$, let $\tau_{\sigma,i}$ be its own saved clock and set*

$$p_{\sigma,i} = V^TMv_{\sigma,i}, \qquad m_{\sigma,i} = \kappa \sin(b^Tx_{\sigma,i}) - \mu B_{y,\sigma,i} \cos(b^Tx_{\sigma,i}).$$

The vectors $x_{\sigma,i}, v_{\sigma,i}$ are the observed full 64-coordinate vectors, assembled directly from the recorded w, θ, w_t, θ_t fields. At the fixed panel counts $n \in \{1, 2, 4\}$, define $T_{\sigma,n} = \tau_{\sigma,n} - \tau_{\sigma,0}$ and

$$A_{\sigma,n} = \frac{p_{\sigma,n} - p_{\sigma,0}}{T_{\sigma,n}} + \frac{d}{T_{\sigma,n}} \sum_{i=0}^{n-1} \frac{\tau_{\sigma,i+1} - \tau_{\sigma,i}}{2}(m_{\sigma,i} + m_{\sigma,i+1}).$$

All twelve saved clocks start exactly at zero and satisfy, as exact rational identities, $T_{\sigma,2} = 2T_{\sigma,1}$ and $T_{\sigma,4} = 4T_{\sigma,1}$. Their actual-clock Lagrange weights at zero are therefore $(8/3, -2, 1/3)$; their sum is one, their first two time moments vanish, and their absolute sum is five.

For each of the six prescribed signed pairs, let x_g^{prep} be that configuration's own prepared rounded full-coordinate position. With the fixed normalization $S = (10^{-3})/(6\,10^{-12}) = 500000000/3$, the stored-data label is

$$\ell_g = -\frac{S}{2} \sum_{\sigma\in\{+,-\}} \left(\frac{8}{3}A_{\sigma,1} - 2A_{\sigma,2} + \frac{1}{3}A_{\sigma,4}\right) - SV^TK_0x_g^{\mathrm{prep}}.$$

Exact rational native matrix arithmetic and Arb arithmetic at 256 bits enclose all six labels, using every signed trace's actual saved times in both the slope and trapezoid. No clock, prepared position, or sign is substituted from another configuration.

Only after the completed label process and its output hash are recorded, the evaluator reads the declared stiffness coefficient. The saved readbacks give $a_E = E_{\mathrm{mod}}/(5\,10^6) = 53/50$ in these cases. Against the declared initial formula

$$f_g^{\mathrm{decl}} = (a_E - 1)SV^TK_0x_g^{\mathrm{prep}},$$

the normalized maximum component errors obey the strict bounds

configuration	*coarse diagnostic*	*fine diagnostic*
original shifted	$9.469\,10^{-8}$	$2.239\,10^{-8}$
direction 2	$9.842\,10^{-8}$	$2.325\,10^{-8}$
direction 3	$7.478\,10^{-7}$	$1.768\,10^{-7}$

Each table entry is an exact rational upper bound on $\|\ell_g - f_g^{\mathrm{decl}}\|_\infty$. All six pairs use the same previously fixed panel counts; no pair or window is selected by error.

Proof. The receiver binds all twelve observation files, the prepared full-coordinate positions, and the native geometry by source hashes. It verifies the saved time and node identities and the native full field/DOF mapping. Binary64 observations and prepared positions are interpreted as exact saved numbers; the native mass is the exact sum $M_0 + M_r$. For every signed clock, the rational Lagrange calculation uses the three actual durations and verifies the constant, linear, and quadratic moment identities before computing the interval label. The label subprocess completes with exit code zero and writes an immutable hashed result before the evaluator subprocess is allowed to parse E_{mod}. The second process evaluates the declared formula and certifies the six displayed rational relaxations.

This result certifies arithmetic on stored finite observations and a subsequent comparison with a declared initial constitutive formula. The vanishing time moments are algebraic identities; no bound on a continuous-time extrapolation remainder, quadrature defect, native solver error, observation error, or unknown physical force follows from them. The records remain development and paired discretization diagnostics. No existing label, dataset purpose, or data split is changed; no training archive is exported, and no new training or native solve is performed.

The initial allowance and its recorded-state part. Use all 64 free transverse displacement/rotation coordinates in their fixed native ordering, and write $E_{\mathrm{el},0}(x) = x^T K_0 x/2$, $\lambda_F = \ell/E_s$ and $d = V^T b$. For the physical signed trajectories define the scaled full-state quantities

$$\begin{aligned} P_\sigma(t) &= \lambda_F V^T M v_\sigma(t), \\ m_\sigma(t) &= \lambda_F d\{\kappa \sin(b^T x_\sigma(t)) - \mu B_{y,\sigma}(t)\cos(b^T x_\sigma(t))\}, \\ g(x) &= \lambda_F V^T \nabla(E_{\mathrm{el},0} + \mathcal{W}_*)(x), \qquad R(x) = \lambda_F V^T D_*(x), \\ s_\sigma(t) &= g(x_\sigma(t)) + R(x_\sigma(t))v_\sigma(t). \end{aligned}$$

The full equation gives $\dot P_\sigma = -s_\sigma - m_\sigma$. Thus the momentum and magnetic terms use precisely the normalization of Proposition S6. At the prepared rounded position x_{prep}, put

$$f_0 = \lambda_F V^T K_0 x_{\mathrm{prep}}, \qquad f_* = g(x_{\mathrm{prep}}) - f_0.$$

Let $\widehat{x}_\sigma, \widehat{v}_\sigma$ denote the actual saved binary64 first-row full-coordinate values, rather than the intended initial conditions. Define

$$\widehat{\delta}_0 = \tfrac{1}{2}\sum_{\sigma=\pm}\{g(\widehat{x}_\sigma) + R(\widehat{x}_\sigma)\widehat{v}_\sigma\} - g(x_{\mathrm{prep}}), \qquad I_0^{\mathrm{rec}} = \|\widehat{\delta}_0\|_\infty.$$

This quantity alone does not bound the physical initial allowance. For example, suppose the physical initial states lie within $\delta_{x,\sigma}, \delta_{v,\sigma}$ of their saved values in the fixed coordinate norm. If g, R have Lipschitz bounds L_g, L_R on the relevant position neighborhoods and $\|R\|_\infty \le \overline{R}$ there, then a sufficient allowance is

$$I_0 = I_0^{\mathrm{rec}} + U_0, \qquad U_0 = \tfrac{1}{2}\sum_{\sigma=\pm}\{L_g\delta_{x,\sigma} + L_R\delta_{x,\sigma}\|\widehat{v}_\sigma\|_\infty + \overline{R}\delta_{v,\sigma}\}.$$

Indeed, insert and subtract the recorded-state forces and use $R(x)v - R(\widehat{x})\widehat{v} = R(x)(v-\widehat{v}) + (R(x) - R(\widehat{x}))\widehat{v}$. This proves $\|S(0) - f_0 - f_*\|_\infty \le I_0$ without imposing exact position or opposite-velocity symmetry.

Exact reception for the declared linear family. For the saved evaluator family,

$$\nabla(E_{\mathrm{el},0}+\mathcal{W}_*)(x)=a_E K_0 x, \qquad D_*=a_C D_0, \qquad D_0=\alpha_0 M_0,$$

and both damping matrices are constant. The existing generators fix the mesh and linear geometry, use the mass-proportional coefficient `alpha_mass`, and set the stiffness-proportional coefficient to zero. The rotary patch changes inertia only. The frozen native matrix construction independently defines $D_0=\alpha_0 M_0$ before runtime rounding. Here the readbacks are $a_E=53/50$ and $a_C=5754906970707699/5004266931050173$. Put $G=\lambda_F a_E V^T K_0$ and $C=\lambda_F a_C V^T D_0$. The recorded defect is the exact rational vector

$$\widehat{\delta}_0=G\left(\frac{\widehat{x}_+ +\widehat{x}_-}{2}-x_{\mathrm{prep}}\right)+C\frac{\widehat{v}_+ +\widehat{v}_-}{2}.$$

If componentwise physical initial-error radii $\rho_{x,\sigma},\rho_{v,\sigma}\in\mathbb{R}^{64}_{\geq 0}$ are supplied, a sharper directly computable sufficient term is

$$U_0=\left\|\frac{|G|}{2}(\rho_{x,+}+\rho_{x,-})+\frac{|C|}{2}(\rho_{v,+}+\rho_{v,-})\right\|_\infty .$$

These radii have not been certified for the current physical observations.

Exact native rational arithmetic on all twelve saved numerical initial states gives the following strict rational upper bounds. Every sign and time-step record is received separately, with no projection or replacement by x_{prep}.

	I_0^{rec}		δ_{prep}
configuration	coarse	fine	
original shifted	$9.871\,10^{-15}$	$2.861\,10^{-15}$	$1.606\,10^{-18}$
direction 2	$2.782\,10^{-14}$	$1.390\,10^{-14}$	$1.907\,10^{-18}$
direction 3	$1.125\,10^{-13}$	$5.730\,10^{-14}$	$3.931\,10^{-18}$

The last column is a distinct preparation-transfer allowance. For each configuration, the frozen exact modal preparation $q=\ell z$ gives $x_{\mathrm{ideal}}=V\ell z$, whereas the stored label uses its own rounded x_{prep}. For the declared correction force,

$$\delta_{\mathrm{prep}}=\|\lambda_F V^T(a_E-1)K_0(x_{\mathrm{prep}}-V\ell z)\|_\infty .$$

Both time-step records share this preparation allowance. Adding it transfers the force-error budget from the rounded configuration to the ideal jet point; a point representation of an interval-certified label also requires its certified arithmetic radius. Arb comparison shows that each present fixed three-window discrepancy is strictly greater than its corresponding I_0^{rec} upper bound. Consequently the recorded initial pairing defect cannot by itself explain those discrepancies. This comparison does not bound U_0 or assign the remaining discrepancy to a particular source.

Endpoint momentum accuracy required by the force budget. For the FW2 majorant of Proposition S6 to fit a force tolerance η, its nonnegative allowances must satisfy

$$\epsilon_p\leq\frac{2T}{11}\left(\eta-I_0^{\mathrm{rec}}-U_0-\frac{B_3T^3}{3}-\frac{5B_m h^2}{12}-5\epsilon_m\right),$$

with a nonnegative bracket. This is a requirement on that certified error budget, not an assertion that cancellations in the actual error are impossible. For every current record the exact first saved duration and coefficient are

$$T=\frac{5902958103587057}{75557863725914323419136}\ \mathrm{s},$$
$$\frac{2T}{11}=\frac{5902958103587057}{41556825049252877805248}\ \mathrm{s}\simeq 1.420454545\,10^{-8}\ \mathrm{s}.$$

After subtracting the recorded I_0^{rec} alone, a force tolerance 10^{-7} therefore leaves an optimistic scaled endpoint-momentum ceiling of approximately $1.42045\,10^{-15}$; the uninstantiated terms can only reduce it. No continuous solver or observation accuracy at that scale has been certified by this calculation.

Derivative formulas without an asserted derivative bound. For the declared full equation set $A = a_E K_0$, $D = a_C D_0$, $\theta = b^T x$, $q_\theta = b^T v$, $r_\theta = b^T a$, and $m = \kappa \sin\theta - \mu B_y \cos\theta$. Then

$$
\begin{aligned}
a &= -M^{-1}(Ax + Dv + bm),\\
j &= -M^{-1}(Av + Da + b\dot{m}), \qquad k = -M^{-1}(Aa + Dj + b\ddot{m}),\\
\dot{m} &= (\kappa\cos\theta + \mu B_y \sin\theta) q_\theta - \mu \dot{B}_y \cos\theta,\\
\ddot{m} &= (\kappa\cos\theta + \mu B_y \sin\theta) r_\theta + (-\kappa\sin\theta + \mu B_y \cos\theta) q_\theta^2\\
&\quad + 2\mu \dot{B}_y \sin\theta\, q_\theta - \mu \ddot{B}_y \cos\theta,\\
s_\sigma''' &= \lambda_F V^T (A j_\sigma + D k_\sigma).
\end{aligned}
$$

Here $a = x''$, $j = x'''$, and $k = x''''$. A bound on these formulas over certified trajectory neighborhoods would supply B_3 and, from $\lambda_F d\ddot{m}$, the magnetic second-derivative allowance. No such neighborhood bounds are supplied by the saved initial-state arithmetic or by the coarse–fine records.

S5.2 A fixed-window time-refinement diagnostic

To test the time-discretization sensitivity of the stored-data receiver in S7, two additional native COMSOL solves were declared before execution. They retain the third preparation $q = (0, 0, 1/2000)\,\mathrm{m}$ and the prescribed velocities $\dot{q} = \pm 100 q$. The source model, material parameters ($E = 5.3 \times 10^6\,\mathrm{Pa}$ and mass-damping coefficient $0.6389230223455499\,\mathrm{s}^{-1}$), drive, spatial mesh, initial-field expressions, generalized-α settings and tolerances were retained. Only the internal step was reduced from 3.90625×10^{-8} to $1.953125 \times 10^{-8}\,\mathrm{s}$. Both signs retain the same 257 requested output times over $20\,\mu\mathrm{s}$, with output spacing $7.8125 \times 10^{-8}\,\mathrm{s}$.

For every sign and internal step, the fixed receiver uses output indices $(1, 2, 4)$. Its own recorded increasing clock is interpreted exactly; the three actual durations satisfy the dyadic relations and yield the predeclared weights $(8/3, -2, 1/3)$. No window, weight or model parameter was selected from these results. An independent 256-bit Arb receiver reconstructed the projected momentum and magnetic quadrature from the saved displacement, rotation, velocity and drive observations. Material coefficients enter only the subsequent evaluator comparison.

Let $\widehat{f}_h$ denote the paired conservative correction label, and let $f_* = (\ell/E_s)(a_E - 1)K_{03} q$ be the correction in the declared material family, with $a_E = 53/50$. This diagnostic uses the ideal full-coordinate reference $x_{\mathrm{prep}} = Vq$; the preceding fixed stored receiver uses the prescribed rounded full-coordinate position. All three time steps are re-received under the same ideal convention and therefore are not bitwise the earlier rounded-reference quantity. An exact rational calculation bounds the corresponding change in the normalized total reference force by 6.944×10^{-17}. The finite-data arithmetic enclosures give the following upper bounds, rounded upward:

internal step h (s)	$\|\widehat{f}_h - f_*\|_\infty$
7.8125×10^{-8}	7.477611×10^{-7}
3.90625×10^{-8}	1.767437×10^{-7}
1.953125×10^{-8}	3.398021×10^{-8}

The maximum component of the difference between consecutive labels decreases from approximately 5.710174×10^{-7} to 1.427635×10^{-7}, a ratio of approximately 4.000. The finest discrepancy is approximately 5.20 times smaller than that at the previous fine step. Thus the previous observed discrepancy is not a demonstrated numerical floor.

The prescribed initial fields are fixed, but their realized initial states are retained rather than assumed exact. At the three steps, the signed-pair position differences are approximately 1.5625×10^{-14}, 7.8125×10^{-15} and 3.9062×10^{-15} in the native coordinate convention; the velocity antisymmetry defects are approximately 8.1400×10^{-8}, 4.0700×10^{-8} and 2.0350×10^{-8}. These changes can contribute to the observed discrete sensitivity. Both new native solves and the arithmetic receiver terminated with actual exit code zero; a separate 512-bit direct endpoint calculation encloses its results inside the

256-bit reception intervals. This experiment supplies a three-step diagnostic for one declared material and one preparation. It does not instantiate the continuous bounds in S6, separate all error sources, identify unknown constitutive physics, or certify a KAM transfer. No training or checkpoint update was performed.

S5.3 A posteriori enclosures for the recorded-state native IVPs

The preceding time-step comparisons do not certify the error of a continuous trajectory. To supply such a bound for a specified mathematical model, consider

$$M\ddot{x} + D\dot{x} + Kx + b\,g(b^T x, t) = 0, \qquad g(\theta, t) = \kappa \sin\theta - \mu B(t)\cos\theta,$$

where $M = M_0 + M_r$, $K = (53/50)K_0$, $D = \alpha_* M_0$ and $b = e_{64}$. The mass and stiffness entries are interpreted as the exact rational values of the saved finite-element matrices. The drive is the exact decimal law $B(t) = 10^{-4}\cos(11.41692154656479\,t)$. The coefficient α_* is the saved binary value of the material readback. These conventions specify a mathematical IVP; they do not identify the solver's compiled transcendental evaluations with exact arithmetic.

For each sign, let Q be the piecewise cubic Hermite curve through the saved full-coordinate positions and velocities, with $\dot{Q}$ its derivative. The comparison solution starts at that sign's own saved numerical state. Define

$$\begin{aligned} R_Q &= M\ddot{Q} + D\dot{Q} + KQ + b\,g(b^T Q, t), \\ r^2 &= (x-Q)^T K (x-Q) + (\dot{x}-\dot{Q})^T M (\dot{x}-\dot{Q}), \\ c_M^2 &= b^T M^{-1} b, \qquad c_K^2 = b^T K^{-1} b. \end{aligned}$$

Lemma S8. Residual-to-state enclosure. *Suppose M, K are symmetric positive definite, D is symmetric positive semidefinite, and $|B(t)| \le A$. On a time cell, assume $\sqrt{R_Q^T M^{-1} R_Q} \le \bar{\rho}$. Put $L_g = \kappa + \mu A$ and $a = L_g c_M c_K/2$. If $r(t_i) \le r_i$, then throughout that cell*

$$r(t_i + \tau) \le e^{a\tau} r_i + \bar{\rho}\frac{e^{a\tau} - 1}{a}, \qquad 0 \le \tau \le h_i. \tag{CT1}$$

The initial value is $r_0 = 0$ for the recorded-state IVP specified above.

Proof. Write $e = x - Q$. The error equation gives, almost everywhere,

$$r\dot{r} = -\dot{e}^T D\dot{e} - (b^T\dot{e})\{g(b^T x, t) - g(b^T Q, t)\} - \dot{e}^T R_Q.$$

The global force bound $|g(\theta, t) - g(\eta, t)| \le L_g|\theta - \eta|$ and the two energy-dual inequalities yield

$$\begin{aligned} |b^T e| \le c_K\|e\|_K, \qquad |b^T\dot{e}| \le c_M\|\dot{e}\|_M, \\ \|e\|_K\|\dot{e}\|_M \le r^2/2, \qquad |\dot{e}^T R_Q| \le r\bar{\rho}. \end{aligned}$$

Thus $\dot{r} \le ar + \bar{\rho}$. A regularized norm establishes the comparison also at $r = 0$. Since Q and $\dot{Q}$ are continuous across the saved nodes, integrating this inequality gives (CT1) without a jump contribution at cell boundaries.

Whole-cell residual reception. Use the degree-nine rational polynomial obtained by replacing the magnetic term in R_Q by

$$b\left\{\kappa(\theta - \theta^3/6) - \mu A(1 - \omega_B^2 t^2/2)(1 - \theta^2/2)\right\}, \qquad \theta = b^T Q.$$

Its Bernstein coefficient vectors β_k on each cell give the uniform mass-dual bound $\max_k \sqrt{\beta_k^T M^{-1}\beta_k}$, by convexity of that norm. If $|\theta| \le \Theta$ and $0 \le \omega_B t \le 1$, a valid additional mass-dual remainder is

$$c_M\left\{\frac{\kappa\Theta^5}{120} + \frac{\mu A}{24}\left(\Theta^4 + (\omega_B t_{i+1})^4\right)\right\}. \tag{CT2}$$

Indeed, the sine remainder is at most $\Theta^5/120$; the product of the two cosine approximations has error at most $(\Theta^4 + (\omega_B t)^4)/24$, since $|1 - \omega_B^2 t^2/2| \le 1$. All polynomial operations and matrix identities are rational. Interval square roots and exponentials supply the outward bounds in (CT1).

For the two half-fine direction-3 records, this reception verifies all eight cells of the fixed four-panel prefix. Exact positive elimination pivots establish definiteness of M, K, M_0. The final radii are bounded by 2.589070×10^{-13} and 2.688128×10^{-13} in the norm of Lemma S8. The growth constant satisfies $a < 479.200675\,\mathrm{s}^{-1}$. These bounds hold for the continuous mathematical IVPs, throughout each cell, rather than only at sampled residual values.

Limits of a generic endpoint estimate. At a saved node the scaled momentum error is at most

$$C_P r(t_i), \qquad C_P = (\ell/E_s) \max_j \sqrt{(V^T M V)_{jj}}.$$

Substitution into the fixed receiver gives a paired momentum allowance less than 0.035010, which exceeds the intended 10^{-7} force tolerance. This is a valid but insufficient upper bound, rather than a lower bound on the actual data error. A direct projected-force reception below retains the spatial and temporal cancellations discarded by this generic estimate.

The certificate concerns solutions starting at the saved numerical states. It supplies neither an error bound for physical measurements, nor a verification of the compiled COMSOL equation or its dense output. A physical initial-state discrepancy requires an additional nonzero initial allowance in (CT1). No learned potential jet or physical KAM transfer follows from the trajectory enclosure alone.

S5.4 Conditional force-derivative bounds from a trajectory tube

Consider the declared native full-coordinate system

$$M\ddot{x} + Kx + D\dot{x} + b\,g(b^T x, t) = 0, \qquad g(\theta, t) = \kappa \sin\theta - \mu B(t) \cos\theta,$$

where $x \in \mathbb{R}^{64}$, $b = e_{64}$, $M = M_0 + M_r$, $K = (53/50)K_0$ and $D = \alpha_* M_0$. The saved material readback gives the exact binary coefficient $\alpha_* = 5754906970707699/9007199254740992$. In particular its ratio to the saved nominal damping coefficient is $5754906970707699/5004266931050173$, rather than an assumed exact $23/20$. The specified drive is

$$B(t) = 10^{-4} \cos(11.41692154656479\, t).$$

Here the displayed decimal drive defines an exact continuous law; the helper also accepts explicit rational overrides if a different literal convention is adopted. This convention does not assert exact evaluation of that law by the numerical solver.

Put $v = \dot{x}$, $a = \ddot{x}$, $j = x^{(3)}$ and $k = x^{(4)}$. Differentiating the smooth declared equation gives

$$\begin{aligned} a &= -M^{-1}(Kx + Dv + bg), \\ j &= -M^{-1}(Kv + Da + b\dot{g}), \\ k &= -M^{-1}(Ka + Dj + b\ddot{g}). \end{aligned}$$

Let $\omega = b^T v$ and $\alpha = b^T a$. Then

$$\begin{aligned} \dot{g} &= (\kappa \cos\theta + \mu B \sin\theta)\omega - \mu B' \cos\theta, \\ \ddot{g} &= (\kappa \cos\theta + \mu B \sin\theta)\alpha + (-\kappa \sin\theta + \mu B \cos\theta)\omega^2 \\ &\quad + 2\mu B' \sin\theta\, \omega - \mu B'' \cos\theta. \end{aligned}$$

Consequently the normalized projected restoring-and-dissipative force $s = (\ell/E_s)V^T(Kx + Dv)$ and the separately treated magnetic force $m = (\ell/E_s)V^T b\, g$ satisfy

$$s^{(3)} = (\ell/E_s)V^T(Kj + Dk), \qquad m'' = (\ell/E_s)V^T b\, \ddot{g}.$$

Subtracting a constant nominal force from s does not change its third derivative.

Projection before interval bounds. Bounding all coordinates of j and k separately can discard useful spatial cancellation. For an arbitrary fixed rational λ, define

$$z = \begin{pmatrix} x \\ v \end{pmatrix}, \quad A_\lambda = \begin{pmatrix} 0 & I \\ -M^{-1}(K + \lambda b b^T) & -M^{-1}D \end{pmatrix}, \quad u = \begin{pmatrix} 0 \\ -M^{-1}b \end{pmatrix}, \quad r = g - \lambda\theta.$$

Then $z' = A_\lambda z + ur$ and, for every constant row c,

$$cz^{(n)} = cA_\lambda^n z + \sum_{q=0}^{n-1} cA_\lambda^{n-1-q} u\, r^{(q)}, \qquad 1 \le n \le 3. \tag{DB1}$$

The helper first forms these rational rows exactly. Its default $\lambda = \kappa$ puts the linear magnetic stiffness in A_λ. With $r_\theta = \kappa\cos\theta + \mu B\sin\theta - \lambda$ and $r_{\theta\theta} = -\kappa\sin\theta + \mu B\cos\theta$,

$$r' = r_\theta\omega - \mu B'\cos\theta, \qquad r'' = r_\theta\alpha + r_{\theta\theta}\omega^2 + 2\mu B'\sin\theta\,\omega - \mu B''\cos\theta.$$

The magnetic derivative is recovered as $g'' = \lambda\alpha + r''$. The quantity α itself follows from (DB1) with $c = (0, b^T)$ and $n = 1$; hence this procedure is not circular. For the three rows $c_i = (\ell/E_s)(V^TK, V^TD)_i$, (DB1) with $n = 3$ directly encloses $s_i^{(3)}$. Optional coordinate rows provide a, j, k when needed. Near zero, the evaluation uses $\cos\theta - 1 = -2\sin^2(\theta/2)$ and the degree-seven polynomial for $\sin\theta - \theta$, with rigorous remainder $|\theta|^9/9!$, to retain the linear cancellation.

Required tube and its exact dual bound. On each time segment, the caller must independently certify

$$(x - \widetilde{x})^T K(x - \widetilde{x}) + (v - \widetilde{v})^T M(v - \widetilde{v}) \le R^2. \tag{DB2}$$

The radius R is a norm radius, not an energy. If an energy error $\frac{1}{2}(\delta x^T K\delta x + \delta v^T M\delta v) \le E$ is supplied, one may use $R = \sqrt{2E}$. Exact positive LDL pivots verify that M and K are positive definite. For a row $c = (c_x, c_v)$, define

$$\rho(c)^2 = c_x K^{-1} c_x^T + c_v M^{-1} c_v^T.$$

The Cauchy–Schwarz inequality in this energy metric gives

$$|c(z - \widetilde{z})| \le R\rho(c). \tag{DB3}$$

For a polynomial center $\widetilde{z}(u)$, $0 \le u \le 1$, the scalar polynomial $c\widetilde{z}(u)$ is formed before bounding its range. Its exact Bernstein coefficients enclose that range; (DB3) supplies the additional tube interval. Interval sine and cosine, the exact drive derivatives and (DB1) then produce componentwise bounds $B_{3,i}$ for $s_i^{(3)}$ and $B_{m,i}$ for m_i''. For the paired force $S = (s_+ + s_-)/2$, the bound is $(B_{3,i,+} + B_{3,i,-})/2$. Bounds over several segments are obtained by their maxima. A common magnetic bound for both signs is obtained by taking the maximum over signs, components and segments.

The method supplies the derivative inputs required by S6 conditional on (DB2). A cubic Hermite curve through the saved observations does not establish (DB2), and neither its interpolation residual nor a coarse–fine difference alone certifies the initial error or a continuous trajectory tube. No numerical tube radius is certified in this component.

S5.5 Reception of the continuous derivative bounds

The completed energy-tube calculation encloses the two declared native full-coordinate IVPs on the first four saved output panels. Each IVP starts from its own saved initial state. Its uniform cell radii use the metric $\delta x^T K\delta x + \delta v^T M\delta v$ with $K = (53/50)K_0$ and $M = M_0 + M_r$. The received all-prefix radius of each cell is applied to the entire cell; a terminal radius is not mistaken for an endpoint-only bound.

Applying the frozen procedure in S5.4 to these verified tubes gives the following componentwise bounds for the paired third derivative, rounded upward:

$$(B_{3,1}, B_{3,2}, B_{3,3}) \le (2.993942 \times 10^{10},\ 4.270068 \times 10^{10},\ 6.523766 \times 10^{12}).$$

For each component, the maximum is first taken over all four cells of each sign and the two signed bounds are then averaged. A common bound on the normalized projected magnetic second derivative, over all cells, signs and components, is

$$B_m \le 2.604809 \times 10^{7}.$$

The actual signed clocks are received separately and their dyadic relations are verified. For their fixed first-panel duration $T \simeq 7.8125 \times 10^{-8}\,\mathrm{s}$, the derivative contributions to S6 satisfy

$$\frac{B_3 T^3}{3} \le 1.036925 \times 10^{-9}, \qquad \frac{5 B_m h^2}{12} \le 6.624371 \times 10^{-8},$$

where h is the largest actual saved panel width. Their sum is at most 6.728064×10^{-8}.

The tube producer and derivative receiver both terminated with actual exit code zero. This receives B_3 and B_m for the stated mathematical IVPs; it does not certify physical measurement error or the compiled or interpolated COMSOL trajectory. Moreover, the generic energy-to-momentum observation allowance remains approximately 0.03501, so the generic endpoint estimate alone does not establish a force tolerance of 10^{-7}. The derivative calculation neither sets the initial pairing allowance to zero nor claims that the full instantaneous-force error bound has been closed.

S5.6 Receiving a fixed label through projected force averages

Proposition S9. Reception through a projected force average. *Use all 64 free transverse displacement/rotation coordinates in the fixed native ordering. Let M, K be constant symmetric positive definite matrices, let D be constant symmetric positive semidefinite, let V_j denote column j of V, and let $\lambda_F = \ell / E_s > 0$. For the declared evaluator family these matrices are*

$$M = M_0 + M_r, \qquad K = a_E K_0, \qquad D = a_C D_0 = \alpha_{\mathrm{mass}} M_0,$$

with the native sum and the declared coefficients interpreted exactly. For each sign $\sigma \in \{+, -\}$, let x_σ be the solution of the declared recorded-state initial-value problem

$$M x_\sigma'' + D x_\sigma' + K x_\sigma + b\, m_\sigma(b^T x_\sigma, t) = 0, \qquad m_\sigma(\theta, t) = \kappa \sin\theta - \mu B_{y,\sigma}(t) \cos\theta.$$

Its initial position and velocity are, by definition, that sign's own saved first-row values. This defines a mathematical initial-value problem and does not identify the physical initial state with a saved measurement.

Let Q_σ be the piecewise cubic Hermite interpolant of that sign's saved positions and velocities, with Q_σ' interpolating the saved velocities. Use each sign's actual saved clock, and assume that the received exact clock identities are

$$t_{\sigma,0} = 0, \quad t_{\sigma,1} = T, \quad t_{\sigma,2} = 2T, \quad t_{\sigma,4} = 4T, \qquad T > 0.$$

The interior time $t_{\sigma,3}$ need only lie strictly between $2T$ and $4T$; it need not equal $3T$. Write $I_{\sigma,i} = [t_{\sigma,i}, t_{\sigma,i+1}]$ and $h_{\sigma,i} = t_{\sigma,i+1} - t_{\sigma,i}$ for $0 \le i \le 3$. Define

$$e_{x,\sigma} = x_\sigma - Q_\sigma, \qquad e_{v,\sigma} = x_\sigma' - Q_\sigma', \qquad r_\sigma^2 = e_{x,\sigma}^T K e_{x,\sigma} + e_{v,\sigma}^T M e_{v,\sigma}.$$

In particular, $e_{x,\sigma}(0) = e_{v,\sigma}(0) = 0$. Suppose certified nonnegative numbers $R_{\sigma,i}, J_{\sigma,i}$ satisfy

$$r_\sigma(t) \le R_{\sigma,i}, \qquad \int_0^t r_\sigma(s)\, ds \le J_{\sigma,i} \quad (t \in I_{\sigma,i}).$$

A sufficient choice from whole-cell energy bounds alone is $J_{\sigma,i} = \sum_{q=0}^{i} h_{\sigma,q} R_{\sigma,q}$.

Define the averaging kernel and the normalized nonmagnetic force by

$$\mathcal{W}_T(t)=\begin{cases}7/4, & 0\le t<T,\\ -11/12, & T\le t<2T,\\ 1/12, & 2T\le t\le 4T,\end{cases}\qquad \mathfrak{s}(z,w)=\lambda_F V^T(Kz+Dw).$$

Thus $k=(7/4,-11/12,1/12,1/12)$ *gives the value of* $\mathcal{W}_T$ *on the four saved cells, regardless of the location of* $t_{\sigma,3}$*. Set*

$$\begin{aligned}C_{Q,\sigma}&=\frac{1}{T}\sum_{i=0}^{3}k_i\int_{I_{\sigma,i}}\mathfrak{s}(Q_\sigma,Q'_\sigma)\,dt,\\ C_{\mathrm{true},\sigma}&=\frac{1}{T}\sum_{i=0}^{3}k_i\int_{I_{\sigma,i}}\mathfrak{s}(x_\sigma,x'_\sigma)\,dt,\\ C_Q&=\tfrac12(C_{Q,+}+C_{Q,-}),\qquad C_{\mathrm{true}}=\tfrac12(C_{\mathrm{true},+}+C_{\mathrm{true},-}).\end{aligned}$$

The integrands defining C_Q *are polynomials. For the received rational native matrices, normalization, clocks and saved values,* C_Q *is therefore an exact rational polynomial integral.*

Let $\widehat{F}$ *be the already fixed observation-only raw force receiver formed from projected momentum increments and magnetic trapezoids with averaging weights* $(8/3,-2,1/3)$*. Its label process is completed and hashed before declared evaluator coefficients are read. Let* x_{prep} *be a fixed reference preparation, either its own rounded position or the exact ideal reference* Vq*. Use the same reference for both the nominal subtraction and the declared target, and put*

$$f_0=\lambda_F V^T K_0 x_{\mathrm{prep}},\qquad \widehat{Y}=\widehat{F}-f_0,\qquad \Delta_Q=\widehat{Y}+f_0-C_Q=\widehat{F}-C_Q.$$

Changing the reference requires only the exact known nominal adjustment to $\widehat{Y}$*; it does not change the fixed raw label* $\widehat{F}$*. The same fixed raw receiver must be enclosed when receiving* Δ_Q*. In particular, if directed Arb arithmetic produces the interval* $\mathbf{\Delta}_{Q,j}$*, use* $d_{Q,j}=\sup\{|z|:z\in\mathbf{\Delta}_{Q,j}\}$ *as the upper bound on* $|\Delta_{Q,j}|$*. A point representation of an interval-certified label must retain its certified arithmetic radius in this enclosure.*

For each projected component define

$$\begin{aligned}A_{D,j}&=(V_j^TDM^{-1}DV_j)^{1/2},\\ A_{KM,j}&=(V_j^TKM^{-1}KV_j)^{1/2},\\ L_j&=(V_j^TKV_j+V_j^TDM^{-1}DV_j)^{1/2},\\ B_{\sigma,i,j}&=\min\{L_jR_{\sigma,i},\,A_{D,j}R_{\sigma,i}+A_{KM,j}J_{\sigma,i}\},\\ \Gamma_j&=\frac{\lambda_F}{2T}\sum_{\sigma\in\{+,-\}}\sum_{i=0}^{3}|k_i|h_{\sigma,i}B_{\sigma,i,j}.\end{aligned}$$

Then $|(C_Q-C_{\mathrm{true}})_j|\le\Gamma_j$*.*

Suppose, in addition, that

$$S(t)=\tfrac12\sum_{\sigma\in\{+,-\}}\mathfrak{s}(x_\sigma(t),x'_\sigma(t))\quad\textit{satisfies}\quad S\in C^3([0,4T]),\qquad\sup_{[0,4T]}\|S'''\|_\infty\le B_3.$$

The fixed saved-data receiver obeys the componentwise bound

$$\left|\widehat{Y}_j-(S_j(0)-f_{0,j})\right|\le|\Delta_{Q,j}|+\Gamma_j+\frac{B_3T^3}{3}\le d_{Q,j}+\Gamma_j+\frac{B_3T^3}{3}.$$

For the declared conservative correction at the chosen reference, define

$$f_*^{\mathrm{prep}}=\lambda_F V^T(K-K_0)x_{\mathrm{prep}},\qquad \delta_0^{\mathrm{rec}}=S(0)-\lambda_F V^TKx_{\mathrm{prep}},\qquad I_0^{\mathrm{rec}}=\|\delta_0^{\mathrm{rec}}\|_\infty.$$

Since the recorded-state initial values are the saved numbers, δ_0^{rec} is exactly computable for this declared linear family and its supplied exact reference. In particular,

$$|\widehat{Y}_j - f_{*,j}^{\mathrm{prep}}| \le d_{Q,j} + \Gamma_j + \frac{B_3 T^3}{3} + |\delta_{0,j}^{\mathrm{rec}}|,$$
$$\|\widehat{Y} - f_*^{\mathrm{prep}}\|_\infty \le \max_j (d_{Q,j} + \Gamma_j) + \frac{B_3 T^3}{3} + I_0^{\mathrm{rec}}.$$

Proof. The dual norm of the linear functional $(e_x, e_v) \mapsto V_j^T(Ke_x + De_v)$ in the product energy norm gives

$$|V_j^T(Ke_x + De_v)| \le \left(V_j^T K V_j + V_j^T D M^{-1} D V_j\right)^{1/2} \left(e_x^T K e_x + e_v^T M e_v\right)^{1/2} = L_j r.$$

Also $e_x' = e_v$ and $e_x(0) = 0$, so Cauchy–Schwarz in the mass norm yields

$$|V_j^T K e_x(t)| = \left| \int_0^t V_j^T K e_v(s)\, ds \right| \le A_{KM,j} \int_0^t r(s)\, ds,$$
$$|V_j^T D e_v(t)| \le A_{D,j} r(t).$$

Both estimates apply to the same error. On a whole cell their minimum is therefore the stated $B_{\sigma,i,j}$. Integrating against the absolute value of the kernel, then taking the half-sum of the two signs, proves $|(C_Q - C_{\mathrm{true}})_j| \le \Gamma_j$.

For every continuous vector function s, direct collection of the three averaging intervals gives

$$\frac{8}{3}\frac{1}{T}\int_0^T s(t)\, dt - 2\frac{1}{2T}\int_0^{2T} s(t)\, dt + \frac{1}{3}\frac{1}{4T}\int_0^{4T} s(t)\, dt = \frac{1}{T}\int_0^{4T} \mathcal{W}_T(t) s(t)\, dt.$$

Consequently C_{true} is the three-window average of S. The Peano-kernel bound of Proposition S6 therefore gives $\|C_{\mathrm{true}} - S(0)\|_\infty \le B_3 T^3/3$. Now use the exact decomposition

$$\widehat{Y} - (S(0) - f_0) = \Delta_Q + (C_Q - C_{\mathrm{true}}) + (C_{\mathrm{true}} - S(0)).$$

The componentwise triangle inequality proves the reception bound. Finally, $S(0) - f_0 = f_*^{\mathrm{prep}} + \delta_0^{\mathrm{rec}}$ gives both prepared-target bounds. The stated interval reception includes the finite arithmetic radius in $d_{Q,j}$ and hence preserves these inequalities.

This organization receives the fixed momentum-and-trapezoid label directly against a continuous nonmagnetic force average. Its Δ_Q already includes the discrepancy of that complete saved-data receiver from the Hermite force average. Separate endpoint-momentum, sampled-magnetic-force and magnetic-quadrature allowances are therefore not added again to this bound. A valid B_3 certificate for the same declared trajectories is still required.

The recorded initial defect is distinct from an unknown physical initial-state allowance. For example, if the same declared force law applies and independently certified componentwise physical initial radii are $\rho_{x,\sigma}, \rho_{v,\sigma}$, a sufficient additional term is

$$U_0 = \left\| \frac{\lambda_F |V^T K|}{2}(\rho_{x,+} + \rho_{x,-}) + \frac{\lambda_F |V^T D|}{2}(\rho_{v,+} + \rho_{v,-}) \right\|_\infty, \qquad I_0 \le I_0^{\mathrm{rec}} + U_0.$$

Here absolute values of matrices are componentwise, and I_0 denotes the physical initial force defect. No such radii, and hence no value of U_0, are established by this proposition. In particular, $e_x(0) = e_v(0) = 0$ for a recorded-state mathematical initial-value problem does not establish $U_0 = 0$ for a physical observation. A claim about a physical trajectory additionally requires initial-state, input and equation/model-discrepancy bounds that apply to that same trajectory; the declared recorded-state tubes alone do not supply them.

The scope is postlabel certification for the declared recorded-state initial-value problems using evaluator coefficients. The statement does not identify a general unknown constitutive force, certify a native compiled equation or its dense output, or establish a complex-network advantage. It changes neither the fixed saved raw label nor any data purpose or training split.

S5.7 Received continuous force bound at the direction-3 configuration

The fixed half-fine direction-3 receiver is now received by Proposition S9. The raw force label was already fixed by the saved observations before this certification. An independent 512-bit recomputation overlaps each of its earlier interval enclosures. The declared material coefficients enter only the subsequent certificate. They are not supplied as labels during that observation-only construction.

The reference preparation in this reception is the ideal point $\bar{x} = V(0, 0, 1/2000)^T$. Accordingly the same nominal force is subtracted there, and the recorded initial allowance includes the whole difference of the two saved initial states from this reference. The two mathematical solutions start at their respective saved first rows; no physical measurement is declared exact.

The components of the force-error budget are listed below. All displayed values are rounded upwards. The quantity $d_{Q,j}$ bounds the complete fixed raw-label minus Hermite nonmagnetic-average discrepancy; it is not a separately fitted residual. The tube term Γ_j uses the minimum of the two rigorously valid projected estimates in the proposition. Totals are rounded upward from the exact component sums; separately rounded displayed entries need not sum to the displayed total.

j	$d_{Q,j}$	Γ_j	$B_{3,j}T^3/3$	$\lvert\delta^{\mathrm{rec}}_{0,j}\rvert$	Total
1	2.999876×10^{-9}	1.502520×10^{-9}	4.758743×10^{-12}	9.950403×10^{-15}	4.507164×10^{-9}
2	4.277506×10^{-9}	6.079664×10^{-10}	6.787091×10^{-12}	1.418104×10^{-14}	4.892274×10^{-9}
3	3.434153×10^{-8}	9.209175×10^{-10}	1.036925×10^{-9}	2.889657×10^{-14}	3.629940×10^{-8}

The largest resulting normalized force-error bound is 3.629940×10^{-8}. Thus this one fixed receiver satisfies the declared 10^{-7} force tolerance for the two recorded-state mathematical IVPs.

This conclusion is stronger than a comparison of saved values with an algebraic material formula: it includes a whole-cell continuous trajectory enclosure and a third-derivative temporal-bias allowance. It remains a postlabel certificate for the specified material family. It supplies no physical measurement/model-discrepancy bound. The eight additional jet-design positions are received separately in Section S5.8. This two-IVP result does not validate a newly trained checkpoint, identify an arbitrary unknown quartic potential, or transfer the candidate-only KAM result to the physical target. The generic endpoint-based 0.035010 upper bound is retained as a valid negative result; the direct projected reception proves a different, sharper inequality for the same fixed data.

S5.8 Actual observations at all eleven design configurations

The eight additional positions of the exact fourth-order design were implemented in native COMSOL. Each position uses $q = \ell z^{(i)}$ and the two prescribed velocities $\dot{q} = \pm 25 q\,\mathrm{s}^{-1}$. The existing three positions retain their own earlier prescribed velocities and their finest available saved records. A fixed internal step of 1.953125×10^{-8} s is used for the sixteen new solves; each exports 257 full-field samples over $20\,\mu$s with the same source-decimal cosine drive and fixed shifted material. Exact preparation arithmetic bounds every new prescribed initial energy by 5.322879×10^{-12} J. This is a feasibility bound for the prescribed initial values, not a claim that their numerical realization has zero error. All sixteen native processes completed successfully, and the source MPH remained unchanged.

The observation-only label process receives each sign's own saved clock, positions, velocities and magnetic field. It uses the same fixed three windows and weights at every node, subtracting the nominal force at the exact ideal design point. The declared material coefficients do not enter the force labels. The generator material is public in its protocol, so this is not a hidden-target or blind test. Each label is then independently received against its own continuous recorded-state IVPs by Proposition S9. The resulting maximum normalized force allowances, rounded upwards, are

Node	Configuration	Force allowance
0	original	2.274829×10^{-8}
1	direction2	2.363062×10^{-8}
2	direction3	3.629940×10^{-8}
3	jet_node03	4.582830×10^{-8}
4	jet_node04	4.582371×10^{-8}
5	jet_node05	2.530279×10^{-8}
6	jet_node06	3.630746×10^{-8}
7	jet_node07	3.629510×10^{-8}
8	jet_node08	4.582309×10^{-8}
9	jet_node09	1.576921×10^{-8}
10	jet_node10	3.631862×10^{-8}

All eleven selected records meet the fixed 10^{-7} tolerance. The older direction-3 step-1 and step-2 bounds remain approximately 7.531×10^{-7} and 1.797×10^{-7} and do not meet that tolerance. They are retained rather than removed by a posteriori window selection. The three existing finest records were selected by the smallest saved internal step before calculating these bounds. The new eight groups have only their single predeclared step.

All eleven groups are development/training configurations. Neither the 257 time samples nor the two signs at a position are declared independent configuration samples. The smaller $25q$ velocity also reduces damping excitation relative to the earlier $100q$ directions; no improvement in damping identifiability is inferred solely from the added position count.

S6 From prepared forces to a low-order potential jet

Set $\ell = 10^{-3}\,\mathrm{m}$, $E_s = 6\,10^{-12}\,\mathrm{J}$, and $w(z) = W(\ell z)/E_s$. Here W is the conservative correction: the known nominal elastic and prescribed magnetic forces are removed with the same normalization. Thus observations approximate the positive gradient $\nabla_z w$, rather than the negative mechanical force. A prepared point means the restriction $x = V\ell z$ of the full native state; departures from this prescribed state belong to the label error.

Proposition S10. Three-point symmetric Hessian reception. *Suppose explicitly that $w(z) = z^T H z/2$ with a constant symmetric matrix H. Let*

$$z^{(0)} = (a,b,c)^T = \left(\frac{1}{2}, \frac{1256189}{200000000}, \frac{26951}{12500000}\right)^T, \quad z^{(1)} = (0, \tfrac{1}{2}, 0)^T, \quad z^{(2)} = (0, 0, \tfrac{1}{2})^T.$$

For $h = (H_{11}, H_{22}, H_{33}, H_{12}, H_{13}, H_{23})^T$, stack the three blocks

$$A(z) = \begin{pmatrix} z_1 & 0 & 0 & z_2 & z_3 & 0 \\ 0 & z_2 & 0 & z_1 & 0 & z_3 \\ 0 & 0 & z_3 & 0 & z_1 & z_2 \end{pmatrix}$$

into $A \in \mathbb{Q}^{9\times 6}$. This matrix has rank six. For $L = (A^T A)^{-1} A^T$, any finite labels $y = Ah + e$ with separately received component bounds $|e_m| \le \eta_m$ satisfy

$$|(Ly - h)_i| \le \sum_{m=1}^{9} |L_{im}| \eta_m.$$

The Hessian in physical q coordinates is $E_s \ell^{-2} H$.

Proof. The third and second axis points determine the off-diagonal entries, H_{22} and H_{33}; then $a \neq 0$ determines H_{11} from the first component at $z^{(0)}$. Thus A has full column rank and $LA = I_6$. Apply L to $y = Ah + e$ and use the triangle inequality. The certificate also verifies these identities exactly using all nine scalar observations.

The numerical application uses the existing fine-step signed labels at these three configurations. Each label is obtained from its actual finite observation window; it is not replaced by the force at the initial point. Its arithmetic radius is added to its individual received force error before applying L. The available numerical errors were evaluated against the declared

quadratic material family. They are therefore not general measurement-error certificates for an arbitrary unknown physical law. Proposition S10 allows arbitrary symmetric H only conditional on independently valid errors for that H. Without the quadratic assumption, third- and higher-order contributions cannot be silently absorbed into the Hessian.

Proposition S11. An eleven-point receiver for the complete fourth jet. *Assume $w \in C^5$ on a neighborhood of $[-1/2, 1/2]^3$, with $\nabla w(0) = 0$. Fix the additive gauge $w(0) = 0$, and write*

$$w(z) = \sum_{2 \le |\alpha| \le 4} c_\alpha z^\alpha + R_5(z).$$

Append the following eight positions to the preceding three:

$$\begin{array}{cccc} (\frac{1}{2}, -\frac{1}{4}, -\frac{1}{2}), & (-\frac{1}{2}, 0, -\frac{1}{2}), & (-\frac{1}{4}, \frac{1}{2}, -\frac{1}{4}), & (0, -\frac{1}{4}, \frac{1}{2}), \\ (\frac{1}{2}, \frac{1}{4}, \frac{1}{2}), & (-\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}), & (-\frac{1}{2}, \frac{1}{4}, \frac{1}{4}), & (-\frac{1}{2}, -\frac{1}{4}, \frac{1}{2}). \end{array}$$

For the 31 ordinary monomials of degrees two through four, define

$$G_{(i,j),\alpha} = \alpha_j (z^{(i)})^{\alpha - e_j}, \qquad J = (G^T G)^{-1} G^T.$$

The term is zero when $\alpha_j = 0$. Exact rational arithmetic gives $\operatorname{rank} G = 31$ and $JG = I_{31}$. Eleven configurations attain the necessary lower bound $3m \ge 31$.

If $y_{ij} = \partial_j w(z^{(i)}) + e_{ij}$, $|e_{ij}| \le \eta_{ij}$, and $|\partial_j R_5(z^{(i)})| \le \tau_{ij}$, then

$$|(Jy - c)_\alpha| \le \sum_{i,j} |J_{\alpha,(i,j)}| (\eta_{ij} + \tau_{ij}). \tag{S48}$$

For a common componentwise fifth-derivative bound $|D^\nu w| \le M_5$ for all $|\nu| = 5$ on the cube, one may use

$$\tau_{ij} = \frac{M_5}{24} \|z^{(i)}\|_1^4.$$

Proof. The exact 33×31 matrix and its 31×33 left inverse are stored in the certificate, which verifies $JG = I_{31}$ without a floating rank threshold. The gradient Taylor expansion gives $y = Gc + t + e$. Multiplication by J proves (S48). For each $\partial_j w$, apply the third-degree Taylor formula with fourth-order remainder along the segment from 0 to z. The multinomial identity gives $\sum_{|\nu|=4} |z|^\nu / \nu! = \|z\|_1^4 / 24$. This derives a gradient-tail bound directly; it does not differentiate an unquantified scalar remainder.

The maximal exact left-inverse row sums, rounded upward here, are 39.124439, 52.294639, and 125.879240 within degrees two, three, and four. They multiply a uniform scalar force-error cap at this fixed half-unit design. The old symmetric 33-configuration design has the smaller constants $25890/6589$, 4, and $104/9$ on the same cube, while also receiving three potentially nonzero linear coefficients. The new design minimizes configuration count under the explicit zero-gradient assumption, at the cost of greater deterministic noise amplification. Using the three existing points plus all eight equal-radius corners would not suffice: that matrix has rank 30, with null polynomial $z_1^4 - z_1^2/2$.

At the preceding three-configuration stage, only the first three positions had observations. That stage did not supply the other eight labels or their error caps. The actual reception below now supplies them for the declared quadratic family; a general unknown target still requires its own fifth-derivative bound. The certificate gives each coefficient's separate contribution from the existing nine scalar error bounds, from future errors, and from M_5. A new label procedure, including finite-window extrapolation, may supply different bounds at each position without changing this exact left inverse.

For consistency with a subsequent jet-to-normal-form calculation, $D^\alpha w(0) = \alpha! c_\alpha$: in particular, $H_{ii} = 2c_{2e_i}$ and $H_{ij} = c_{e_i + e_j}$ for $i \ne j$. The physical derivative is multiplied by $E_s \ell^{-|\alpha|}$. For a fixed candidate, its known signed finite-design tail alias can be subtracted as in the existing force-to-jet receiver; this never removes an unknown target tail or an observation error. The resulting interval jet must still satisfy the spectral, canonical-normalization, Birkhoff nondegeneracy and quantitative analytic KAM conditions uniformly. Neither full design rank nor this three-position quadratic reception is

itself a KAM conclusion or an identification of a full 64-coordinate unknown potential. The construction supplies the explicit local-jet interface required between prepared-force learning and those later gates.

As an initial-energy feasibility witness only, $q = \ell z$ and $\dot{q} = \pm 25 q\,\mathrm{s}^{-1}$ at the proposed positions give exact full-field preparations $x = Vq$, $v = V\dot{q}$. Under the already declared material interval, their initial energy is below $0.90(6\,10^{-12}\,\mathrm{J})$. At this earlier preparation stage, the velocity was only a feasibility witness: relative to $\pm 100q$, its initial linear damping-force signal is four times smaller and its kinetic energy is sixteen times smaller. A subsequent experimental choice must also meet a separate dissipation-excitation and observation error budget. No COMSOL solve or realized initial-state guarantee is included in this preparation calculation.

S7 Full-coordinate observability of the four-feature material jet

This section concerns an additional, explicitly stated target class. A general unknown potential on $\mathbb{R}^{64}$ need not factor through the four inputs used by the network. We suppose here that its material correction has the local form

$$W(x) = E_s \Phi(u(x), e(x)), \quad u(x) = Bx/\ell, \quad e(x) = E_0(x)/E_s, \quad E_0(x) = \tfrac{1}{2} x^T K_0 x, \tag{S49}$$

where $\ell = 1/1000$ m and $E_s = 3/500000000000$ J are exact decimal normalizations, $BV = I_3$, $K_3 = V^T K_0 V$, and K_0 is positive definite. The local function Φ is real analytic, has no constant or linear u terms, and is otherwise unrestricted. These are target assumptions, not conclusions from the finite observations. The selected network has this factorization by construction, with

$$\begin{aligned} \Phi_\theta(u, e) = c_E e + \frac{E_{\mathrm{net}}}{E_s} \Big[& N_\theta(Du, \lambda e) - N_\theta(0) \\ & - D_{s_q} N_\theta(0) Du \Big], \\ D = \mathrm{diag}(\ell/Q_j), \qquad & \lambda = E_s/E_{\mathrm{net}}. \end{aligned} \tag{S50}$$

Here Q_j, E_{net} keep their saved binary values interpreted as exact numbers. No equality of these binary scales with ℓ, E_s is asserted. The invertible constant rescaling does not change any dimensions below. The original 918-coordinate checkpoint and the newly refitted 914-coordinate comparison models share this feature geometry; no KAM certificate for one set of weights is transferred to another.

Assign weight one to each u_j and weight two to e. Denote by $\mathcal{P}_{2:4}$ the ordinary polynomial space spanned by $u^\alpha e^b$ with $2 \le |\alpha| + 2b \le 4$. It has

$$\dim \mathcal{P}_{2:4} = (6+1) + (10+3) + (15+6+1) = 42. \tag{S51}$$

These are coefficients of the material correction, before restoring the known elastic and magnetic potentials. Weighted degree is physical degree after substitution in (S49). Let

$$a(u) = \frac{\ell^2}{2E_s} u^T K_3 u, \qquad A(u, e) = e - a(u). \tag{S52}$$

On the trial space $x = V\ell u$, one has $e = a(u)$.

Proposition S12. The eleven unobserved feature-jet coordinates. *The restriction map*

$$\mathcal{R} : \mathcal{P}_{2:4} \longrightarrow \mathbb{R}[u_1, u_2, u_3]_{2:4}, \qquad \mathcal{R}P = P(u, a(u)),$$

is surjective of rank 31. *Its kernel is exactly*

$$\ker \mathcal{R} = A\,\mathcal{P}_{\le 2}, \qquad \mathcal{P}_{\le 2} = \mathrm{span}\{1, u_i, u_i u_j\ (i \le j), e\}, \tag{S53}$$

and has dimension 11. *In particular, complete knowledge of the restricted material jet does not identify the full-coordinate jet even inside this four-feature target class.*

Proof. The monomials with $b = 0$ give all 31 degree-two through degree-four polynomials, proving surjectivity. Polynomial division in the variable e by the monic polynomial $e - a(u)$ gives $P = AQ + P(u, a(u))$. Since a is weighted homogeneous of degree two, the quotient has weighted degree at most two; conversely every displayed AQ belongs to $\mathcal{P}_{2:4}$ and restricts to zero. Multiplication by the nonzero polynomial A is injective, and the quotient space has $1 + 3 + 6 + 1 = 11$ basis elements.

The existing eleven-node energetic design receives projected forces $V^T \nabla W(V \ell u_i)$ and the 31 coefficients obtained from those same force labels. Its exact 33-by-31 gradient design has a retained left inverse. Consequently it receives $\mathcal{R}P$, rather than all 42 coordinates of P. Every AQ has identically zero restricted potential and restricted gradient, not merely a zero value at these eleven points. This observation concerns the point/jet energy objective; it does not say that arbitrary AQ is invisible to all full-field trajectories or to the separate weak-residual loss.

Proposition S13. What complete forces at the old nodes could add. *Put $q_i = \ell u_i$ and*

$$n_i = (K_0 V - B^T K_3) q_i. \tag{S54}$$

Suppose the existing projected gradient design has rank 31. *Let $J = \{i : n_i \neq 0\}$, and let r be the rank of the evaluation matrix of all ten ordinary quadratic polynomials in u at the nodes in J. Then the design that evaluates the complete* 64*-coordinate gradients of $E_s P$ at the same nodes has rank exactly $31 + r$. If $r = 10$, its one-dimensional kernel is*

$$\operatorname{span}\{A^2\}. \tag{S55}$$

Proof. Vanishing of all complete forces implies vanishing of the projected forces, hence $P = AQ$ by the rank-31 assumption. Write $Q = c_0 + c^T u + u^T C u + c_e e$, with an ordinary quadratic convention for the six independent entries. At a trial-space node, $A = 0$ and

$$\nabla_x (E_s A Q) = Q(u_i, a(u_i)) n_i.$$

The scalar polynomials $Q(u, a(u))$ fill the entire ten-dimensional quadratic polynomial space; the map from the eleven coordinates of Q to that space has kernel $\operatorname{span}\{A\}$. The normal constraints therefore add exactly r independent rows. When $r = 10$, Q is a multiple of A, proving the last assertion.

The polynomial $A(x)^2$ has zero value and complete gradient on the whole trial space, while its full-coordinate physical fourth jet is nonzero. This is an observability statement. A nonzero quartic term does not by itself prove either singular or nonsingular Birkhoff twist; its effect on KAM nondegeneracy needs a separate normal-form calculation. Nor does exact algebraic rank imply usable reconstruction at the available noise level. The current strict force allowances cover the three projected channels, not 64 certified force components. Existing full-field position and velocity arrays are therefore a possible input to a new force receiver, not already received complete-force labels.

Proposition S14. Eleven normal labels, or one after full-force reuse. *There is a nonzero direction w satisfying $Bw = 0$ and $V^T K_0 w = 0$. Put $k_w = w^T K_0 w > 0$. On*

$$x = V \ell u + t w, \qquad e = a(u) + \delta, \qquad \delta = t^2 k_w / (2E_s) > 0, \tag{S56}$$

the normalized directional energy-gradient label is

$$\mathcal{L}_w W(u, t) := \frac{w^T \nabla W(x)}{t k_w} = \partial_e \Phi(u, a(u) + \delta). \tag{S57}$$

For an exactly weighted-degree-four target P, subtract any known restricted representative and write its remaining part as AQ. Take the ten quadratic interpolation nodes

$$0, \quad \pm h e_i \ (i = 1, 2, 3), \quad h(e_i + e_j) \ (i < j), \qquad h \neq 0,$$

at one positive offset δ_1, and one additional label at $u = 0$, $\delta_2 > 0$, $\delta_2 \neq \delta_1$. These eleven labels identify all eleven coefficients of Q. The absolute determinant of their design matrix, with the ordinary quadratic basis, is

$$16|h|^{15}|\delta_2 - \delta_1| > 0. \tag{S58}$$

Eleven scalar linear labels are necessary when only the restricted 31 coefficients have been received. If instead the old complete-force design has rank 41, one label (S57) at $u = 0$ and any $\delta > 0$ completes its rank to 42.

Proof. The stacked constraints on w have at most six independent rows in dimension 64, so their common kernel has dimension at least 58. Positive definiteness gives $k_w > 0$. Expansion of E_0 gives (S56). Along w, the coordinate u is constant and $D_w e = tk_w/E_s$, proving (S57). For $Q = c_0 + c^T u + u^T C u + c_e e$,

$$\partial_e (AQ)\big|_{e=a(u)+\delta} = c_0 + c^T u + u^T C u + c_e\{a(u) + 2\delta\}.$$

At fixed δ_1 the first ten rows receive an ordinary quadratic. The ten-node quadratic evaluation determinant has absolute value $8|h|^{15}$: the three signed-axis pairs give $2|h|^3$ each and the three mixed nodes give $|h|^2$ each. Subtracting the $u = 0$ first row from the last row isolates $2(\delta_2 - \delta_1)c_e$; the known $a(u)$ column is a combination of the quadratic columns. This proves (S58). The dimension lower bound is elementary rank–nullity. In the rank-41 case the remaining term is cA^2, whose normalized normal derivative at $u = 0$ is $2c\delta$.

This direction also proves that the 42 feature polynomials give 42 distinct physical jets: substitution $e = a(u) + k_w t^2/(2E_s)$ is injective, by its nonzero highest power of t^2. All native matrices are rational source objects, so a rational w can be constructed. Rational nonzero t_1, t_2 with different squares give positive rational offsets and an exact rational interpolation design; taking a square root is unnecessary for its construction.

Finite labels require a separately bounded remainder. For an analytic target write $\Phi = P + R$, where the physical terms of R start at degree five. At nonzero interpolation nodes, normal observations receive $\partial_e P + \partial_e R$, not the jet derivative alone. If received measurement/force-reconstruction errors are η_i and separate valid tail allowances are $\rho_i \geq |\partial_e R(u_i, a(u_i) + \delta_i)|$, then for the eleven-label design matrix D_N the coefficient error satisfies

$$|\widehat{q} - q| \leq |D_N^{-1}|(\eta + \rho).$$

Uncertainty of an already received restricted representative contributes its own additional propagated label allowance. Equivalently one can receive all 42 rows together with their joint error matrix. Analyticity alone supplies no numerical ρ_i. The prior $R = 0$ is a genuine finite-degree target restriction, and must not be inferred from the network architecture or from a restricted $M_5 = 0$ statement.

For example, set $s = t\sqrt{k_w/(2E_s)}$ and consider the dimensionless lifted remainder $\widetilde{R}(u, s) = R(u, a(u) + s^2)$. If every ordered fifth partial of the lifted target is bounded by M_5 on the relevant convex segments, its Taylor remainder obeys

$$|\partial_e R(u, a(u) + s^2)| \leq \frac{M_5(\|u\|_1 + |s|)^4}{48|s|}, \qquad s \neq 0.$$

This follows by Taylor's formula for the derivative in s, followed by division by $2s$. This is an additional target bound on the lifted four-dimensional domain, not a candidate-only derivative bound or a bound on the three-dimensional restriction.

Finally, the stipulated scalar family $W_* = \beta E_0$ corresponds to $\Phi_* = \beta e$. Its global form fixes the otherwise unobserved directions by assumption; current loss-to-parameter and family KAM certificates are valid precisely with that source prior. Relaxing it requires the additional observable rows and tail/error allowances above, then an independently received full-coordinate spectral and cubic-corrected twist criterion. This section proves neither that the new labels are already available nor that their eventual errors meet such a KAM tolerance.

Exact-source instance for the current full-coordinate model. For the retained native rational source, the elastic matrix has 64 positive exact LDL pivots, $BV = I_3$, and $V^T K_0 V = K_3$ exactly. The retained eleven nodes and their exact gradient

left inverse receive the rank-31 restriction in Proposition S12. An exact-source calculation gives

$$\begin{aligned} \operatorname{rank}(K_0V - B^TK_3) &= 3, \\ n_i \neq 0 \quad (i = 1, \dots, 11), & \\ \operatorname{rank}\big[u_i^\alpha\big]_{i=1,\dots,11,\ |\alpha|\le 2} &= 10. \end{aligned} \tag{S59}$$

Consequently the 704 complete-force rows at those same nodes would have rational rank 41, with kernel $\operatorname{span}\{A^2\}$. One additional normalized normal-gradient row at $u = 0$ and a positive off-trial energy offset raises the rank to 42. These statements describe the design, not an already received set of complete-force observations.

The rank calculation does not use a floating-point singular-value threshold. All source rationals are reduced modulo each of the two primes $p_1 = 1000000007$ and $p_2 = 1000000009$; every used denominator is nonzero in each field. The retained 41-by-41 full-force minors have residues 518511633 and 46316120, respectively. After the new normal row, the retained 42-by-42 minors have residues 541265412 and 980655374. A nonzero modular minor proves that the corresponding rational minor is nonzero. The exact nonzero vector A^2 annihilated by all 704 old rows supplies the matching upper bound 41; the augmented matrix has only 42 columns. The exact restriction and alias identities, the active quadratic evaluation rank 10, and the normal-geometry rank 3 were also checked.

For a constructive normal design, exact rational row reduction of $[B; V^TK_0]$ gives rank 6, so its common kernel has dimension 58. A retained rational w satisfies both zero constraints and $k_w = w^TK_0w > 0$. Taking $h = 1/4$, $t_1 = 2^{-16}$, and $t_2 = 2t_1$ gives

$$\delta_1 = \frac{k_w t_1^2}{2E_s} > 0, \qquad \delta_2 = 4\delta_1, \qquad |\det D_N| = 16h^{15}(\delta_2 - \delta_1) > 0.$$

The exact direction, all interpolation rows and their inverse, the selected modular minors, and the exact full-force design are retained as reproducibility artifacts. Physical admissibility of these new configurations and useful force-reconstruction precision remain to be received. No new COMSOL solution, force measurement, or training run is part of this calculation.

Thus reuse of complete forces at the old configurations, if they can be received with valid errors, could replace ten of the eleven new scalar labels in Proposition S14. The existing certified allowances still cover only the three projected force channels. Neither the rank result nor the retained full-field position/velocity histories supplies the missing complete-force error bounds or the analytic target-tail allowances. Membership of an unknown material in the four-feature class remains a separate model assumption, and no unknown-target KAM conclusion follows from these rank certificates alone.

S8 Normal-force bounds from the full-field observations

Use the exact source normalizations $\ell = 1/1000$ and $E_s = 3/500000000000$, and the four-feature notation $u = Bx/\ell$, $e = x^TK_0x/(2E_s)$, $a(u) = \ell^2u^TK_3u/(2E_s)$, and $A = e - a(u)$. For a fixed old node $q_i = \ell u_i$, define

$$x_i = Vq_i, \quad n_i = (K_0V - B^TK_3)q_i, \quad d_i = M^{-1}n_i, \quad c_i = d_i^Tn_i > 0, \quad r_i = d_i/c_i. \tag{S60}$$

The exact source has $n_i \neq 0$ at all eleven retained nodes. Since $B = (V^TMV)^{-1}V^TM$ and $BV = I$, one has $V^Tn_i = 0$, $Bd_i = 0$, and $r_i^TK_0x_i = 1$. Here r_i is a normalized force functional in the stored SI coordinate convention; it is not an ordinary SI displacement direction to be added to the state. The scalar label $r_i^T\nabla W$ is dimensionless. These are identities for the exact source matrices, not identities asserted for every rounded implementation buffer.

Proposition S15. A normal force component. *Suppose a material correction has the polynomial decomposition*

$$W(x) = E_s\{R(u) + A(u,e)Q(u,e)\},$$

where R is any restricted representative. Then

$$r_i^T\nabla W(x_i) = Q(u_i, a(u_i)). \tag{S61}$$

In particular, the representative contributes exactly zero. Writing $Q = S(u) + cA$ identifies an ordinary quadratic S with ten coordinates; the coefficient of A^2 remains unobservable on the trial space.

Proof. At x_i, $A = 0$ and $E_s \nabla_x A = n_i$, whereas $\nabla_x R = B^T \nabla_u R/\ell$. The identities $B r_i = 0$ and $r_i^T n_i = 1$ give the result. The substitution $e = a(u)$ maps the eleven-dimensional Q space onto the ten-dimensional ordinary quadratic space with kernel $\mathrm{span}\{A\}$.

For an arbitrary fixed direction r, consider the declared mathematical initial-value problem

$$M\ddot{x} + D\dot{x} + Kx + b\, g(b^T x, t) = 0, \quad K = \tfrac{53}{50} K_0, \quad D = \alpha M_0, \tag{S62}$$

with the same exact source-decimal magnetic drive and coefficients as the retained full-coordinate Hermite certificates. Each exact solution starts at its own saved first state. This is the declared simulation reference; a tube for this equation is not a certificate for a general unknown constitutive law or for the compiled COMSOL equation.

Let T be the first saved time step and keep the already fixed Richardson windows $(T, 2T, 4T)$. With endpoint weights $(-7/4, 8/3, -1, 1/12)$ at nodes $(0, 1, 2, 4)$ and cell weights $w = (7/4, -11/12, 1/12, 1/12)$, form the observation-only signed label

$$L_r^\sigma = -\frac{1}{T} \sum_{j \in \{0,1,2,4\}} a_j r^T M v^\sigma(t_j) - \sum_{k=0}^{3} \frac{w_k h_k}{2T}\, r^T b\{g_{\mathrm{saved}}(t_k) + g_{\mathrm{saved}}(t_{k+1})\}. \tag{S63}$$

Only saved positions, velocities, times and magnetic inputs, and known nominal source matrices enter this label. The pair label for the material component is

$$y_i = \tfrac{1}{2}(L_{r_i}^+ + L_{r_i}^-) - r_i^T K_0 x_i = \tfrac{1}{2}(L_{r_i}^+ + L_{r_i}^-) - 1.$$

The label uses no material correction coefficient. No old projected-force allowance is reused.

Let Q^σ denote the retained cubic Hermite interpolant and let R_k^σ be its previously certified all-prefix energy-tube radius:

$$\|(x - Q, \dot{x} - \dot{Q})\|_{K,M} \le R_k^\sigma.$$

This full-coordinate tube is independent of the output projector. Define

$$L_r^2 = r^T K r + r^T D M^{-1} D r, \qquad A_{D,r}^2 = r^T D M^{-1} D r, \tag{S64}$$

$$A_{KM,r}^2 = r^T K M^{-1} K r, \qquad J_k^\sigma = \sum_{j \le k} h_j R_j^\sigma. \tag{S65}$$

The newly calculated directional allowance is

$$\Gamma_r^\sigma = \sum_k \frac{|w_k| h_k}{T} \min\{L_r R_k^\sigma,\, A_{D,r} R_k^\sigma + A_{KM,r} J_k^\sigma\}. \tag{S66}$$

Both bounds follow by Cauchy–Schwarz in the certified energy metric; the second also uses $x(0) - Q(0) = 0$ and integration of the velocity error. All gains in this expression are recomputed for r_i.

Let C_r^σ be the exact weighted integral of $r^T(KQ^\sigma + D\dot{Q}^\sigma)$. Let $B_{3,r}^\sigma$ bound the third time derivative of $r^T(Kx + D\dot{x})$ on the retained tubes. The existing generic full-coordinate derivative recurrence is evaluated with the new row $[r^T K, r^T D]$; its old three projected outputs are not used. Average signed quantities with an overbar and put

$$I_{0,r} = \left| r^T \{K(\bar{x}_0 - x_i) + D\bar{v}_0\} \right|. \tag{S67}$$

The fixed receiver then gives

$$|y_i - \tfrac{3}{50}| \le \eta_i := |\bar{L}_r - \bar{C}_r| + \bar{\Gamma}_r + \bar{B}_{3,r} T^3/3 + I_{0,r} \tag{S68}$$

for the declared reference equation. Magnetic sample and quadrature discrepancies are already included in the raw-minus-Hermite term. The exact initial discrepancy is retained even when the saved signed states nearly cancel; it is not set to zero by the intended preparation. Arithmetic enclosure radii are included when a saved midpoint is used.

For the eleven ordinary quadratic evaluations, retain a deterministic exact left inverse J_N of the eleven-by-ten evaluation matrix. For a genuinely degree-four member of the feature target class, the coefficient enclosure is

$$|\widehat{s} - s| \le |J_N|\eta.$$

If the target is analytic rather than exactly degree four, independently certified normal tail allowances must be added before this propagation. The reference-equation certificate above does not provide such unknown target bounds. A finite bound may also be too large to identify small nonlinear coefficients; such rows remain in the result without changing the projector, time window, or retained configurations.

For another target equation, the observation-only label formula remains well defined, but its error certificate must be replaced by a tube for that equation, valid directional force/tube sensitivity and third-time- derivative bounds, and an initial-state allowance for that target. These quantities are not inherited from the declared linear material reference merely because the observation files are the same.

Bounds at all eleven configurations. The normal-force construction uses both velocity signs and the fixed $(1, 2, 4)$ windows at all eleven prescribed nodes. Directional gains, third-time-derivative bounds and initial-state allowances are computed for each r_i using the full-coordinate Hermite tubes of the declared reference IVPs. The original three projected-force error bounds are not reused for these new directions.

Node	Normal material label y_i	Certified error bound η_i
0	4.206169974	4.211977
1	0.059962646	$3.788056\,10^{-5}$
2	0.059999578	$4.428778\,10^{-7}$
3	0.059999468	$5.584729\,10^{-7}$
4	0.059999465	$5.613862\,10^{-7}$
5	0.059999416	$6.126460\,10^{-7}$
6	0.059999579	$4.416293\,10^{-7}$
7	0.059999575	$4.459154\,10^{-7}$
8	0.059999468	$5.581601\,10^{-7}$
9	0.059999634	$3.839009\,10^{-7}$
10	0.059999581	$4.398915\,10^{-7}$

Table S1. Dimensionless normal-force labels on the existing records. Labels are rounded displays; error bounds are rounded upward from the certified exact rational bounds and include arithmetic midpoint radii. All rows are retained. Their error certificate concerns the declared reference IVPs, whose material normal value is $3/50$.

Node 0 is a particularly weak normal channel: $c_0 \simeq 2.063\,10^{-18}$. Its endpoint-momentum and sampled-magnetic contributions are approximately -633648.03 and 633653.23 before the nominal subtraction of one. These large additive values explain the cancellation in this normalization; they are not separate noise bounds. The certified error decomposition at that node is $|\bar{L} - \bar{C}| \simeq 4.15706$, $\bar{\Gamma} \simeq 0.0238312$, $\bar{B}_3 T^3/3 \simeq 0.0310829$, and $I_0 \simeq 4.13\,10^{-7}$. Thus the raw-minus-Hermite term dominates this certificate. The bound alone does not establish an actual-error lower bound. Separately, comparison with the declared reference value gives a posterior label discrepancy of approximately 4.14617; this comparison was not used to choose a window or a projector.

Bounds for all ten quadratic coordinates. The original fixed first-lexicographic left inverse is retained. It gives maximum coefficient radii by ordinary degree $0, 1, 2$ of approximately 1.77923, 17.7908, and 56.9309. The predetermined refinement rule is to enumerate every ten-node subset of the eleven-node evaluation matrix, skip only exact singular matrices, propagate each valid coefficient interval, and intersect all resulting intervals. All eleven subsets are exactly nonsingular in this instance. Each interval is a necessary consequence of the same simultaneous label constraints, so their intersection remains an enclosure. No preferred subset is selected by its error, and all eleven observations remain in the construction.

For the ordinary quadratic coefficients of $S(u) = Q(u, a(u))$, the maximum intersection half-widths satisfy

$$r_S^{(0)} < 1.3700\,10^{-4}, \qquad r_S^{(1)} < 1.5880\,10^{-4}, \qquad r_S^{(2)} < 9.7100\,10^{-4}.$$

These are radii about the certified, generally biased coefficient centres. They are not bounds on the coefficients' absolute values. For example, the degree-two coefficient intervals together give the distinct absolute bound $1.8780\,10^{-3}$ in this normalization. The constant interval contains $3/50$, and every nonconstant interval contains zero, as required for the declared linear material reference.

This calculation turns the old full-field records into additional directional force information for that reference equation and an explicit ten-coordinate coefficient interface. The A^2 coordinate remains unobserved on every old trial configuration. An unknown four-feature analytic target additionally needs target-valid trajectory and force-error bounds and a third-order remainder allowance for its normal-force function S (or an equivalent lifted-potential tail bound). The vanishing remainder of the declared scalar material family is a source prior, not a conclusion for an unknown target. No interval here certifies a full-coordinate Birkhoff-twist tolerance or transfers a KAM result to an unknown material.

S9 Trajectory information about the last feature-jet coordinate

Keep the exact source definitions $u = Bx/\ell$, $e = x^T K_0 x/(2E_s)$, $a(u) = \ell^2 u^T K_3 u/(2E_s)$, and $A = e - a(u)$. At each of the eleven old ideal initial configurations, retain the fixed normal functional r_i constructed without trajectory outcomes, with $Br_i = 0$ and $r_i^T K_0 V q_i = 1$. It is not adjusted at later times. Write the weighted-four material polynomial as

$$P(u, e) = R(u) + A(u, e)S(u) + c_A A(u, e)^2,$$

where R has the 31 restricted coefficients and S is an ordinary quadratic with ten coefficients.

Proposition S16. Cancellation along a full-coordinate trajectory. *For every full-coordinate state x, not only the initial trial plane,*

$$r_i^T \nabla_x (E_s P) = J_i(x)\{S(u(x)) + 2c_A A(x)\}, \qquad J_i(x) = r_i^T K_0 x. \tag{S69}$$

Proof. $D_{r_i} u = 0$ and $E_s D_{r_i} A = r_i^T K_0 x$. Apply the product rule. Both $D_{r_i} R$ and $D_{r_i} S$ vanish.

Thus uncertainties in the 31 coefficients of R do not contaminate this normal observable. On a trajectory that leaves the trial plane, the remaining coefficient has sensitivity $2A(x)J_i(x)$. Its pointwise nonzero value, or a nonzero numerical matrix rank, does not establish a coefficient bound at the available error level.

For a prescribed horizon $T_n = t_n - t_0$, define $\mathcal{W}_n f = T_n^{-1} \int_{t_0}^{t_n} f(t)\,dt$. The original three horizons have $n = j, 2j, 4j$. Keep both the three individual averages and their prescribed combination

$$\mathcal{R}_j = \tfrac{8}{3}\mathcal{W}_j - 2\mathcal{W}_{2j} + \tfrac{1}{3}\mathcal{W}_{4j}.$$

All signed records and all predetermined nodes are retained. For a full-coordinate equation with linear damping αM_0, the weak observation has the linear form

$$Y_{i,n} = \sum_{|\gamma| \le 2} G_{i,n,\gamma} s_\gamma + H_{i,n} c_A + D_{i,n}\alpha, \tag{S70}$$

where, using the same average or fixed combination on each term,

$$G_{i,n,\gamma} = \mathcal{W}_n[J_i(x)u(x)^\gamma], \quad H_{i,n} = 2\mathcal{W}_n[A(x)J_i(x)], \quad D_{i,n} = \mathcal{W}_n[r_i^T M_0 \dot{x}].$$

Here Y consists of the negative momentum difference, the known magnetic integral and the known nominal elastic integral. In a reference-only branch with a stipulated known $D = \alpha M_0$, subtract the last term. If α is unknown, it remains a twelfth column. Pairing initial velocities does not remove its integral on an entire trajectory. State-dependent learned damping is a different target class and is not represented by this scalar nuisance column.

All polynomial columns can be integrated on a cubic Hermite centre: J and u have time degree three, A has degree six, and Ju^γ and $2AJ$ have degree at most nine. Saved binary64 observations are interpreted exactly before any rounding. An outward interval evaluation of these integrals certifies arithmetic for the saved Hermite centre, not closeness to a physical trajectory.

What a valid trajectory tube would add. Suppose on a cell a target-valid tube gives $\|x - Q\|_K \le R$ and $\|\dot{x} - \dot{Q}\|_M \le R$. The nominal operator $C = K_0 - B^T K_3 B$ gives $A(x) = x^T C x/(2E_s)$. With centre bounds $U_k \ge |u_k(Q)|$, $J_Q^{\max} \ge |J(Q)|$ and $A_Q^{\max} \ge |A(Q)|$, put

$$\epsilon_{u_k} = R\|(B_k/\ell)K^{-1/2}\|, \quad \epsilon_J = R\|r^T K_0 K^{-1/2}\|,$$
$$\epsilon_A = \frac{R\sup\|K^{-1/2}CQ\| + \frac{1}{2}R^2\|K^{-1/2}CK^{-1/2}\|}{E_s}.$$

These yield the following cellwise errors:

$$|\Delta(Ju^\gamma)| \le \epsilon_J \prod_k (U_k + \epsilon_{u_k})^{\gamma_k} + J_Q^{\max} \sum_k \gamma_k \epsilon_{u_k} (U_k + \epsilon_{u_k})^{\gamma_k - 1} \prod_{l \ne k} (U_l + \epsilon_{u_l})^{\gamma_l},$$
$$|\Delta(2AJ)| \le 2\{\epsilon_A J_Q^{\max} + \epsilon_J (A_Q^{\max} + \epsilon_A)\}.$$

Zero-exponent derivative terms are omitted. Integrate with absolute values of the fixed window weights. Endpoint momentum errors, magnetic integration/sample errors, nominal elastic errors and, when appropriate, damping errors must be certified as well. A three-component force allowance cannot substitute for these new functional errors.

For one known-damping row, let H_Q be the centre sensitivity, ϵ_H its tube/arithmetic error, and let E include observation error and all propagated S-coefficient and column uncertainties. A finite coefficient bound requires a separated denominator:

$$|H_Q| > \epsilon_H, \qquad |c_A| \le \frac{|Y_Q - G_Q \widehat{s}| + E}{|H_Q| - \epsilon_H}.$$

The sign or numerical rank of H_Q alone is insufficient. For an unknown scalar damping coefficient, the augmented columns must also separate c_A from α at the certified precision.

The old verified tubes cover only the first four saved cells of each selected record and only the declared reference equation. Longer predetermined windows may be examined for saved-data geometry, but do not inherit those tube/error values. Likewise an analytic four-feature target needs a separately bounded normal-force remainder beyond $S + 2c_A A$; a general unknown 64-dimensional potential need not have this feature factorization. None of these target assumptions follows from the existing twenty-two simulation records.

Finite intervals on the first four cells. Only the $j = 1$ windows have the certified first-four-cell tube and functional-error bounds. Among the fixed rows, 34 known-damping rows have a separated interval denominator. Intersecting every such valid interval gives

$$-12.872332490981 \le c_A \le 12.872689597309. \tag{S71}$$

When a single unknown constant α is retained, all 946 row pairs are examined, of which 355 have a separated interval determinant. Their simultaneous necessary intervals give

$$-16.009760737420 \le c_A \le 16.009085357190. \tag{S72}$$

The corresponding constant-damping interval is $-798.726283550263 \le \alpha \le 800.418421268464$ in s^{-1}. These outer decimal enclosures are displays of exact rational interval endpoints. They are not confidence intervals. Every valid row or row pair enters the intersection; an inseparable row is retained without dividing by an interval containing zero. Independent interval reconstruction gives the same exact intersections.

The intervals are finite for the declared reference trajectory problem but remain far wider than the A^2 twist tolerance in Supplementary Section S16. The longer fixed windows $j > 1$ supply saved-Hermite geometry only; no target-valid trajectory or force-error bound is assigned to them. Thus later off-plane motion supplies algebraic sensitivity to A^2 without establishing its smallness or an unknown material transfer.

S9.1 Composition of observation, preparation, and jet allowances

The finite-window receiver approximates a force at the prepared rounded full state x_i^{prep}, whereas the polynomial design uses the exact point $\bar{x}_i = V\ell z^{(i)}$. This distinction must be retained. Let

$$F_c(x) = (\ell/E_s)V^T\nabla\mathcal{W}_*(x), \qquad w(z) = \mathcal{W}_*(V\ell z)/E_s.$$

Then $\nabla_z w(z^{(i)}) = F_c(\bar{x}_i)$. Suppose a separate preparation allowance satisfies

$$\|F_c(x_i^{\mathrm{prep}}) - F_c(\bar{x}_i)\|_\infty \le d_i.$$

For example, $d_i = L_{c,i}\|x_i^{\mathrm{prep}} - \bar{x}_i\|_\infty$ suffices if F_c has the corresponding Lipschitz constant on the joining segment. This allowance concerns the correction force: the nominal force is already subtracted at the actual prepared position in the finite label.

Corollary S17. A conditional physical-observation to jet interface. *For every design position, suppose the hypotheses and the common-clock form of Proposition S6 hold, with valid constants for that position. Let r_i enclose the arithmetic error of the reported label midpoint, and define*

$$b_i = I_{0,i} + \frac{B_{3,i}T_i^3}{3} + \frac{5B_{m,i}h_i^2}{12} + 5\epsilon_{m,i} + \frac{11\epsilon_{p,i}}{2T_i} + d_i + r_i.$$

Then each reported component $\hat{y}_{ij}$ approximates $\partial_j w(z^{(i)})$ with absolute error at most b_i. Under the fifth-derivative and equilibrium assumptions of Proposition S11, the same exact left inverse gives

$$|(J\hat{y} - c)_\alpha| \le \sum_{i,j} |J_{\alpha,(i,j)}| \left(b_i + \frac{M_5}{24}\|z^{(i)}\|_1^4\right).$$

For independently computed candidate coefficients $c_{\theta,\alpha}$, a sufficient candidate-to-target allowance is this right-hand side plus $|c_{\theta,\alpha} - (J\hat{y})_\alpha|$. If the two signs use different clocks, their individual clock-dependent constants must replace the common-clock terms before this composition.

Proof. Apply the finite-window inequality to the prepared-state target, then the preparation and arithmetic triangle inequalities. This gives the componentwise force allowance b_i at the exact design node. Substitute it into (S48); the final statement is another scalar triangle inequality. No statistical or optimizer property is required for these deterministic implications.

The stored-data interval radii certify r_i. Comparisons against a declared constitutive formula do not by themselves certify $\epsilon_{p,i}, \epsilon_{m,i}, B_{3,i}, B_{m,i}$ or M_5 for an unknown physical target. A recorded-state initial-defect calculation also does not supply a true-state measurement bound. All eleven design positions and their stated error and tail allowances are required for this receiver. Their actual reception for the declared quadratic family is supplied below; a general unknown physical target still requires its own tail and measurement bounds. Once admitted, these ordinary coefficient intervals still have to pass the separate spectral, Birkhoff, analytic-domain and KAM gates; they do not inherit the candidate-only KAM conclusion automatically. The KAM verifier receives the total potential jet after restoring the known nominal elastic and prescribed magnetic coefficients, with the physical factors $E_s\ell^{-|\alpha|}$ applied consistently.

For the declared linear material family, Proposition S9 provides an alternative to the separate endpoint and magnetic budgets. At a reference design node where that proposition has been instantiated, the force error may instead be received componentwise as

$$b_{ij} = d_{Q,ij} + \Gamma_{ij} + \frac{B_{3,ij}T_i^3}{3} + |\delta_{0,ij}^{\mathrm{rec}}|,$$

provided the nominal subtraction and target use the same reference node. Any additional reported-midpoint radius must also be retained. The componentwise b_{ij} replaces b_i in the left-inverse sum above; the fifth-derivative tail and the other design-node requirements remain. The preceding numerical example received only the direction-3 pair. The next subsection receives every required node separately. No physical measurement allowance is supplied by either reception.

S9.2 Instantiated eleven-node coefficient intervals and their limits

The actual label midpoints $\hat{y} \in \mathbb{R}^{33}$ and their componentwise continuous-force and arithmetic allowances η now instantiate the exact left inverse in Proposition S11. For the declared simulation family $\mathcal{W}_*(x) = \beta x^T K_0 x/2$, the reduced correction is quadratic, its origin gradient vanishes and $M_5 = 0$. These are explicit constitutive-family assumptions, not properties inferred from eleven point observations. In that family,

$$c_* = J\hat{y} + e_c, \qquad |(e_c)_\alpha| \le r_\alpha := \sum_{i,j} |J_{\alpha,(i,j)}|\eta_{ij}.$$

Every one of the 31 received intervals contains the corresponding independently evaluated declared coefficient. Their maximum radii within degrees two, three and four are bounded respectively by

$$4.282817 \times 10^{-7}, \qquad 8.141379 \times 10^{-7}, \qquad 1.808799 \times 10^{-6}.$$

The known magnetic quartic has maximum normalized ordinary coefficient approximately 5.363314×10^{-9}. After restoring those prescribed coefficients, all fifteen total quartic coefficient intervals contain zero. This is an information and sign-ambiguity diagnostic; it is not itself a calculation of the Birkhoff twist determinant. In particular, finite design rank and small force error do not establish the uniform twist and analytic conditions for an unknown physical target. For a general C^5 target, the independent fifth-derivative tail in the preceding receiver must also be restored.

S10 A coherent finite-data ambiguity in the quartic material jet

The following result concerns identifiability from the received finite supervision interface. It does not replace the stipulated quadratic constitutive family used in the target-family theorem. The actual interface contains eleven configurations, thirty-three force components, and thirty-one ordinary potential coefficients.

Write $q = \ell z$, with $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$ J. Let z_i be the eleven exact design nodes, $y \in \mathbb{R}^{33}$ the saved binary64 correction-force centers interpreted as exact rationals, and $\eta \in (0,\infty)^{33}$ their original component-specific received allowances. Let G be the sampled gradient matrix for all ordinary monomials of degrees two through four. Its saved rational left inverse J satisfies $JG = I_{31}$. The received coefficient center and radius are, without an independence assumption,

$$c^{\mathrm{obs}} = Jy, \qquad r^{\mathrm{obs}} = |J|\eta.$$

The prescribed magnetic potential remains $\kappa(1 - \cos(d^T q))$ with the exact native positive κ and native port vector d. The material correction alone is denoted by W.

Proposition S18. One common admissible error vector with vanishing quartic twist. *For the received data, there is a parameter $\widetilde{\beta} \in [0, 0.1]$ such that the entire material correction*

$$\widetilde{W}(q) = \frac{\widetilde{\beta}}{2} q^T K_{03} q + \frac{\kappa}{24}(d^T q)^4 \tag{S73}$$

has zero origin gradient and identically zero fifth material derivative, and satisfies all thirty-three original force intervals and all thirty-one original coefficient intervals simultaneously. In particular, the coefficient inequalities arise from the same error vector as the force inequalities.

The conservative Hamiltonian

$$\widetilde{H}(q,p) = \tfrac{1}{2} p^T M_3^{-1} p + \tfrac{1}{2} q^T K_{03} q + \widetilde{W}(q) + \kappa(1 - \cos(d^T q))$$

has a positive definite quadratic part and satisfies $\widetilde{H}_3 = \widetilde{H}_4 = 0$ exactly. Consequently its degree-four Birkhoff twist is the zero matrix. Thus the finite force/jet intervals, even together with the stated zero-gradient and $M_5 = 0$ material-tail conditions, do not uniformly imply the nondegenerate degree-four local KAM criterion in a material class admitting quartic corrections.

Proof. All calculations in this proof use exact rational arithmetic. Let $a \in \mathbb{Q}^{31}$ be the ordinary coefficient vector of $\ell^2 z^T K_{03} z/(2E_s)$. For $|\alpha| = 4$, put

$$b_\alpha = \frac{\kappa \ell^4}{E_s \alpha!} d^\alpha,$$

and set all other entries of b to zero. The ordinary coefficients and sampled gradients of (S73) are $c(\beta) = \beta a + b$ and $f(\beta) = G(\beta a + b)$, respectively. For each force component form

$$|\beta(Ga)_m + (Gb)_m - y_m| \le \eta_m.$$

For each coefficient also form

$$|\beta a_\alpha + b_\alpha - (Jy)_\alpha| \le (|J|\eta)_\alpha.$$

When the multiplier of β is nonzero, each inequality gives an exact closed rational interval; when it vanishes, the residual inequality is checked directly. Intersecting all sixty-four constraints and $[0, 1/10]$ gives a nonempty interval. Its exact endpoints are retained in the machine-readable result. A decimal enclosure, wider than those exact endpoints, is

$$[\beta_-, \beta_+] \subset [0.05999999779279021,\ 0.05999999877671872].$$

Choose its exact rational midpoint as $\widetilde{\beta}$. Direct substitution verifies

$$\max_m \frac{|f_m(\widetilde{\beta}) - y_m|}{\eta_m} < 0.982670319892135, \qquad \max_\alpha \frac{|c_\alpha(\widetilde{\beta}) - (Jy)_\alpha|}{(|J|\eta)_\alpha} < 0.236855585657115.$$

For the same vector $e = f(\widetilde{\beta}) - y$, exact multiplication verifies $c(\widetilde{\beta}) - Jy = Je$, using $JG = I$. The coefficient constraints therefore preserve all cross-component correlations. They are not selected independently from their marginal intervals.

The polynomial $\widetilde{W}$ has no linear term and degree at most four, so $\nabla\widetilde{W}(0) = 0$ and $D^5\widetilde{W} \equiv 0$. The exact Sylvester principal minors verify $K_{03} \succ 0$, and $1 + \widetilde{\beta} > 0$. Thus

$$\widetilde{H}_2 = \tfrac{1}{2} p^T M_3^{-1} p + \tfrac{1}{2} q^T [(1 + \widetilde{\beta}) K_{03} + \kappa d d^T] q$$

is positive definite. Finally the Taylor expansion of the retained cosine gives

$$\widetilde{H} = \widetilde{H}_2 + \frac{\kappa}{720} (d^T q)^6 + O(\|q\|^8).$$

Both cubic and quartic homogeneous polynomials vanish before any linear canonical transformation. The cubic homological correction and the quartic resonant projection are therefore zero. The degree-four Birkhoff twist is exactly zero.

This is a coherent ambiguity of the finite deterministic supervision interface. It is not a claim that the alternative material reproduces all saved COMSOL time histories, is the actual physical material, or is realizable by the fixed network architecture. It does not contradict the target-family KAM theorem under the stronger hypothesis $W_* = \beta E_0$, which excludes the quartic term in (S73). A vanishing degree-four twist also does not imply absence of invariant tori: higher-order nondegeneracy may be relevant. The result identifies the particular criterion that cannot be inferred uniformly from this interface.

The executable certificate performs 1,289 exact checks. The protocol, source hashes, full common interval, coherent error vector, and actual successful child-process exit are retained together. No network update or new native solve is used in this construction.

S11 Exact finite-node ambiguity without a target-tail prior

The preceding quartic ambiguity uses the original nonzero force allowances and retains a zero fifth material derivative. A separate construction explains why improving finite-node precision alone does not replace an independent target-tail

assumption in an unrestricted analytic class. Let $a_1, \dots, a_m \in \mathbb{R}^3 \setminus \{0\}$ be a finite set of configuration nodes and set

$$\Phi(z) = \prod_{i=1}^{m} \frac{\|z - a_i\|_2^2 \|z + a_i\|_2^2}{\|a_i\|_2^4}, \qquad \Delta w(z) = \frac{\kappa \ell^4}{24 E_s} (d^T z)^4 \Phi(z).$$

The squared norms in this expression are real polynomials and continue to complex arguments as sums of squares, without complex conjugation.

Proposition S19. An even analytic correction invisible to all retained forces. *The polynomial Δw is even and nonnegative on real configurations. Its gradient vanishes exactly at every a_i and $-a_i$. Its homogeneous terms of degrees zero through three vanish, and its fourth homogeneous term is $\kappa \ell^4 (d^T z)^4/(24E_s)$. Thus replacing the material correction $\beta q^T K_{03} q/2$ by*

$$W_{\mathrm{alt}}(q) = \tfrac{\beta}{2} q^T K_{03} q + E_s \Delta w(q/\ell)$$

preserves all these sampled forces and the positive equilibrium Hessian while cancelling the prescribed magnetic quartic. Its cubic and quartic total Hamiltonian terms, and therefore its degree-four Birkhoff twist, vanish. The alternative material remains real entire and even.

Proof. Each factor is an even polynomial, equals one at the origin, and has zero gradient there. At its own signed node, its value and gradient are zero because a squared Euclidean distance and its gradient vanish there. The product rule proves the node identities, including in the presence of other factors. Consequently $\Phi(z) = 1 + O(\|z\|^2)$, and

$$E_s \Delta w(q/\ell) = \tfrac{\kappa}{24} (d^T q)^4 + O(\|q\|^6).$$

This cancels the fourth term in $\kappa(1 - \cos(d^T q))$. There is no cubic term, so the cubic homological correction also vanishes. The positive quadratic part is unchanged. Every factor is nonnegative on real arguments; hence the material addition is nonnegative there. Polynomiality gives the claimed entire continuation.

For the eleven actual nonzero rational nodes, exact arithmetic verifies all twenty-two signed-node identities and the fifteen ordinary quartic coefficient cancellations. The resulting polynomial has degree 48. Independent univariate restrictions also exhibit nonzero fifth derivatives inside the configuration cube. Thus this example has $M_5 \neq 0$ on that cube and is explicitly outside the stipulated zero-tail material family. Its sampled correction forces agree exactly with those for $\beta = 3/50$, whose compatibility with the saved force allowances is checked separately. Its higher-order tail aliases the fourth coefficient under the finite receiver: if b is its fourth coefficient vector and h its sampled higher-order force tail, then $Jh = -b$. The receiver's independent target-tail term cannot be omitted even if the retained forces were measured exactly.

This construction concerns the information in finite pointwise forces. It does not assert that the alternative material generates the saved transient records or belongs to the fixed network class. Neither this example nor a zero degree-four twist proves absence of invariant tori; they delimit inference by the particular nondegenerate quartic criterion.

S12 A finite received-jet loss for the unchanged complex model

Let $\mathcal{A} = \{\alpha \in \mathbb{N}_0^3 : 2 \le |\alpha| \le 4\}$, so $|\mathcal{A}| = 6 + 10 + 15 = 31$. All coefficients below are ordinary monomial coefficients; $D^\alpha w(0) = \alpha!\, c_\alpha$. Write $q = \ell z$, $\ell = 1/1000$, and $E_s = 3/500000000000$. The conservative correction of the retained complex model is

$$W_\theta(q) = c_E E_0(q) + E_{\mathrm{net}} \left[N_\theta(q/Q_*, E_0(q)/E_{\mathrm{net}}) - N_\theta(0) - \sum_{j=1}^{3} \gamma_{\theta,j} q_j / Q_{*,j} \right],$$

where $E_0(q) = \frac{1}{2} q^T K_{03} q$ and $\gamma_{\theta,j} = \partial_j N_\theta(0)$. Here N_θ is the real readout of the existing split-tanh complex network, and E_{net} and Q_* are the saved checkpoint scales. Define $c_{\theta,\alpha}$ by the degree-four Taylor polynomial of $w_\theta(z) = W_\theta(\ell z)/E_s$. The prescribed magnetic potential is kept in the full Hamiltonian, but is not added to these correction coefficients or to the correction labels.

Let $\widehat{c} = Jy$ and $r \geq 0$ be the already received exact centers and radii from the fixed eleven-node force design. Assume that its received deterministic statement is

$$|c_{*,\alpha} - \widehat{c}_\alpha| \leq r_\alpha, \qquad \alpha \in \mathcal{A}. \tag{JG1}$$

For the current receipt, the zero target fifth-derivative tail and $\nabla W_*(0) = 0$ follow from the explicit declared family $W_* = \beta E_0$. They are family priors, not conclusions learned from the eleven positions. A generic unknown target instead needs its own centering and fifth-derivative hypotheses in (JG1).

Fix $s_\alpha = \max\{r_\alpha, 10^{-8}\} > 0$ before any optimizer update, and define the original-center mathematical objective

$$L_{\mathrm{jet}}(\theta) = \frac{1}{31} \sum_{\alpha \in \mathcal{A}} \left(\frac{c_{\theta,\alpha} - \widehat{c}_\alpha}{s_\alpha} \right)^2. \tag{JG2}$$

The training inputs in (JG2) are the received centers and radii. No known-zero cubic or quartic coefficient is substituted as a label.

Proposition S20. Received-jet fitting bound. *For every parameter vector of the same model, (JG1)–(JG2) imply*

$$|c_{\theta,\alpha} - c_{*,\alpha}| \leq |c_{\theta,\alpha} - \widehat{c}_\alpha| + r_\alpha \leq s_\alpha \sqrt{31 L_{\mathrm{jet}}(\theta)} + r_\alpha \quad (\alpha \in \mathcal{A}). \tag{JG3}$$

If a reported joint score is $L_{\mathrm{joint}} = \lambda_{\mathrm{jet}} L_{\mathrm{jet}} + L_{\mathrm{other}}$ with fixed $\lambda_{\mathrm{jet}} > 0$ and $L_{\mathrm{other}} \geq 0$, then L_{jet} in the last expression may be replaced by $L_{\mathrm{joint}}/\lambda_{\mathrm{jet}}$.

Proof. The first inequality is the triangle inequality and (JG1). Each nonnegative summand in $31 L_{\mathrm{jet}}$ is at most their sum, which gives the second inequality after taking square roots. Finally $L_{\mathrm{jet}} \leq L_{\mathrm{joint}}/\lambda_{\mathrm{jet}}$. No descent, convexity, optimizer convergence or independent sampling assumption is needed. A point-force loss by itself cannot replace L_{jet} in this argument.

The component residual in the first bound of (JG3) is normally tighter than the bound derived from a single aggregate loss. In a final source-bound check, let $[a_\alpha, b_\alpha]$ enclose the exact-real coefficient of the actual saved weights. A rigorous computable bound is

$$|c_{\theta,\alpha} - c_{*,\alpha}| \leq \Delta_\alpha + r_\alpha, \qquad \Delta_\alpha = \max\{|a_\alpha - \widehat{c}_\alpha|, |b_\alpha - \widehat{c}_\alpha|\}. \tag{JG4}$$

For fixed exact positive scales, the rational expression $\overline{L} = (1/31) \sum_\alpha (\Delta_\alpha / s_\alpha)^2$ is an upper bound for (JG2). Thus it also validates the aggregate version of (JG3). Directly receiving the candidate coefficients in (JG4) requires no candidate fifth-order remainder. Candidate remainder and analytic-domain bounds may still be needed for later claims about functions or Hamiltonian transport beyond the finite jet.

The Torch polynomial implementation is optimizer arithmetic. Its float64 value is not itself a certified upper bound for (JG2). If it uses rounded centers $\widetilde{c}_\alpha$ and fixed rounded positive scales $\widetilde{s}_\alpha$, the corresponding exact-real loss yields instead

$$|c_{\theta,\alpha} - c_{*,\alpha}| \leq \widetilde{s}_\alpha \sqrt{31 \widetilde{L}_{\mathrm{jet}}} + r_\alpha + |\widetilde{c}_\alpha - \widehat{c}_\alpha|.$$

This statement still presupposes the exact-real candidate coefficient; final Arb reception supplies that link despite optimizer rounding of operators, ℓ, E_s and network calculations. Recording the actual normalization and rounding centers prevents an unreported change of loss.

At present the received maximum fourth-order coefficient radius is 1.808799×10^{-6} as an upward decimal bound. Even zero loss therefore does not guarantee that every received quartic error bound is below 10^{-6}. The jet is a deterministic transformation of the same eleven force records, not an independent additional dataset. Reducing (JG2) retains the full complex energy and dissipation heads; it neither replaces the architecture by a scalar prior nor supplies a physical-target spectral, twist or KAM certificate.

S12.1 Precision-weighted calibration of the final complex readout

For fixed hidden layers, let $h_\psi(s) \in \mathbb{C}^{12}$ be the last hidden representation. Its real scalar energy readout is $N_{a,b,\psi}(s) = \mathrm{Re}[(a+ib)^T h_\psi(s) + b_0]$. The centering operation in W_θ is linear in this readout: it removes its constant and its three linear configuration terms, while retaining the response through the quadratic energy feature. Consequently, with the saved feature scales and fixed hidden layers,

$$W_\theta(q) = \sum_{k=0}^{24} \xi_k \Phi_{\psi,k}(q), \qquad \xi = (c_E, a_1, \dots, a_{12}, b_1, \dots, b_{12}).$$

The output bias cancels under centering and is held fixed as a gauge. The $\Phi_{\psi,k}$ include the nominal energy skip and the centered real and imaginary hidden readouts, with their actual signs.

Proposition S21. A finite linear fitting block. *For fixed ψ, the 33 normalized force components at the eleven design nodes and the 31 normalized correction coefficients form a linear map $A_\psi \xi \in \mathbb{R}^{64}$. If $\hat{y}$ denotes the force midpoints, $\tilde{c}$ the saved jet centers, and $\eta_i > 0$ and $\tilde{s}_\alpha > 0$ the fixed received scales, the numerical block objective is the weighted least-squares expression*

$$\left\| D^{-1} \left(A_\psi \xi - \begin{pmatrix} \hat{y} \\ \tilde{c} \end{pmatrix} \right) \right\|_2^2, \qquad D = \mathrm{diag}(\eta_1, \dots, \eta_{33}, \tilde{s}_1, \dots, \tilde{s}_{31}). \tag{PW1}$$

This is a block of the unchanged complex model. It changes neither its function graph nor the number of stored real parameters.

Proof. The centered potential is linear in ξ. Differentiation at a fixed configuration and extraction of a Taylor coefficient are linear operations, so stacking their values gives $A_\psi \xi$. Positive diagonal scaling gives (PW1). No independence of the 64 rows is asserted: the jet rows are deterministic transformations of the same 33 force observations.

The implemented step uses float64 automatic differentiation to form A_ψ and a single least-squares solve, without a ridge penalty. Its numerical rank and condition estimate are diagnostics, not an exact-rank or exact-optimality certificate. The method was selected after the disclosed earlier fitting results; it is not a blind test. The step fixes the already trained hidden layers and the complete dissipation head. It performs one readout-block solve and zero gradient updates; the preceding full-layer training remains part of its documented history.

The force scales in (PW1) are observation-error allowances. The preceding gradient stage instead normalized point-force residuals by the observed signal RMS while normalizing coefficients by their much smaller received errors. The scores of these two objectives are therefore not directly comparable. Each term of (PW1) is the square of a residual divided by its scale, namely $(r/s)^2$.

The proof of fit is independent of the numerical solver. At the ideal exact design node, let an outward enclosure $[f_i^-, f_i^+]$ contain the force component defined by the saved weights and exact native operators. Set $\delta_i^f = \max\{|f_i^- - \hat{y}_i|, |f_i^+ - \hat{y}_i|\}$. If $|f_{*,i} - \hat{y}_i| \le \eta_i$, then

$$|f_{\theta,i} - f_{*,i}| \le \delta_i^f + \eta_i.$$

The corresponding coefficient statement is (JG4), with the appropriate saved-center rounding allowance. In particular, lying inside a data interval does not mean equality with its unknown target. Neither finite-node fit nor a small coefficient loss alone implies an analytic norm bound, a physical-target twist condition, or a complex-network advantage.

S13 A local KAM criterion for the actual conservative network

Consider the conservative restriction of the full-observation model,

$$H_N(q,p) = \tfrac{1}{2} p^\mathsf{T} A^{-1} p + U_N(q), \qquad A = A^\mathsf{T} > 0. \tag{S74}$$

Here $q, p \in \mathbb{R}^3$, and U_N includes the elastic, prescribed magnetic, and learned potential terms. The centering implemented in the network gives $\nabla U_N(0) = 0$. Dissipation and the time-dependent transverse magnetic input are set to zero in (S74).

The statement below concerns this restriction as its own Hamiltonian system. It does not assume that the restriction is an invariant subsystem of the full finite-element dynamics.

Proposition S22. Local criterion based on the selected weights. *Suppose the real-coordinate continuation of U_N is holomorphic near $q = 0$, and suppose the following conditions hold for the same weights:*

1. *$K_N = D^2U_N(0)$ is positive definite;*
2. *the positive generalized frequencies of (K_N, A) obey $k \cdot \omega_N \neq 0$ for all integer k with $0 < |k|_1 \leq 4$;*
3. *the degree-four Birkhoff normal form is*

$$h_N(I) = \omega_N \cdot I + \tfrac{1}{2} I^{\mathsf{T}} T_N I, \qquad \det T_N \neq 0. \tag{S75}$$

Then every neighborhood of $(q, p) = (0, 0)$ contains a positive-measure family of invariant Lagrangian three-dimensional tori of H_N with Diophantine frequencies. Neither a specified nonzero torus radius nor a specified frequency is asserted.

Proof. Positivity gives a real linear canonical change of coordinates to $H_2 = \sum_{j=1}^{3} \omega_{N,j}(X_j^2 + Y_j^2)/2$. Let $a_j = (X_j + iY_j)/\sqrt{2}$, so that $\{a_j, \bar{a}_k\} = -i\delta_{jk}$ and $I_j = a_j\bar{a}_j$. For a homogeneous monomial $a^\alpha \bar{a}^\beta$,

$$\{H_2, a^\alpha \bar{a}^\beta\} = i\,\omega_N \cdot (\alpha - \beta) a^\alpha \bar{a}^\beta.$$

The second condition solves every cubic homological equation and all quartic equations except the action monomials $\alpha = \beta$. For $H_3 = \sum h_{\alpha\beta} a^\alpha \bar{a}^\beta$, choose

$$\chi_3 = \sum_{|\alpha|+|\beta|=3} \frac{i h_{\alpha\beta}}{\omega_N \cdot (\alpha - \beta)} a^\alpha \bar{a}^\beta.$$

Thus $\{H_2, \chi_3\} = -H_3$. Under the corresponding time-one canonical pullback the effective quartic term is

$$H_{4,\mathrm{eff}} = H_4 + \tfrac{1}{2}\{H_3, \chi_3\}.$$

Its action projection gives (S75); a quartic generating function removes its other monomials. These finite polynomial flows are analytic and locally invertible for sufficiently small (X, Y). Consequently

$$H_N \circ \Psi = h_N(I) + R_5(X, Y), \qquad R_5 = O(\|(X, Y)\|^5). \tag{S76}$$

All normal-form transformations and the remainder in this equation refer to H_N, including its learned cubic terms.

Fix action and time units before the following estimates. More explicitly, for fixed $S_0, \omega_0 > 0$, replace (X, Y) by $(X, Y)/\sqrt{S_0}$, H by $H/(S_0\omega_0)$, and time t by $\omega_0 t$. The resulting equations are canonical, and the strict conditions above are invariant under this fixed rescaling. Use the same symbols for these dimensionless variables. Fix a balanced action box, for example $1 < J_j < 2$, and put $I = \rho^2 J$. This substitution is used to describe shrinking domains; Hamilton's equations and the following KAM application still use the physical canonical actions I. On a slightly larger complex action box and a fixed angle strip $|\operatorname{Im}\theta| < s$, a branch of $\sqrt{I_j}$ is analytic, and

$$X_j = \sqrt{2I_j}\cos\theta_j, \qquad Y_j = -\sqrt{2I_j}\sin\theta_j$$

has $\{\theta_j, I_k\} = \delta_{jk}$ and size $O(\rho)$. Analyticity of (S76) therefore supplies constants $C, M, c_h > 0$, independent of sufficiently small ρ, such that the perturbation has bound $F_\rho = C\rho^5$, the action Hessian of h_N has norm at most M, and the frequency parameterization is analytic on a complex neighborhood of width $h_\rho = c_h\rho^2$. The last assertion follows directly from the affine inverse $I = T_N^{-1}(\omega - \omega_N)$ and $\det T_N \neq 0$. Choose c_h sufficiently small, using $\|T_N^{-1}\|$ and the distance of the balanced box from $I_j = 0$, so that all complex centers $I_0(\omega)$ together with their action offsets $|\delta I| < r_\rho$ remain in the larger complex action box. This ensures that the square-root branches, fixed angle strip, and bound $C\rho^5$ are valid uniformly for every frequency parameter in that neighborhood.

Apply Theorems A and B of Pöschel (2001), using $\tau > 2$, $\nu = \tau + 1$, fixed $s \le 1$, and

$$\alpha_\rho = a\rho^{2+\beta}, \qquad 0 < \beta < \tfrac{1}{2}, \qquad r_\rho = \sqrt{C/M}\,\rho^{5/2}, \qquad a > 0. \tag{S77}$$

In the frequency-parameter formulation the local perturbation includes both the Taylor quadratic part of h_N and R_5, with bound $Mr_\rho^2 + F_\rho = 2C\rho^5$. The theorem's sufficient smallness inequality is met if

$$C\rho^{1-2\beta} \le \frac{\gamma^2 a^2 s^{2\nu}}{4M}, \tag{S78}$$

where $\gamma > 0$ depends only on dimension and τ. Indeed (S78) is exactly $F_\rho \le \gamma^2 \alpha_\rho^2 s^{2\nu}/(4M)$. The remaining requirements $\alpha_\rho s^\nu \le h_\rho$, $r_\rho = o(\rho^2)$, and $r_\rho, s, h_\rho \le 1$ also hold after decreasing ρ. Here $1 - 2\beta > 0$ is essential.

For completeness the Diophantine parameter set is nonempty with positive measure on these shrinking domains. The frequency image is an affine image of a box of side ρ^2. A resonance slab for k has thickness at most $C_k \alpha_\rho/|k|^{\tau+1}$ in frequency space, with a uniform geometric factor depending on T_N. Summing the cross-sectional estimates over integer $k \neq 0$ converges when $\tau > 2$. Including the boundary layer, the excluded fraction is at most a constant times $\alpha_\rho/\rho^2 = a\rho^\beta$, which tends to zero. The Lipschitz estimates in Theorem B preserve positive measure for the torus family. Transforming back by the canonical analytic changes proves the claim in every neighborhood of the equilibrium.

Consequences and limits for learning. This criterion permits changes in the learned quadratic, cubic, and quartic terms. It requires their recomputation from the actual selected network, including the cubic contribution to the quartic normal form. The finite nonresonance checks concern the linear frequencies only through order four; the Diophantine frequencies of the resulting tori are selected from the nearby nonlinear frequency image. No all-integer Diophantine property of ω_N is inferred from a finite search.

The three strict conditions are open conditions on the degree-four jet. A rigorous force-to-jet receiver can therefore supply a sufficient learning criterion: an enclosure of the admissible degree-four jets must pass positivity, the finite frequency gaps, and the twist test uniformly. An independent analytic continuation hypothesis is still needed for the remainder. Such a receiver need not bound a large quadratic change by the much smaller reference magnetic quartic; the quadratic change is included in the new spectrum.

An interval test on the actual network is a posterior criterion for that model. Its success alone does not prove that the physical target has the same low-order jet, that an optimizer will find such weights, or that a finite-data loss guarantees its conditions. A jet penalty based on the prescribed magnetic law is a stated physical prior when the observations cannot resolve those coefficients. It must not be reported as experimental identification of an unknown nonlinearity.

For a specified finite action range the constants in (S78), the normal-form domain, and the chosen Diophantine parameter set must be bounded on that range. The local constant C bounds the remainder after the cubic and quartic canonical transformations; a bound for the original potential's Taylor remainder cannot be substituted for it without controlling these transformations. The local existence conclusion does not by itself certify such a range, identify an individual trajectory as a torus, or transfer a torus to the full damped, driven finite-element model. These limitations also prevent interpreting this result as a control-safety theorem.

S13.1 Received coefficient-guided fit and calibrated checkpoint

Starting from the initial eleven-configuration fit, 200 complete outer cycles update the energy branch by one strong-Wolfe LBFGS iteration and the dissipation branch by one Adam step with the conservative contribution detached. There are 432 energy closure evaluations and 200 dissipation Adam updates. Every complex weight layer changes in both its real and imaginary coordinates. The selected cycle is 200. The coefficient loss drops from approximately 50743.76 to 4.399638, but the maximum point-force residual rises to 2.434500×10^{-4}. Independent coefficient evaluation bounds the largest declared-family quartic discrepancy by 3.342145×10^{-6}. This stage improves the initial high-order error but does not satisfy all received intervals.

The precision-weighted step (PW1) is then applied to those saved hidden layers. It performs one 25-coordinate linear solve with no ridge penalty. Only c_E and the final complex energy weight change; all other saved coordinates remain bitwise

identical to the warm start. The full two-branch graph retains its 918 real parameters. The selected skip is approximately 0.06002256805 and the Euclidean norm of its 24 real neural readout coordinates is approximately 0.0017053353, compared with 0.29265564 before calibration. This change is a final-readout calibration of a previously trained network, not a new full-layer gradient run.

Arb evaluation at 512 bits receives this exact saved graph with exact native operators at the ideal rational nodes. All 33 force residual intervals lie inside their individual received allowances, and all 31 coefficient intervals lie inside the original data intervals. The learner's once-rounded coordinates are separately recorded and are not used as substitutes for the ideal-node check. The maximum finite-node residual has upward bound 1.797797×10^{-8}; the maximum residual-to-allowance ratio is below 0.790301. The corresponding maximum candidate-to-target force bound, using the triangle inequality and the admitted observation error, is 5.023184×10^{-8}. These statements concern the eleven configurations and the explicitly admitted target family.

The strict exact-real upper bound for the actual saved-center coefficient loss is

$$\overline{L}_{\mathrm{jet}} < 1.082009 \times 10^{-2}.$$

Its computation uses rational endpoint bounds on an independent outward expansion, not the float64 training loss. The original data centers and the exported training centers are kept distinct, with the latter's exact rounding allowance included. The maximum normalized coefficient-error bounds are

Degree	Data-based triangle bound	Declared-family comparison
2	4.616863×10^{-7}	3.635946×10^{-8}
3	8.174276×10^{-7}	1.768735×10^{-8}
4	2.066906×10^{-6}	3.908709×10^{-8}

The middle column follows from the data intervals and (JG4). The final column additionally evaluates the explicitly declared quadratic simulation target and is an evaluator comparison, not a training label or a guarantee for an unknown constitutive law. The quartic discrepancy has thus decreased from more than 6.345877×10^{-4} for the initial point-force fit to less than 3.908709×10^{-8} after the disclosed coefficient-guided and final-readout stages.

The calibrated candidate's three-coordinate conservative restriction is pole-free on $|z_j| \le 1/64$ and has frequencies approximately

$$(11.73745102,\ 72.26615415,\ 326.09695871)\ \mathrm{rad\,s^{-1}}.$$

Its cubic-corrected quartic Birkhoff twist has two positive and one negative eigenvalues and determinant near -5.271108×10^{19}, strictly excluding zero. Its positive Hessian and finite-order nonresonance satisfy the qualitative local criterion of Proposition S22 for the same saved conservative restriction. No specified finite torus radius or physical-target KAM transfer follows from this candidate-specific calculation.

Finite data also admit a scalar correction. For the comparison subfamily $W(q) = \zeta E_0(q)$, force and coefficient constraints are linear inequalities in an unknown ζ. Exact rational intersection of all 33 force and 31 coefficient intervals is nonempty. Its endpoints are approximately 0.0599999981693 and 0.0600000004402. An exact rational midpoint witness satisfies all 64 inequalities. No material value is assigned to ζ in this calculation. The zero cubic and quartic terms here define this comparison subfamily; they are not extra labels supplied to the complex network. Consequently the received finite data cannot require a nonlinear correction or establish a complex-network advantage. This limitation persists after the substantial improvement in the calibrated network's coefficient fit. All stages use the same development data, without an independent validation or probability claim.

S14 An exact-source KAM certificate for the full-coordinate learned Hamiltonian

This section treats the finite conservative Hamiltonian defined by the saved network weights and the exact interpretation of their mechanical sources. Let $x \in \mathbb{R}^{64}$ collect the 32 native transverse displacement and rotation pairs, and let $p \in \mathbb{R}^{64}$ be the canonical momenta. The source entries of M_0, M_r, K_0 are interpreted as their exact saved binary values, whereas V is

the exact rational basis retained in the source operator record. The sum and inverse in

$$M = M_0 + M_r, \qquad P = \operatorname{diag}(Q_1, Q_2, Q_3)^{-1}(V^{\mathsf{T}}MV)^{-1}V^{\mathsf{T}}M \tag{S79}$$

are exact mathematical operations. The positive feature scales Q_a and energy scale E are the exact binary values in the same saved checkpoint; they are not redefined by rounded decimal expressions. Let $N(l, e)$ denote the real energy readout of the realification of that fixed complex network. With $b = e_{64}$, define

$$\begin{aligned}
\mathcal{H}_N(x, p) &= \frac{1}{2}p^{\mathsf{T}}M^{-1}p + U_N(x), \\
U_N(x) &= (1 + c_E)E_0(x) + \kappa[1 - \cos(b^{\mathsf{T}}x)] \\
&\quad + E\left[N\left(Px, \frac{E_0(x)}{E}\right) - N(0, 0) - N_l(0, 0) \cdot Px\right], \qquad E_0(x) = \frac{1}{2}x^{\mathsf{T}}K_0x.
\end{aligned} \tag{S80}$$

Thus all 64 transverse configuration coordinates remain in the Hamiltonian; no invariance of $\operatorname{range}(V)$ is assumed. This is the exact-source mathematical graph with the same saved weights as the trained model. Numerical evaluation uses rounded matrix buffers and floating-point operations, so bitwise equivalence with every runtime operation is not part of this statement. The reproducibility manifest identifies the saved weights, source arrays, and interval certificates used below.

The actual quadratic spectrum and analytic domain. All derivatives of N in this section are evaluated at $(l, e) = (0, 0)$. The centering in (S80) gives $\nabla U_N(0) = 0$, while the energy feature retains its quadratic contribution. In particular,

$$H := U_N''(0) = fK_0 + EP^{\mathsf{T}}N_{ll}P + \kappa bb^{\mathsf{T}}, \qquad f = 1 + c_E + N_e > 0. \tag{S81}$$

The source-bound spectral certificate establishes $M > 0$, $H > 0$, and 64 simple generalized eigenvalues in disjoint positive intervals. Taking their positive square roots gives the actual neural frequencies ω_i and the strict finite-order bound

$$|k \cdot \omega| \geq \frac{545883796425}{281474976710656} > 0, \qquad k \in \mathbb{Z}^{64}, \quad 0 < |k|_1 \leq 4. \tag{S82}$$

These frequencies include the learned Hessian correction in (S81).

Realification makes the network analytic in the real configuration coordinates. Complexifying those real coordinates, the analytic-domain certificate excludes every hidden $\tanh$ pole throughout

$$|w_j| < 2^{-40}\,\mathrm{m}, \qquad |\theta_j| < 2^{-40}\,\mathrm{rad}, \qquad j = 1, \ldots, 32.$$

For example, the first- and second-layer imaginary variations from their real centers are bounded by 6.597×10^{-8} and 5.018×10^{-8}, respectively, both strictly below $\pi/2$. The remaining affine operations, centering polynomial, mechanical polynomial, and cosine introduce no poles. The kinetic term is entire in p. This polydisc supplies an analytic neighborhood; its component radius is not a certified torus-amplitude bound.

Lemma S23. Enclosures of the actual modal projections. *Let S_0 be a square real proposal matrix, with its saved binary entries interpreted exactly, and put*

$$G = S_0^{\mathsf{T}}MS_0, \qquad A = S_0^{\mathsf{T}}HS_0, \qquad g = \|G - I\|_\infty < 1.$$

For each column define

$$\mu_i = \frac{A_{ii}}{G_{ii}}, \qquad d_i = (A - \mu_i G)e_i, \qquad \epsilon_i = \frac{\|d_i\|_2}{\sqrt{(1 - g)G_{ii}}}.$$

The interval $[\mu_i - \epsilon_i, \mu_i + \epsilon_i]$ contains a generalized eigenvalue. If it excludes every certified spectral interval except the one for mode i, intersecting the two intervals yields a refined enclosure J_i of that eigenvalue. If

$$\gamma_i = \operatorname{dist}\left(\mu_i, \bigcup_{j \neq i} J_j\right) > 0,$$

the sign of an actual M-normalized eigenvector s_i can be chosen so that

$$\left\| s_i - \frac{S_0 e_i}{\sqrt{G_{ii}}} \right\|_M \le \frac{\sqrt{2}\epsilon_i}{\gamma_i}. \tag{S83}$$

For every real row h^T this also gives

$$\left| h^\mathsf{T} s_i - \frac{h^\mathsf{T} S_0 e_i}{\sqrt{G_{ii}}} \right| \le \frac{\|h^\mathsf{T} S_0\|_2}{\sqrt{1-g}} \frac{\sqrt{2}\epsilon_i}{\gamma_i}. \tag{S84}$$

Proof. Symmetry gives $\|G - I\|_2 \le g$ and therefore $G \ge (1-g)I > 0$. In particular, S_0 is invertible. Set $u_i = S_0 e_i/\sqrt{G_{ii}}$ and $r_i = (H - \mu_i M)u_i$. Then

$$r_i^\mathsf{T} M^{-1} r_i = \frac{d_i^\mathsf{T} G^{-1} d_i}{G_{ii}} \le \frac{\|d_i\|_2^2}{(1-g)G_{ii}}.$$

Expansion in an M-orthonormal eigenbasis bounds the distance of μ_i from the spectrum by ϵ_i. The interval exclusion test identifies the enclosed eigenvalue because the previously certified spectrum is complete and simple. In this eigenbasis the sum of the squared coefficients of u_i outside mode i is at most ϵ_i^2/γ_i^2. Choose the sign of s_i to make its coefficient nonnegative. The inequality $1 - \sqrt{1-t} \le t$ for $0 \le t \le 1$ then proves (S83). Finally,

$$h^\mathsf{T} M^{-1} h = (h^\mathsf{T} S_0) G^{-1} (S_0^\mathsf{T} h) \le \frac{\|h^\mathsf{T} S_0\|_2^2}{1-g},$$

and Cauchy–Schwarz in the mass metric proves (S84).

The cubic-corrected twist in the same actual chart. Let $S = (s_1, \dots, s_{64})$, $\Omega = \operatorname{diag}(\omega_i)$, and $L = S\Omega^{-1/2}$. Since $S^\mathsf{T} M S = I$ and $S^\mathsf{T} H S = \Omega^2$, the change

$$x = LX, \qquad p = MLY\Omega = MS\Omega^{1/2}Y \tag{S85}$$

is canonical and gives $\mathcal{H}_2 = \sum_i \omega_i (X_i^2 + Y_i^2)/2$. Apply Lemma S23 to the three rows of P and to b^T, and divide by the positive enclosures of $\sqrt{\omega_i}$. This encloses the actual projections

$$Z_{ai} = (PL_i)_a, \qquad t_i = b^\mathsf{T} L_i.$$

The nominal stiffness in this chart follows from the actual Hessian identity:

$$R_{ij} := L_i^\mathsf{T} K_0 L_j = \frac{\delta_{ij}\omega_i - E\sum_{a,b=1}^3 Z_{ai} N_{ab} Z_{bj} - \kappa t_i t_j}{f}. \tag{S86}$$

Thus only these four projected rows are needed to propagate the full modal uncertainty. Define

$$v_i = \sum_{a=1}^3 N_{ae} Z_{ai}, \qquad Q_{ij} = \sum_{a,b=1}^3 N_{abe} Z_{ai} Z_{bj}.$$

Differentiating (S80) gives the actual cubic derivative tensor and the fourth-derivative pairs:

$$\begin{aligned} C_{ijk} &= v_i R_{jk} + v_j R_{ik} + v_k R_{ij} + E \sum_{a,b,c=1}^3 N_{abc} Z_{ai} Z_{bj} Z_{ck}, \\ D_{iijj} &= \frac{N_{ee}}{E}(R_{ii}R_{jj} + 2R_{ij}^2) + Q_{ii}R_{jj} + R_{ii}Q_{jj} + 4Q_{ij}R_{ij} \\ &\quad + E \sum_{a,b,c,d=1}^3 N_{abcd} Z_{ai} Z_{bi} Z_{cj} Z_{dj} - \kappa t_i^2 t_j^2. \end{aligned} \tag{S87}$$

Here C, D are derivative tensors, not ordinary polynomial coefficients. Consequently the direct quartic action Hessian has diagonal entries $D_{iiii}/8$ and off-diagonal entries $D_{iijj}/4$.

To include the cubic correction, put $a_i = (X_i + iY_i)/\sqrt{2}$ and $I_i = a_i\bar{a}_i$. If $\mathcal{H}_3 = \sum h_{\alpha\beta}a^\alpha\bar{a}^\beta$, the finite-order bound (S82) permits the choice

$$\chi_3 = \sum_{|\alpha|+|\beta|=3} \frac{ih_{\alpha\beta}}{\omega\cdot(\alpha-\beta)}a^\alpha\bar{a}^\beta, \qquad \{\mathcal{H}_2,\chi_3\} = -\mathcal{H}_3.$$

The twist T is defined, in this same chart, by

$$\frac{1}{2}I^{\mathsf{T}}TI = \Pi_{\text{action}}\left[\mathcal{H}_4 + \frac{1}{2}\{\mathcal{H}_3,\chi_3\}\right]. \tag{S88}$$

The action projection retains all repeated-index terms and their cross contributions. In particular, the actual network is not treated as an even potential with vanishing cubic jet.

Proposition S24. Certified nondegeneracy of the full-coordinate twist. *For the saved learned Hamiltonian* (S80)*, the outward calculation encloses the actual T in* (S88)*. For $\mathcal{T} = \Omega^{-1}T\Omega^{-1}$ and the saved exact-binary preconditioner B_0, it proves*

$$\|I - B_0\mathcal{T}\|_\infty < \frac{2353}{10^9} < 1. \tag{S89}$$

Hence T is nonsingular.

Proof. The four-feature source jet is enclosed using outward ball arithmetic at 384-bit precision. The modal residual test succeeds for all 64 modes. Its certified Gram defect is below 1.248×10^{-14}, its mass-normalized residuals are below 1.347×10^{-5}, and the eigenvector distances in (S83) are below 7.472×10^{-10}. The resulting projection enclosures and the identities (S86)–(S87) enclose the modal cubic tensor and all quartic pairs. The finite cubic homological sum in (S88), including its repeated-index terms, is evaluated with outward arithmetic and strictly nonzero divisor enclosures. This yields an enclosure of the full actual $\mathcal{T}$. The preconditioner is only a proposal: the final calculation encloses every entry of $I - B_0\mathcal{T}$ and bounds every absolute row sum to prove (S89). The Neumann criterion makes $B_0\mathcal{T}$ invertible, so $\mathcal{T}$ and then T are invertible. No definiteness of T is needed.

Theorem S25. Local KAM tori of the fixed full-coordinate learned Hamiltonian. *Every open neighborhood of $(x,p) = 0$ contains a family of real-analytic invariant Lagrangian 64-tori of* (S80) *whose union has positive 128-dimensional Lebesgue measure. The original Hamiltonian flow on each torus is conjugate to a Diophantine rigid rotation.*

Proof. The analytic domain, positive quadratic spectrum, and finite-order nonresonance established above permit real-analytic canonical normalization through degree four. With the actual cubic correction retained, the result is

$$\mathcal{H}_N\circ\Phi = \omega\cdot I + \frac{1}{2}I^{\mathsf{T}}TI + R_5(X,Y), \qquad I_i = \frac{X_i^2+Y_i^2}{2}, \qquad R_5 = O(\|(X,Y)\|^5). \tag{S90}$$

The finite canonical maps are analytic and invertible near the equilibrium, and Proposition S24 establishes the required invertibility of T.

Fix action and time units for the following scaling argument. Choose a bounded positive action domain $\overline{B} \Subset (0,\infty)^{64}$ with fixed analytic buffers and a fixed complex angle strip. The square roots in $X_i = \sqrt{2I_i}\cos\vartheta_i$, $Y_i = -\sqrt{2I_i}\sin\vartheta_i$ then have uniform analytic branches. For $I = aJ$ with $a > 0$, set

$$\mathcal{H}_a(J,\vartheta) = a^{-2}(\mathcal{H}_N\circ\Phi)(aJ,\vartheta) = \frac{\omega}{a}\cdot J + \frac{1}{2}J^{\mathsf{T}}TJ + f_a(J,\vartheta), \qquad \|f_a\| = O(a^{1/2}). \tag{S91}$$

The scaling of actions is conformally symplectic; the Hamiltonian $\mathcal{H}_a$ uses the constant time change $\sigma = at$. It preserves invariant sets and the Lagrangian property.

The integrable frequency map $J \mapsto \omega/a + TJ$ has the fixed invertible derivative T. Its image is a translation of TB. Hence the analytic widths, Hessian bounds, inverse Hessian bounds, and geometric resonance-strip estimates can be chosen independently of a on an interior action domain. Apply the parameterized Theorems A–B of Pöschel (2001) with $\tau = 64 > 63$ and Diophantine constant $\gamma_a = a^{1/8}$. After balancing the local action radius against the perturbation,

$r_a = O(a^{1/4})$, their sufficient smallness condition is met because

$$\frac{\|f_a\|}{\gamma_a^2} = O(a^{1/4}) \longrightarrow 0.$$

More explicitly, the perturbation including the local integrable quadratic Taylor term is $O(a^{1/2})$, whereas $\gamma_a r_a = O(a^{3/8})$. Thus the corresponding smallness ratio is $O(a^{1/8}) \to 0$, and the condition comparing γ_a with the fixed complex frequency width also holds. The excluded fraction of the translated frequency domain tends to zero with γ_a; the Lipschitz parameter estimates in Theorem B preserve a positive-measure torus family. No all-integer Diophantine condition on the equilibrium vector ω is required.

Returning to original time multiplies every torus frequency by $a > 0$, which preserves its Diophantine property with a correspondingly smaller positive constant. The action scaling multiplies volume by $a^{64} > 0$, and the canonical maps preserve volume. The resulting torus union therefore has positive original phase-space measure. Its Cartesian size is $O(\sqrt{a})$, so choosing a sufficiently small places it in any prescribed open neighborhood of the equilibrium. This proves the assertion using the general fifth-order remainder; an even-potential sixth-order remainder is unnecessary.

The conclusion concerns the finite 64-coordinate transverse conservative mathematical graph with the same saved network weights. It does not include the additional axial coordinates, damping or forcing, the continuous PDE, or an unknown physical material. It supplies neither a computable nonzero torus radius nor Lyapunov stability of every nearby trajectory. It also does not establish finite-data identification, transfer to a physical target, or superiority of complex over real network representations.

S15 A full-coordinate robustness criterion at the saved neural model

The base Hamiltonian in this section is the same verified 918-parameter exact-source conservative graph, with all 64 configuration coordinates. Its equilibrium Hessian, canonical chart, finite-order divisors, and cubic-corrected twist are verified inputs. This differs from the earlier three-coordinate material-family calculation. No tolerance from that calculation is used here.

Write the verified chart as $x = LX$ and $H_2 = \frac{1}{2}\sum_{j=1}^{64} \omega_j (X_j^2 + Y_j^2)$. With $a_j = (X_j + iY_j)/\sqrt{2}$, one has $\{a_j, \bar{a}_k\} = -i\delta_{jk}$ and $I_j = a_j \bar{a}_j$. Let $\|P\|_1$ be the sum of absolute ordinary coefficients of a homogeneous polynomial in $(a, \bar{a})$. Let $d_3 > 0$ be a verified lower bound on all cubic homological divisors, $h_3 \geq \|H_3\|_1$, and $t \geq \|T^{-1}\|_2$ for the verified full 64×64 Birkhoff twist. In particular H_3 is not assumed zero.

Consider a real analytic conservative perturbation $\Delta U = \Delta U_3 + \Delta U_4 + O(|x|^5)$ with no terms of degree at most two. Thus the equilibrium Hessian and the canonical chart remain exactly the verified ones. Suppose its canonical polynomials satisfy $\|\Delta H_3\|_1 \leq \epsilon_3$ and $\|\Delta H_4\|_1 \leq \epsilon_4$.

Proposition S26. A criterion retaining the nonzero neural cubic. *If*

$$t\left[2\epsilon_4 + \frac{18 h_3 \epsilon_3 + 9\epsilon_3^2}{d_3}\right] < 1, \tag{S92}$$

then the perturbed full-coordinate quartic twist is nonsingular. Together with real analyticity and the inherited fixed Hessian and order-four nonresonance, this gives the same qualitative local elliptic KAM conclusion: every open neighborhood of the equilibrium contains a positive-measure union of invariant Lagrangian 64-tori.

Proof. Use the same pullback convention as in the verified base certificate: $\{H_2, \chi_3\} = -H_3$ and $\{H_2, \Delta\chi_3\} = -\Delta H_3$. The change in the effective quartic is

$$\Delta H_4^{\mathrm{eff}} = \Delta H_4 + \tfrac{1}{2}\big(\{H_3, \Delta\chi_3\} + \{\Delta H_3, \chi_3\} + \{\Delta H_3, \Delta\chi_3\}\big).$$

For homogeneous degrees m, n, the canonical complex bracket satisfies $\|\{P, Q\}\|_1 \leq mn\|P\|_1\|Q\|_1$. The divisor bound gives $\|\chi_3\|_1 \leq h_3/d_3$ and $\|\Delta\chi_3\|_1 \leq \epsilon_3/d_3$. Balanced quartic projection does not increase the coefficient one-norm.

Extraction of its action Hessian has Frobenius norm at most twice that one-norm. Consequently

$$\|\Delta T\|_2 \le 2\epsilon_4 + (18h_3\epsilon_3 + 9\epsilon_3^2)/d_3.$$

Condition (S92) makes $\|T^{-1}\Delta T\|_2 < 1$, so $T + \Delta T$ is invertible. No term of degree at least five changes the Hessian, the finite-order divisors or this twist. Apply the already verified finite-dimensional local elliptic KAM construction. The resulting neighborhood may depend on the higher-order analytic remainder; no common numerical radius is claimed.

The 13 *cubic and* 22 *quartic feature directions.* Use the exact source normalizations $\ell = 10^{-3}$, $E_s = 6 \times 10^{-12}$ and

$$u = Bx/\ell, \quad e_s = x^T K_0 x/(2E_s), \quad a(u) = \ell^2 u^T K_3 u/(2E_s), \quad A = e_s - a(u).$$

The saved neural fourth input is $e_{\rm net} = x^T K_0 x/(2E_{\rm net}) = (E_s/E_{\rm net})e_s$, where $E_{\rm net}$ is the fixed saved binary64 neural energy scale. The homogeneous cubic feature directions are the ten monomials in u of degree three and the three monomials Au_i. The homogeneous quartic directions are the fifteen monomials in u of degree four, the six monomials Au_iu_j with $i \le j$, and A^2. There are seven separate quadratic directions, namely six u_iu_j and A; uncertainty in those directions is excluded by the present fixed-Hessian hypothesis and requires an additional spectral analysis.

Let $U_i \ge \|u_i(LX)\|_1$ and $C_A \ge \|A(LX)\|_1$ in the same canonical complex convention. Set

$$h_k(U) = \sum_{|\alpha|=k} \prod_i U_i^{\alpha_i}, \quad C_3 = E_s\{h_3(U) + C_A h_1(U)\}, \quad C_4 = E_s\{h_4(U) + C_A h_2(U) + C_A^2\}.$$

Specifically write $\Delta U_3 = E_s[\delta R_3 + A\delta S_1]$ and $\Delta U_4 = E_s[\delta R_4 + A\delta S_2 + \delta c_A A^2]$. Dimensionless bounds η_3, η_4 on the ordinary coefficients inside these respective brackets imply $\epsilon_3 \le C_3\eta_3$ and $\epsilon_4 \le C_4\eta_4$. These constants are computed anew from the verified 64-mode chart; they are not derivatives or three-mode constants.

How the constants are enclosed. The saved modal intervals give $Z = PL$ and $v = b^T L$. All derivatives below are taken at the origin in the saved four-feature jet $N(s_1, s_2, s_3, e_{\rm net})$, with $s = Px$. Write $f = 1 + c_E + N_4(0)$, where $N_4 = \partial N/\partial e_{\rm net}$, and $N_{ss} = (N_{ij})_{i,j\le 3}$, where $N_{ij} = \partial^2 N/(\partial s_i \partial s_j)$. The exact-source Hessian identity gives

$$R = L^T K_0 L = \{\mathrm{diag}(\omega) - E_{\rm net} Z^T N_{ss} Z - \kappa v^T v\}/f.$$

Thus C_A is bounded by $E_s^{-1}\sum_{i,j}|(R - \ell^2 U_X^T K_3 U_X)_{ij}|$, where $(U_X)_{ij} = Q_i Z_{ij}/\ell$; and $\|u_i(LX)\|_1 = \sqrt{2}\sum_j |(U_X)_{ij}| \le U_i$. The coefficient one-norm of the neural H_3 is bounded by substituting these verified modal intervals into the physically weighted-degree-three terms of the saved four-feature jet. Magnetic cosine contributes no cubic term at the origin.

Let $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_{64})$ and $N_T = \Omega^{-1} T \Omega^{-1}$. A fixed approximate inverse J with $\|JN_T - \mathrm{Id}\|_\infty \le d < 1$ gives

$$\|T^{-1}\|_2 \le \|T^{-1}\|_\infty \le \frac{\|J\|_\infty}{(1-d)\omega_{\min}^2}.$$

This uses symmetry of the exact inverse T^{-1}. All source binary parameters, the exact rational K_3, modal intervals and arithmetic bounds retain their original identities.

A simultaneous cubic and quartic sufficient neighborhood. For the exact-source 918-parameter graph with checkpoint prefix `f51bcea12c2f`, outward computation on its verified 64-mode chart gives $C_3 < 5.539762 \times 10^{13}$, $C_4 < 1.326507 \times 10^{24}$, $h_3 < 1.060037 \times 10^6$, $t < 4.060190 \times 10^{-5}$ and $d_3 > 1.696884$. The exact rational bounds, rather than these rounded displays, yield

$$(\eta_3, \eta_4) = (2^{-57}, 2^{-69}), \qquad t\left[2C_4\eta_4 + \frac{18h_3C_3\eta_3 + 9C_3^2\eta_3^2}{d_3}\right] < 0.357977 < 1. \tag{S93}$$

This is one sufficient neighborhood with an exactly fixed quadratic jet. It includes changes to the nonzero cubic and its homological correction. Supplementary Section S16 gives a direct quartic bound under $\Delta H_3 = 0$; Section S17 combines that response with cubic errors while fixing only H_2.

The seven quadratic directions, a general unknown feature factorization, target-valid observation errors and analytic remainders are separate hypotheses. Neither criterion places the present unknown target inside the stated coefficient neighborhood. An exceeded sufficient bound does not prove absence of invariant tori. No numerical torus radius or dissipative conclusion is assigned.

S16 A direct quartic response with the quadratic and cubic terms fixed

The reference is the exact-source conservative Hamiltonian H_N^{64} of the article, with the same 918 real parameters, checkpoint prefix `f51bcea12c2f`, exact nominal mass and elasticity operators, and complete 64-mode canonical chart. The fourth neural input is E_0/E_{net}; coefficients below use the distinct exact source scale $E_s = 6 \times 10^{-12}$ J. Let $x = LX$, $I_i = (X_i^2 + Y_i^2)/2$, and $\Omega = \mathrm{diag}(\omega_1, \dots, \omega_{64})$. The full reference cubic H_3 and its homological correction are retained. Both H_2 and H_3 are held exactly fixed in this section.

Use $u = Bx/\ell$, $\ell = 10^{-3}$ m, and $A = x^T(K_0 - B^T K_3 B)x/(2E_s)$. Write

$$u_r = v_r^T X, \qquad A = \frac{X^T C_\perp X}{2E_s}, \qquad C_\perp = L^T(K_0 - B^T K_3 B)L.$$

The 22 quartic directions P_j are the fifteen $E_s u^\gamma$ with $|\gamma| = 4$, the six $E_s A u_r u_s$ with $r \le s$, and $E_s A^2$. Define their action-Hessian responses T_j by $\langle P_j \rangle = I^T T_j I/2$, where brackets denote independent angle averages in the fixed canonical chart.

Lemma S27. Exact action-Hessian responses. *For $P = E_s \prod_{m=1}^4 (v_m^T X)$, the response is*

$$(T_P)_{ii} = 3E_s \prod_{m=1}^4 v_{m,i}, \tag{S94}$$

$$(T_P)_{ik} = E_s \sum_{\substack{S \subset \{1,2,3,4\} \\ |S|=2}} \prod_{m \in S} v_{m,i} \prod_{m \notin S} v_{m,k}, \quad i \neq k. \tag{S95}$$

Repeated factors occupy distinct positions in the six-term sum. For $P = E_s A u_r u_s$, the response is

$$(T_P)_{ii} = \tfrac{3}{2}(C_\perp)_{ii} v_{r,i} v_{s,i}, \tag{S96}$$

$$(T_P)_{ik} = \tfrac{1}{2}\{(C_\perp)_{ii} v_{r,k} v_{s,k} + (C_\perp)_{kk} v_{r,i} v_{s,i}\} + (C_\perp)_{ik}(v_{r,i} v_{s,k} + v_{r,k} v_{s,i}), \quad i \neq k. \tag{S97}$$

For $P = E_s A^2$, the response is

$$(T_P)_{ii} = \frac{3(C_\perp)_{ii}^2}{4E_s}, \qquad (T_P)_{ik} = \frac{(C_\perp)_{ii}(C_\perp)_{kk} + 2(C_\perp)_{ik}^2}{2E_s} \ (i \neq k). \tag{S98}$$

Proof. Since $X_i = \sqrt{2I_i}\cos\theta_i$, $\langle X_i^4 \rangle = 3I_i^2/2$ and $\langle X_i^2 X_k^2 \rangle = I_i I_k$ for $i \neq k$. Every other quartic monomial has a zero angle average. A coefficient p of X_i^4 therefore contributes $3p$ to T_{ii}; a coefficient q of $X_i^2 X_k^2$ contributes q to T_{ik}. Expanding the three displayed quartic polynomials gives the formulas, including the off-diagonal factors from the symmetric quadratic form.

Let T_0 be the full reference cubic-corrected twist, $N_0 = \Omega^{-1} T_0 \Omega^{-1}$ and $N_j = \Omega^{-1} T_j \Omega^{-1}$. For the fixed rational inverse proposal J, let $d < 1$ satisfy $\|\mathrm{Id} - JN_0\|_\infty \le d$. Take nonnegative entrywise bounds

$$(G_4)_{ik} \ge \sum_{j=1}^{22} |(JN_j)_{ik}|, \qquad b_4 = \max_i \sum_k (G_4)_{ik}. \tag{S99}$$

Theorem S28. A fixed-H_2, H_3 full-coordinate sufficient criterion. *Consider a real analytic conservative perturbation whose terms through degree three vanish and whose quartic part is $\Delta H_4 = \sum_{j=1}^{22} \delta_j P_j$, with $|\delta_j| \le \eta_4$. If*

$$d + b_4 \eta_4 < 1, \tag{S100}$$

then its full 64×64 *quartic twist is nonsingular. The fixed positive Hessian, finite nonresonance and real analyticity then imply the qualitative local elliptic KAM conclusion for the perturbed* 64*-coordinate Hamiltonian. The higher-order analytic remainder may change the local neighborhood; no numerical torus radius is specified.*

Proof. As H_2 and H_3 are unchanged, both the cubic homological solution and its quartic correction remain unchanged. Thus the normalized perturbed twist is exactly $N_0 + \sum_j \delta_j N_j$. For every matrix entry, $|\sum_j \delta_j (JN_j)_{ik}| \le \eta_4 (G_4)_{ik}$. Taking row sums gives

$$\left\| \mathrm{Id} - J \left(N_0 + \sum_j \delta_j N_j \right) \right\|_\infty \le d + b_4 \eta_4 < 1.$$

The Neumann criterion makes this product invertible, hence also the perturbed twist. The remaining local KAM hypotheses are inherited from the fixed quadratic part and the stated analyticity. Apply Supplementary Theorem S35.

Exact-source enclosure and numerical instance. The modal projections are interval enclosures from the same exact source graph, rather than rounded eigenvectors treated as exact modes. To retain cancellation in the normal quadratic form, let $Z = PL$, $v = b^T L$ and $f = 1 + c_E + N_e(0)$. The source Hessian identity gives

$$C_\perp = \frac{\Omega - E_{\mathrm{net}} Z^T N_{ll}(0) Z - \kappa v^T v}{f} - \ell^2 U_X^T K_3 U_X, \qquad U_X = \mathrm{diag}(Q/\ell) Z. \tag{S101}$$

Here N_{ll} and N_e are derivatives in the saved neural inputs (l, e_{net}) at the origin, with $e_{\mathrm{net}} = E_0 / E_{\mathrm{net}}$. The positive f, the binary feature scales, K_3, modal intervals, baseline twist and fixed preconditioner all refer to this same graph. Outward evaluation of all 22 responses gives

$$\begin{aligned} d &< 2.352888 \times 10^{-6}, & b_4 &< 4.2914981274 \times 10^{10}, \\ \eta_4 &= 2^{-37}, & d + b_4 \eta_4 &< 0.312250 < 1. \end{aligned} \tag{S102}$$

The sum of the 22 separately bounded induced norms is less than $5.0443713467 \times 10^{10}$; summing entrywise before taking the maximum row sum yields the smaller b_4 above. Independent exact interval arithmetic verifies all 4096 entrywise bounds and the same strict criterion. Compared with 2^{-69} in (S93), this quartic coefficient radius is 2^{32} times larger, under the essential additional condition $\Delta H_3 = 0$. It is not an optimal radius.

For a single direction let $g_j \ge \|JN_j\|_\infty$. The half-margin allowance $(1-d)/(2g_j)$ ensures $d + g_j|\delta_j| \le (1+d)/2 < 1$. For $E_s A^2$,

$$g_{A^2} < 3.108542704 \times 10^9, \qquad \frac{1-d}{2 g_{A^2}} \simeq 1.6084669610 \times 10^{-10}. \tag{S103}$$

The ratio is displayed approximately; the exact rational allowance is used in the numerical certificate. It bounds a perturbation about the reference coefficient, rather than the absolute target coefficient. The old-trajectory intervals in (S71)–(S72) are many orders of magnitude too wide to lie in this neighborhood. The seven quadratic directions, changes to H_3, unknown feature factorization, target remainders and physical observation errors are not supplied by the fixed-H_2, H_3 calculation.

S17 A joint full-coordinate cubic and quartic error criterion

Keep the same analytic conservative graph, equilibrium Hessian, canonical chart and order-four nonresonance as in the full-coordinate certificate. Let T denote its cubic-corrected twist, $\Omega = \mathrm{diag}(\omega)$, $N_T = \Omega^{-1} T \Omega^{-1}$, and let J satisfy $\|JN_T - \mathrm{Id}\|_\infty \le d < 1$. This section permits simultaneous errors in the cubic and quartic jets; only the quadratic jet is fixed exactly.

For the 22 dimensionless quartic coefficients in $\Delta U_4 = E_s[\delta R_4 + A \delta S_2 + \delta c_A A^2]$, write T_j for the action-Hessian response to the jth unit coefficient. These responses are linear in ΔU_4 and do not depend on whether the cubic jet is perturbed. Define the entrywise nonnegative matrix

$$G_{ik} = \sum_{j=1}^{22} |(J \Omega^{-1} T_j \Omega^{-1})_{ik}|, \qquad b_4 = \max_i \sum_k G_{ik}.$$

In numerical use each entry and b_4 is replaced by its outward upper bound. Use the same coefficient one-norm convention in complex canonical variables as in Supplementary Section S15. Suppose $\|H_3\|_1 \le h_3$, $\|\Delta H_3\|_1 \le C_3\eta_3$, and every quartic coefficient error has absolute value at most η_4. Let d_3 be the positive cubic divisor lower bound and $\omega_* > 0$ a lower bound on every ω_j.

Proposition S29. Simultaneous cubic and quartic perturbations. *The perturbed full* 64×64 *twist is nonsingular if*

$$d + \frac{\|J\|_\infty}{\omega_*^2 d_3}\left[18h_3C_3\eta_3 + 9(C_3\eta_3)^2\right] + b_4\eta_4 < 1. \tag{S104}$$

Under the stated analytic and fixed-quadratic hypotheses, the local positive-measure 64*-torus conclusion therefore persists.*

Proof. Separate the twist change as $\Delta T = T_4 + T_c$, with $T_4 = \sum_j \delta c_j T_j$ and T_c caused by cubic elimination. The definition of G gives $\|J\Omega^{-1}T_4\Omega^{-1}\|_\infty \le b_4\eta_4$. For any balanced quartic polynomial $Q = \sum_i q_{ii}I_i^2 + \sum_{i<k} q_{ik}I_iI_k$, its action Hessian has diagonal entries $2q_{ii}$ and off-diagonal entries q_{ik}. Each absolute row sum is at most $2\sum_{i\le k}|q_{ik}|$. Thus its matrix infinity norm, as well as its Frobenius norm, is bounded by twice its coefficient one-norm. The cubic homological equation and the canonical bracket estimate give

$$\|T_c\|_\infty \le \{18h_3C_3\eta_3 + 9(C_3\eta_3)^2\}/d_3.$$

The two diagonal factors Ω^{-1} give $\|J\Omega^{-1}T_c\Omega^{-1}\|_\infty \le \|J\|_\infty\omega_*^{-2}\|T_c\|_\infty$. Consequently the left side of (S104) bounds $\|J\Omega^{-1}(T+\Delta T)\Omega^{-1} - \mathrm{Id}\|_\infty$. The Neumann criterion proves invertibility. The quadratic spectrum and finite-order nonresonance are unchanged, so the analytic local elliptic KAM argument applies. Terms of degree at least five may change the size of the neighborhood, but not this twist test.

For the same 918-parameter exact-source graph with checkpoint prefix `f51bcea12c2f`, outward rational arithmetic gives $\eta_3 = 2^{-57}$ and $\eta_4 = 2^{-37}$ with the left side of (S104) below 0.487747. This combines the nonzero-cubic estimate and the directional quartic estimate; it does not assume $\Delta H_3 = 0$. It remains a sufficient conditional neighborhood. Neither the seven quadratic errors nor the stated cubic and quartic tolerances have been established for an unknown target by the present observations. Exceeding this sufficient neighborhood does not imply that target tori are absent.

The separate contributions to the displayed margin are a base defect below 2.352888×10^{-6}, a cubic budget below 0.175497, and a quartic budget below 0.312248. Independent exact rational reconstruction verifies their sum below $0.487747 < 1/2$. The cubic interaction with the nonzero reference cubic is included; it dominates the term quadratic in the cubic perturbation at this coefficient radius. This sufficient neighborhood retains all thirteen cubic and twenty-two quartic feature directions, with no terms of degree at most two. It is a qualitative local statement, not a numerical torus-radius or unknown-physical-target certificate.

S18 A quantitative full-coordinate bridge with a moving quadratic jet

Retain the same saved 918-real-parameter candidate and exact source operators as the full-coordinate certificate. Let $H^0 = p^TM^{-1}p/2 + U^0(x)$ be its certified analytic conservative Hamiltonian on 64 configuration coordinates. Let $S^TMS = I$, $S^TU^0_{xx}(0)S = \Omega^2$, $L = S\Omega^{-1/2}$, and retain the sign-selected real eigenbasis from the received modal certificate. Write H_3^0, H_4^0 for its homogeneous potentials in $x = LX$ and $z = (X+iY)/\sqrt{2}$. All polynomial norms below are coefficient one-norms in the independent formal variables $(z, \bar z)$.

Assume that an analytic real target has the same M, the same equilibrium at zero, and homogeneous differences through degree four of the form

$$\begin{aligned}
\Delta U_2 &= E_s[\delta R_2(u) + \delta a A],\\
\Delta U_3 &= E_s[\delta R_3(u) + A\delta S_1(u)],\\
\Delta U_4 &= E_s[\delta R_4(u) + A\delta S_2(u) + \delta c_A A^2],
\end{aligned}$$

where $u = Bx/\ell$, $A = x^TK_0x/(2E_s) - \ell^2u^TK_3u/(2E_s)$, $\ell = 1/1000$, and $E_s = 3/500000000000$ exactly. Each ordinary coefficient of the respective 7, 13 and 22 terms has absolute value at most η_2, η_3, η_4. Higher homogeneous terms can be

arbitrary analytic terms of degree at least five. These hypotheses concern the target's full-coordinate jet; they do not follow solely from projected observations.

Use received outward constants

$$\|u_i(LX)\|_1 \le U_i, \quad \|A(LX)\|_1 \le a_A, \quad \|H_3^0\|_1 \le h_3, \quad \|H_4^0\|_1 \le h_4, \quad \|\Delta U_r(LX)\|_1 \le C_r\eta_r \quad (r=3,4).$$

Set $C_2 = E_s(\sum_{i\le j} U_iU_j + a_A)$. Let w_-, w_+ bound all reference frequencies below and above, $g>0$ bound every distinct reference squared-frequency separation below, and $d_3, d_4 > 0$ bound all reference nonzero divisors of orders three and at most four, respectively. In particular d_3 also includes the single-mode cubic divisors. Put

$$\begin{aligned} \epsilon &= w_+C_2\eta_2, & v &= \epsilon/w_-,\\ t &= 16\sqrt{\frac{w_+}{w_- - v}}\frac{\epsilon}{g-\epsilon} + \frac{v}{2(w_- - v)}, & s &= 1+t,\\ e_3 &= h_3(s^3-1) + C_3\eta_3 s^3, & d_3' &= d_3 - 3v. \end{aligned}$$

Any outward upper bound may replace t in these expressions.

Let T^0 be the reference cubic-corrected Birkhoff twist and let J be the retained exact-binary preconditioner, with $\|J\Omega^{-1}T^0\Omega^{-1} - I\|_\infty \le d < 1$. The received directional quartic response gives b_4 such that $\|J\Omega^{-1}T_4(\Delta U_4)\Omega^{-1}\|_\infty \le b_4\eta_4$, when ΔU_4 is written in the reference chart.

Proposition S30. Moving-Hessian quantitative criterion. *Suppose*

$$0 \le \epsilon < \min\{w_-^2, g/2\}, \qquad v < w_-, \qquad 4v < d_4, \qquad 3v < d_3,$$

and

$$d + b_4\eta_4 + \frac{\|J\|_\infty}{w_-^2}\left[2(h_4 + C_4\eta_4)(s^4-1) + \frac{9(2h_3e_3 + e_3^2)}{d_3'} + \frac{27h_3^2 v}{d_3 d_3'}\right] < 1. \tag{S105}$$

Then the target Hessian is strictly elliptic with simple frequencies and nonresonance through order four, and its full 64-dimensional cubic-corrected twist is nonsingular. The same analytic local elliptic KAM theorem gives a positive-measure family of invariant Lagrangian 64-tori in every neighborhood of the equilibrium.

Proof. For a homogeneous polynomial in X alone, the substitution $X_j = (z_j + \bar z_j)/\sqrt{2}$ multiplies its coefficient norm by $2^{r/2}$ in degree r: distinct X multiindices have disjoint supports in $(z, \bar z)$. Thus $\|\nabla_X^2 \Delta U_2(LX)\|_2 \le 2\|\Delta U_2(LX)\|_{1,X} \le C_2\eta_2$. For $E = S^T\Delta U_{2,xx}S$ this gives $\|E\|_2 \le w_+C_2\eta_2 = \epsilon$. The min–max characterization of the eigenvalues of the real symmetric matrices Ω^2 and $\Omega^2 + E$ gives $|\lambda_i' - \omega_i^2| \le \epsilon$. Consequently all eigenvalues remain positive and separated, and $|\omega_i' - \omega_i| \le \epsilon/w_- = v$. Every divisor with $0 < |k|_1 \le 4$ changes by at most $4v$.

Let R be the real orthogonal matrix whose ith column is the normalized eigenvector of $\Omega^2 + E$ for λ_i'. Projecting its eigenvalue equation onto the complement of the ith reference coordinate gives

$$\|(R_{ji})_{j\ne i}\|_2 \le \frac{\epsilon}{g-\epsilon} < 1.$$

There is therefore a unique sign choice $R_{ii} > 0$. The column distance to the ith coordinate vector is at most $\sqrt{2}\epsilon/(g-\epsilon) \le 2\epsilon/(g-\epsilon)$. Orthogonality makes the squared distance of row i to the corresponding coordinate row equal to the squared distance of column i, namely $2(1-R_{ii})$. Cauchy–Schwarz in 64 coordinates therefore yields

$$\|R - I\|_\infty \le \frac{16\epsilon}{g-\epsilon}.$$

This explicitly fixes the real phase and does not assume a differentiable eigenvector routine or arbitrarily chosen complex phases.

In the target chart $S' = SR$, $L' = SR(\Omega')^{-1/2}$, define $C = \Omega^{1/2}R(\Omega')^{-1/2}$. The canonical relation between the two charts is $X^0 = CX'$, $Y^0 = C^{-T}Y'$. Since all nonlinear terms are configuration potentials, their change is controlled just by C. Splitting $C - I$ into its $R - I$ part and its diagonal frequency-scaling part proves $\|C - I\|_\infty \le t$. For the latter part use $|\sqrt{\omega_i/\omega_i'} - 1| \le v/[2(w_- - v)]$.

For any homogeneous configuration polynomial P_r, multilinear expansion of $P_r((I + (C - I))X)$ gives

$$\|P_r(CX) - P_r(X)\|_1 \le (s^r - 1)\|P_r\|_1, \qquad \|P_r(CX)\|_1 \le s^r\|P_r\|_1.$$

The same bounds hold in the complex coefficient norm by the degree-dependent factor just proved. Hence the target cubic in its own chart differs from H_3^0 by at most e_3. The direct quartic twist change is the sum of the reference-chart directional response to ΔU_4 and the response to $(H_4^0 + \Delta U_4)(CX) - (H_4^0 + \Delta U_4)(X)$. The latter twist infinity norm is bounded by $2(h_4 + C_4\eta_4)(s^4 - 1)$.

On degree-three polynomials, the homological inverse for ω' has coefficient operator norm at most $1/d_3'$, and its difference from the reference inverse has norm at most $3v/(d_3 d_3')$, coefficient by coefficient. For homogeneous polynomials of degrees r, s, $\|\{f_r, g_s\}\|_1 \le rs\|f_r\|_1\|g_s\|_1$. The action projection is contractive in this norm. The action Hessian of a balanced quartic polynomial has infinity norm at most twice its coefficient norm. Applying these bounds to $\Pi_{\mathrm{action}}\{H_3, \chi_3\}/2$ gives

$$\|T_c' - T_c^0\|_\infty \le \frac{9(2h_3e_3 + e_3^2)}{d_3'} + \frac{27h_3^2 v}{d_3 d_3'}.$$

The last summand controls the moving homological denominators, even when the cubic polynomial itself is unchanged in the common formal chart.

Combining the three twist changes proves that the left side of (S105) bounds $\|J\Omega^{-1}T'\Omega^{-1} - I\|_\infty$. The normalizing Ω here is deliberately the fixed reference matrix; any positive diagonal congruence can test nonsingularity of T'. The Neumann criterion proves T' nonsingular. Analyticity, the verified finite-order nonresonance and the invertible cubic-corrected twist permit the same general fifth-order local KAM reduction as for the reference Hamiltonian. No analyticity norm for an unknown target, finite torus radius, dissipative-flow conclusion or target observation reception is claimed by this finite-jet test.

Same-candidate numerical instance. Outward constants from the same modal, feature-jet and directional-response enclosures give a coefficient neighborhood

$$(\eta_2, \eta_3, \eta_4) = (2^{-86}, 2^{-57}, 2^{-37}).$$

All spectral and divisor inequalities above hold strictly, and the left side of (S105) is below $0.495840 < 1/2$. Here $\epsilon < 5.061 \times 10^{-15}$ and $t < 1.751 \times 10^{-15}$. The quartic norm includes all 22 weighted degree-four network terms and the magnetic cosine quartic. Exact rational recomputation checks both the perturbed criterion and its zero-quadratic-error limit. With the original unrounded constants, the latter agrees exactly with Proposition S29; outwardly shortened constants need only give a larger valid bound.

These are sufficient coefficient tolerances, not observed target errors. For the seven quadratic directions, comparison of complete target and candidate intervals gives a maximum distance of approximately 4.617×10^{-7} among the six restricted quadratic coefficients and 2.636×10^{-4} in the A direction. None fits $\eta_2 = 2^{-86}$. The strict theorem therefore supplies a positive theoretical neighborhood while the unknown-target transfer remains unproved. Its analytic feature and equilibrium hypotheses still require independent support.

S19 From the recorded 31-coefficient loss to a full-coordinate KAM test

Fix the same 918-real-parameter saved candidate used by the full-coordinate reference certificate. Use the exact source scales $\ell = 1/1000$, $E_s = 3/500000000000$, and set

$$u = Bx/\ell, \qquad e = x^T K_0 x/(2E_s), \qquad A = e - \ell^2 u^T K_3 u/(2E_s).$$

Here $BV = I_3$ and $K_3 = V^T K_0 V$, so $A = 0$ on the trial space. For $\nu \in \{\theta, *\}$ assume the material correction is $W_\nu(x) = E_s \Phi_\nu(u, e)$, with Φ_ν real analytic near zero, $\Phi_\nu(0,0) = 0$ and $\partial_u \Phi_\nu(0,0) = 0$. Give u weight one and e weight two. Its weighted Taylor jet in physical degrees two through four, denoted P_ν, has the unique decomposition

$$\begin{aligned} P_\nu &= R_\nu(u) + AS_\nu(u) + c_{\nu,A^2} A^2, \\ R_\nu &= \sum_{2 \le |\alpha| \le 4} R_{\nu,\alpha} u^\alpha, \qquad S_\nu = \sum_{|\gamma| \le 2} S_{\nu,\gamma} u^\gamma. \end{aligned} \tag{S106}$$

There are $31 + 10 + 1 = 42$ ordinary coefficients. Since $A = 0$ on $x = V\ell u$, R_ν is precisely the restricted material jet received by the existing eleven-node design. The terms AS_0, AS_1 and AS_2, with S_j homogeneous of degree j, have physical degrees two, three and four, respectively; A^2 has degree four.

The objective actually used. The recorded coefficient objective is the 31-term saved-center loss

$$\widetilde{L}_{\mathrm{jet}} = \frac{1}{31} \sum_{2 \le |\alpha| \le 4} \left(\frac{k_{\theta,\alpha} - \widetilde{c}_\alpha}{\widetilde{s}_\alpha} \right)^2, \qquad \widetilde{s}_\alpha > 0. \tag{S107}$$

Here $k_{\theta,\alpha}$ denotes its recorded coefficient convention; this is the existing objective, not a 42-term replacement. Require a rigorous upper bound $\overline{L}_{31} \ge \widetilde{L}_{\mathrm{jet}}$ and received bounds

$$|R_{*,\alpha} - \widetilde{c}_\alpha| \le \rho_\alpha, \qquad |R_{\theta,\alpha} - k_{\theta,\alpha}| \le \tau_\alpha.$$

The radius ρ_α includes observation/force-reconstruction, target-tail and original-to-saved-center allowances; τ_α includes any runtime-centering or source-arithmetic transfer. With the main text's exact-source interpretation $k_{\theta,\alpha} = R_{\theta,\alpha}$, one takes $\tau_\alpha = 0$. A float64 loss alone supplies neither a rigorous $\overline{L}_{31}$ nor a transfer bound. The already proved component argument gives

$$|R_{\theta,\alpha} - R_{*,\alpha}| \le E_{R,\alpha} := \widetilde{s}_\alpha \sqrt{31 \overline{L}_{31}} + \rho_\alpha + \tau_\alpha. \tag{S108}$$

For the actual saved-center coefficient contract, its stored radius already includes the original observation radius and center-rounding allowance. Taking that radius as ρ_α and $\tau_\alpha = 0$ in the exact-source convention counts each allowance once. The outward same-weight coefficient evaluation gives $\overline{L}_{31} < 0.010821$. This upper loss bound does not imply that the resulting coefficient errors meet the much smaller KAM tolerances.

A received direct component residual may replace the aggregate term by its tighter bound, retaining the same allowances exactly once.

The additional eleven validation constraints. Independently receive target intervals $I_{*,\gamma} = [l_{*,\gamma}, u_{*,\gamma}]$ for the ten coefficients of S_* and I_{*,A^2} for c_{*,A^2} from additional force or trajectory functionals. Receive candidate intervals $I_{\theta,\gamma}$ and I_{θ,A^2} for the same exact source and saved weights. All force, trajectory, damping-nuisance, arithmetic and analytic-tail errors needed for those target intervals must be included. For any two intervals define

$$d_\times([l,u],[l',u']) = \max\{|l - u'|, |u - l'|\}, \qquad E_{S,\gamma} = d_\times(I_{\theta,\gamma}, I_{*,\gamma}), \quad E_{A^2} = d_\times(I_{\theta,A^2}, I_{*,A^2}).$$

This compares the complete intervals, including their center discrepancy. Here "independently receive" means independently certify the missing constraints; it asserts no statistical independence of samples. Complete forces on the old trial space can receive S but cannot receive A^2; that last coordinate needs a valid off-trial or trajectory sensitivity.

Corollary S31. Conditional 31-loss plus 11-validation composition. *Define*

$$\eta_2 = \max\{\max_{|\alpha|=2} E_{R,\alpha},\ E_{S,0}\},$$
$$\eta_3 = \max\{\max_{|\alpha|=3} E_{R,\alpha},\ \max_{|\gamma|=1} E_{S,\gamma}\},$$
$$\eta_4 = \max\{\max_{|\alpha|=4} E_{R,\alpha},\ \max_{|\gamma|=2} E_{S,\gamma},\ E_{A^2}\}. \tag{S109}$$

Assume the target autonomous conservative Hamiltonian has the same positive mass matrix, equilibrium and known elastic/magnetic potentials as the reference, and satisfies the analytic and feature-factorization hypotheses above. Retain the source-bound reference constants and hypotheses of Proposition S30. If the triple (S109) *satisfies all its strict inequalities, then the target is strictly elliptic with simple linear frequencies, nonresonance through order four, and nonsingular full 64-dimensional cubic-corrected Birkhoff twist. Its local elliptic KAM conclusion follows: invariant Lagrangian 64-tori of positive measure occur in every neighborhood of zero.*

Proof. Each nonnegative summand of $31\widetilde{L}_{\mathrm{jet}}$ bounds its individual squared residual; the triangle inequality proves (S108). The maximum distance between two closed real intervals is $d_\times$, proving the eleven other bounds. Subtract (S106) for the two systems. Sorting its terms by physical degree gives exactly the 7, 13 and 22 coefficient perturbations admitted by the moving-Hessian input, bounded by (S109). That input and its analytic local KAM theorem now apply. No IID sampling, sample independence, optimization convergence or zero-loss premise enters this argument.

A finite positive loss is permitted, but must meet the strict KAM budget. For example, for any triple b_2, b_3, b_4 passing that test, monotonicity of its bounds in nonnegative coefficient errors shows that it suffices that the eleven extra errors meet their corresponding b_j and, for every α, $\rho_\alpha + \tau_\alpha < b_{|\alpha|}$ and

$$\overline{L}_{31} \le \frac{1}{31} \min_{2\le|\alpha|\le4} \left(\frac{b_{|\alpha|} - \rho_\alpha - \tau_\alpha}{\widetilde{s}_\alpha} \right)^2 .$$

This is a conditional budget for the recorded loss, not a newly executed training objective. The present 31-coefficient receipt and incomplete extra validation do not instantiate the corollary. A finite jet is not the full analytic function: higher terms remain analytic and need their own bounds when finite observations are used to receive the target jet. Target force/tail assumptions are not established by fitting the candidate. The conclusion concerns the autonomous conservative system; it gives no KAM statement for the dissipative or driven training trajectories and no specified finite torus radius.

S20 A local KAM criterion for the admitted quadratic target family

Let $I = [\beta_-, \beta_+] \subset (-1, \infty)$ be a compact interval, $A, K_0 \in \mathbb{R}^{3\times3}$ symmetric positive definite, $d \in \mathbb{R}^3$, and $\kappa > 0$. Consider the conservative family

$$H_\beta(q,p) = \tfrac{1}{2} p^T A^{-1} p + \tfrac{1+\beta}{2} q^T K_0 q + \kappa(1 - \cos(d^T q)). \tag{S110}$$

Here the physical-family hypothesis $W_* = \beta E_0$ is stipulated. Compatibility of observations with I does not establish this constitutive hypothesis for an unknown device.

Lemma S32. Uniform finite-order conditions for the quadratic family. *Write $K_\beta = (1+\beta)K_0 + \kappa dd^T$. Suppose that throughout I the generalized eigenvalues of (K_β, A) are simple, positive and ordered as $\lambda_1 < \lambda_2 < \lambda_3$, with $\omega_j = \sqrt{\lambda_j}$. Suppose further that*

$$\inf_{\beta\in I} |k\cdot\omega(\beta)| > 0 \quad (0 < |k|_1 \le 4), \qquad \inf_{\beta\in I} (d^T u_j(\beta))^2 > 0 \quad (j = 1,2,3),$$

where the eigenvectors are A-orthonormal. Then the quartic Birkhoff twist, in physical canonical actions, has inertia $(2,1,0)$ uniformly on I. In particular its determinant is bounded away from zero. Every member of (S110) *satisfies the qualitative local elliptic KAM criterion: sufficiently small neighborhoods of the origin contain a positive-measure family of*

invariant Lagrangian three-tori. This assertion assigns no numerical torus radius or all-integer Diophantine property to the equilibrium frequency vector.

Proof. The Hessian K_β is positive definite because $1+\beta>0$. Choose the signs of the eigenvectors so that $d^T u_j>0$, and let $S=(u_1,u_2,u_3)$. The canonical transformation

$$q=S\,\mathrm{diag}(\omega_j^{-1/2})X,\qquad p=AS\,\mathrm{diag}(\omega_j^{1/2})Y$$

gives $H_2=\sum_j \omega_j(X_j^2+Y_j^2)/2$. There is no cubic term. Put $a_j=(d^Tu_j)^2/\omega_j>0$ and $I_j=(X_j^2+Y_j^2)/2$. The angle averages are $\langle X_j^4\rangle=3I_j^2/2$ and $\langle X_i^2X_j^2\rangle=I_iI_j$ for $i\neq j$. The order-four nonresonance hypothesis makes the resonant projection of $-\kappa(d^Tq)^4/24$ equal to

$$Z_4=-\frac{\kappa}{16}\sum_j a_j^2I_j^2-\frac{\kappa}{4}\sum_{i<j}a_ia_jI_iI_j=\tfrac12 I^TTI,\qquad T=-\frac{\kappa}{8}D(2\mathbf{1}\mathbf{1}^T-I_3)D,\quad D=\mathrm{diag}(a_1,a_2,a_3).$$

The middle matrix has eigenvalues $5,-1,-1$ and inverse $-I_3+(2/5)\mathbf{1}\mathbf{1}^T$. Consequently

$$\det T=-\frac{5\kappa^3}{512}(a_1a_2a_3)^2<0,\qquad T^{-1}=-\frac{8}{\kappa}D^{-1}[-I_3+(2/5)\mathbf{1}\mathbf{1}^T]D^{-1}. \tag{S111}$$

Congruence and the negative scalar give two positive and one negative twist eigenvalues. The potential is entire. Simplicity of the spectrum and nonzero overlaps give analytic canonical coordinates locally in β; the displayed sign convention patches them across I. The compact parameter interval and the strict uniform bounds control the finite normal-form transformations, twist inverse and analytic remainders uniformly. The usual small-action rescaling therefore places each member within the classical nondegenerate KAM theorem. The resulting Diophantine frequencies are selected from the nonlinear frequency image; they need not equal $\omega(\beta)$.

An enclosure of the entire parameter interval. Choose a rational $\beta_c\in I$, solely from the received interval. For $K_c=K_{\beta_c}$, set

$$m=\frac{1+\beta_-}{1+\beta_c},\qquad M=\frac{1+\beta_+}{1+\beta_c}.$$

Since $K_0\preceq K_c/(1+\beta_c)$, $mK_c\preceq K_\beta\preceq MK_c$ for every $\beta\in I$. The generalized min–max principle then gives $m\lambda_j(\beta_c)\le\lambda_j(\beta)\le M\lambda_j(\beta_c)$. Disjoint outward intervals obtained this way label the eigenvalues throughout I.

To avoid substituting center eigenvectors for an interval family, use

$$P_\beta(t)=\det(K_\beta-tA),\qquad N_\beta(t)=d^T\,\mathrm{adj}(K_\beta-tA)d,\qquad (d^Tu_j)^2=-\frac{N_\beta(\lambda_j)}{P'_\beta(\lambda_j)}.$$

This residue identity follows from the symmetric definite spectral decomposition of $(K_\beta-tA)^{-1}$. Exact spectral projectors of $A^{-1}K_\beta$ give the signed eigenvectors

$$u_j=\left[\prod_{k\neq j}\frac{A^{-1}K_\beta-\lambda_kI_3}{\lambda_j-\lambda_k}\right]A^{-1}d\,/\sqrt{(d^Tu_j)^2}.$$

Outward evaluation with both β and all λ_j interval-valued encloses the complete family of canonical charts. Exact identities follow from the spectral theorem; residual intervals containing zero serve as implementation checks rather than proofs of those identities. The executable certificate separately verifies all order-at-most-four divisors, the positive overlap lower bounds and (S111).

The interval I in the numerical instance is the exact scalar compatibility intersection of the received 33 force and 31 coefficient constraints. No material value is assigned to β independently of that interval. This is a target-family certificate under a strong constitutive prior. It supplies no identification of unknown physics, necessity of a neural correction, complex-network advantage, invariant full-field trial subspace, or forced/damped KAM statement.

S20.1 A received numerical instance for the entire target family

The exact compatibility interval received from the 33 force and 31 coefficient inequalities is used unchanged. Its endpoints are approximately 0.0599999981693 and 0.0600000004402; these displays do not replace the exact rational endpoints in the calculation. The matrices are the same native $A = M_3$, $K_0 = K_{03}$, d and exact κ used in the current three-coordinate mechanical restriction. No network weights enter this family calculation.

Corollary S33. Qualitative local KAM for every admitted family member. *Under the explicit constitutive hypothesis $W_* = \beta E_0$, every β in the entire received compatibility interval satisfies Lemma S32. The uniform frequency enclosures in physical radians per second are*

Mode	*Strict enclosing interval*
1	$[11.73745098, 11.73745102]$
2	$[72.26615477, 72.26615488]$
3	$[326.09696048, 326.09696086]$

All three squared modal port overlaps are strictly positive throughout the interval. Every integer combination with $0 < |k|_1 \le 4$ satisfies $|k \cdot \omega| > 11.73745099\ \mathrm{rad\,s^{-1}}$. In the physical-action convention of (S111),

$$-6.217103 \times 10^{17} < \det T(\beta) < -6.217101 \times 10^{17}, \qquad \sigma_{\min}(T(\beta)) > 98709.9.$$

The twist inertia is $(2, 1, 0)$. The target family therefore has the qualitative local invariant-three-torus conclusion of Lemma S32. No numerical action or torus radius is asserted.

Proof. The frozen 512-bit Arb calculation receives the exact beta endpoints from the successful scalar feasibility certificate. Exact rational Sylvester tests verify the mass and the lower-endpoint Hessian. Exact Sturm isolation at the interval-selected dyadic center, followed by the Loewner/min–max enclosure, gives disjoint spectral intervals for every beta. The characteristic-resolvent residues and polynomial spectral projectors are evaluated with both beta and the eigenvalues interval-valued. They receive positive port overlaps and enclose the changing eigenbasis. All 64 plus/minus representatives through order four exclude zero. The analytic twist determinant and inverse formulas then give the stated strict bounds; a separate generic quartic $a, \bar{a}$ expansion checks their factors. The real spectral theorem and the angle-average calculation supply the exact identities. The local elliptic reduction and the classical KAM theorem (Pöschel 2001) give the qualitative conclusion.

The result concerns the admitted conservative family directly. It does not establish perturbative transfer from the learned candidate, unknown physical identification, an invariant full-field trial subspace, or necessity of a neural correction.

S21 Exact finite conditions for the declared full-coordinate family

The matrices in this section are the exact real constants represented by the stored binary64 source entries. Conversion to rational numbers precedes the sum $M = M_0 + M_r$. All 64 free transverse displacement and rotation coordinates are retained, in the declared physical order. Set $b = e_{64}$ and consider only the conservative Hamiltonian

$$H_\beta(x, p) = \tfrac{1}{2} p^T M^{-1} p + \tfrac{1+\beta}{2} x^T K_0 x + \kappa(1 - \cos(b^T x)), \qquad \kappa = \frac{6380294379767596475747514881383}{10633823966279326983230456482242756608}.$$

Let $[\beta_-, \beta_+]$ denote the exact rational intersection recorded in the frozen scalar-family receiver. Its decimal display is approximately $[0.059999998169293786, 0.0600000044017715]$; the decimal values do not define the interval. The reference parameter is $\beta_c = 3/50$.

Proposition S34. Received full-coordinate finite conditions. *For every $\beta \in [\beta_-, \beta_+]$, the origin of H_β is elliptic. Its 64 positive frequencies are simple, all modal tip overlaps are nonzero, and*

$$|k \cdot \omega(\beta)| \ge \frac{2472231}{1073741824} > 0, \qquad k \in \mathbb{Z}^{64}, \quad 0 < |k|_1 \le 4.$$

Consequently its quartic Birkhoff normal form has a nonsingular action Hessian. In the convention $H_{\mathrm{BNF}} = \omega \cdot I + \frac{1}{2} I^T T I + O_6$*, it is*

$$T = -\frac{\kappa}{8} D(2\mathbf{1}\mathbf{1}^T - I_{64})D, \qquad D = \operatorname{diag}\left(\frac{(b^T u_j)^2}{\omega_j}\right),$$

where $u_i^T M u_j = \delta_{ij}$*. Its inertia is* $(63, 1, 0)$ *and*

$$\det T = -127 \left(\frac{\kappa}{8}\right)^{64} \prod_{j=1}^{64} \left(\frac{(b^T u_j)^2}{\omega_j}\right)^2 \neq 0.$$

Proof. Exact banded rational LDL^T factorizations establish $M \succ 0$ and $K_0 \succ 0$. Thus $(1+\beta)K_0 + \kappa b b^T \succ 0$ for $\beta > -1$. The reduced characteristic polynomial $P_0(t) = \det(K_0 - tM)$ and tip cofactor $N_0(t) = \det[(K_0 - tM)_{1:63,1:63}]$ have degree-preserving reductions of degrees 64 and 63 modulo the verified prime 2147483647. Every source denominator is invertible modulo this prime. Their leading coefficients are respectively 1029774066 and 1236496576. Stored coefficient lists of degrees 62 and 63 give an explicit identity $AP_0 + BN_0 = 1$ over the finite field. Hence P_0, N_0 are coprime over $\mathbb{Q}$ and have no common complex root.

For $s = 1 + \beta$, $t = \lambda/s$, rank-one expansion gives

$$P_\beta(\lambda) = s^{64} P_0(t) + \kappa s^{63} N_0(t), \qquad N_\beta(\lambda) = s^{63} N_0(t).$$

Therefore these polynomials have no common root for any $\beta > -1$. At a repeated eigenvalue of a symmetric definite pencil the adjugate vanishes, whereas at a simple eigenvalue it is a nonzero multiple of uu^T. The absence of a common root excludes repeated eigenvalues and zero tip overlap, uniformly for all such parameters.

At β_c, 128 exact rational endpoint inertia computations isolate all 64 positive eigenvalues in disjoint intervals. The floating eigensolver supplies candidate endpoints only. All 64 brackets passed their first declared padding, without an endpoint retry. For $m = (1+\beta_-)/(1+\beta_c)$ and $M_* = (1+\beta_+)/(1+\beta_c)$,

$$m K_{\beta_c} \preceq K_\beta \preceq M_* K_{\beta_c}.$$

The generalized min–max principle and exact dyadic bounds on $\sqrt{m}, \sqrt{M_*}$ transport the brackets to the whole parameter interval. On a grid with denominator 2^{32} every transported frequency is within $\varepsilon = 337625/2^{32}$ of its stored center.

An exact integer enumeration includes repetitions and covers all signed sums with at most four terms: positive-only sums, one against one, two against one, three against one, and two against two. The identical two-pair multisets represent $k = 0$ and are excluded. Sorting all 2080 pair multisets covers the two-against-two minimum by adjacent distinct sums. The 45760 triple multisets cover all three-against-one cases. The smallest center gap is $11239424/2^{32}$, attained by the signed modes $(2, 23, 30) - (37)$. Subtracting 4ε gives $9888924/2^{32} = 2472231/1073741824$. Canonical oscillator normalization and quartic angle averaging then give the displayed twist formula. Its nonsingularity follows from the exact nonzero overlaps, not from a floating determinant.

The conclusion is relative to the 64-coordinate transverse Hamiltonian, whose canonical phase space has dimension 128. It does not assert a 96-dimensional Lagrangian torus in a model retaining axial coordinates. The source matrices define exact constants; equality with compiled simulator operations or with unknown material laws is a separate premise. The local KAM implication is supplied by the following analytic argument.

S22 From fourth-order finite conditions to local KAM tori

Theorem S35. Local elliptic KAM criterion in arbitrary finite dimension. *Let* H *be real analytic and even near an elliptic equilibrium in* $\mathbb{R}^{2n}$*, with* $n \geq 2$*. Suppose its positive linear frequencies satisfy* $k \cdot \omega \neq 0$ *for* $0 < |k|_1 \leq 4$*, and its quartic Birkhoff normal form is*

$$H \circ \Phi = \omega \cdot I + \tfrac{1}{2} I^T T I + R, \qquad \det T \neq 0, \quad R = O(\|Z\|^6).$$

Then every sufficiently small neighborhood of the equilibrium contains a positive-Lebesgue-measure union of invariant Lagrangian n-tori with Diophantine frequencies. No all-order Diophantine hypothesis on the equilibrium frequency vector is required. This is a qualitative local conclusion, with no specified numerical amplitude threshold.

Proof. Finite fourth-order nonresonance allows the quartic homological equation to be solved except for action monomials. There are no cubic terms. The degree-four generating function has an odd Hamiltonian vector field, so its local analytic time-one map is odd. It preserves the evenness of the Hamiltonian; consequently the remainder begins at degree six. This constructs the stated finite normal form without dividing by any higher-order small divisor.

We apply Theorems A and B of Pöschel (2001). Fix $\tau > n-1$ and $\nu = \tau + 1$. The action-offset radius in those theorems will be denoted by r, their frequency parameter radius by h_p, and their Diophantine constant by α. Their smallness requirements are

$$|P|_{r,s,h_p} \le \gamma \alpha r s^\nu, \qquad \alpha s^\nu \le h_p, \qquad r, s, h_p \le 1,$$

where $s > 0$ is an angular analytic strip and $\gamma > 0$ depends only on n, τ. These hypotheses must be verified after accounting for the shrinking nonlinear frequency domain.

Let $0 < \rho \ll 1$ and choose real actions in $\rho(1,2)^n$; use a fixed smaller cube for the frequency parameters. On a complex neighborhood of this cube of action width $c\rho$, and a fixed strip $|\operatorname{Im}\theta| < s$, the oscillator action–angle map is analytic with size $O(\sqrt{\rho})$. The square roots have a fixed analytic branch since the neighborhood stays away from every $I_j = 0$. Consequently $|R| \le C\rho^3$ there. All constants in this proof are fixed before $\rho \to 0$; none is asserted to be numerically small.

Write $h_0(I) = \omega \cdot I + \frac{1}{2} I^T T I$. Since T is invertible, $\varpi = \nabla h_0(I_0) = \omega + T I_0$ is a valid frequency parameter and $I_0 = T^{-1}(\varpi - \omega)$. Its real frequency domain Ω_ρ is a translated parallelepiped of diameter $O(\rho)$ and volume comparable to ρ^n. Take $h_p = c_p \rho$, with $c_p > 0$ sufficiently small, so the complex parameter neighborhood is mapped into the analytic action neighborhood. Translate $I = I_0(\varpi) + J$ for each fixed parameter. The same physical Hamiltonian becomes

$$H = e(\varpi) + \varpi \cdot J + P(J, \theta; \varpi), \qquad P = \tfrac{1}{2} J^T T J + R(I_0 + J, \theta),$$

and therefore $|P|_{r,s,h_p} \le C(r^2 + \rho^3)$.

Choose $r = \rho^{3/2}$ and $\alpha = \rho^{5/4}$, keeping $s \le 1$ fixed. For sufficiently small ρ all domains are valid and

$$\frac{|P|_{r,s,h_p}}{\alpha r s^\nu} \le C_s \rho^{1/4} \longrightarrow 0, \qquad \frac{\alpha s^\nu}{h_p} = \frac{s^\nu}{c_p} \rho^{1/4} \longrightarrow 0.$$

Thus the actual quantitative hypotheses of Theorems A and B hold. In particular, the remainder is compared with the shrinking Diophantine constant and shrinking action radius; nonsingularity of T alone is not used as a substitute for this check.

For completeness, the Diophantine parameter set has positive measure in this shrinking frequency domain. The intersection with a resonance slab $|k \cdot \varpi| < \alpha / |k|_1^\tau$ has measure at most $C\rho^{n-1}\alpha/|k|_1^{\tau+1}$. Summing over $k \neq 0$ converges for $\tau > n-1$. A boundary strip of width $O(\alpha)$ has the same order, and the excluded relative measure is therefore $O(\alpha/\rho) = O(\rho^{1/4})$. Theorem B supplies the bi-Lipschitz parameter correction and the Lagrangian torus embeddings. Its estimates give a parameter correction of size $O(|P|/r) = O(\rho^{3/2}) = o(\rho)$ and a Lipschitz change tending to zero, since $|P|/(\alpha r s^\nu) = O(\rho^{1/4})$. Together with the invertible frequency-to-action map this preserves positive measure for the union of invariant tori of this single Hamiltonian. Their oscillator amplitude is $O(\sqrt{\rho})$ and the analytic canonical map Φ maps them into arbitrarily small neighborhoods of the original equilibrium.

Corollary S36. Declared full64 family. *For each exact $\beta \in [\beta_-, \beta_+]$ in Proposition S34, the conservative Hamiltonian H_β has positive-measure families of invariant Lagrangian 64-tori arbitrarily near its elliptic equilibrium.*

Proof. The cosine Hamiltonian is even and analytic, the exact finite divisor bound solves the fourth-order normal form, and the received twist is nonsingular. Apply Theorem S35 with, for example, $\tau = 64 > 63$. The Diophantine frequencies are selected from the nonlinear frequency image, rather than assigned to the equilibrium vector. The conclusion concerns this declared family and supplies neither a numerical torus radius nor a certificate for unknown material perturbations.

S23 A source-conditioned learning-to-full64 KAM bridge

This section keeps the saved 918-real-parameter network fixed. Its three-configuration-coordinate force evaluations constrain the parameter of the declared *64-coordinate transverse conservative* material family $W_\star = \beta E_0$. This global scalar constitutive assumption is an explicit source-model prior; it does not follow from finite force data. In particular, a three-coordinate neural Hamiltonian is not identified with a 64-coordinate neural Hamiltonian.

Let y_j denote the 33 saved normalized force labels, with deterministic error allowances $\eta_j > 0$, indexed by $j = 1, \dots, 33$. In the declared family their exact targets are βm_j, where m_j is reconstructed from the exact mechanical stiffness and the ideal observation configuration. These 33 observation multipliers are distinct from the 64 modal overlap coefficients used in the twist formula. Thus $|\beta m_j - y_j| \le \eta_j$. The unchanged network graph supplies exact-real values g_j enclosed by received intervals $[\ell_j, u_j]$. Define

$$\delta_j = \max\{|\ell_j - y_j|, |u_j - y_j|\}, \qquad L_F = \frac{1}{33}\sum_{j=1}^{33}\left|\frac{g_j - y_j}{\eta_j}\right|^2, \qquad \overline{L}_F = \frac{1}{33}\sum_{j=1}^{33}\left(\frac{\delta_j}{\eta_j}\right)^2.$$

Then $L_F \le \overline{L}_F$ by interval containment. The normalization in this force-loss interface is η_j; it is not the separate 31-coefficient jet training loss or a fresh validation loss.

Proposition S37. A nonzero force-loss interface. *For every target parameter consistent with the declared family and label allowances, the componentwise and aggregate bounds are*

$$|g_j - \beta m_j| \le \delta_j + \eta_j, \qquad |g_j - \beta m_j| \le \eta_j\big(1 + \sqrt{33\overline{L}_F}\big).$$

More generally, $\overline{L}_F \le \gamma^2/33$ implies the sufficient allowance $\epsilon_j = (1+\gamma)\eta_j$. Each such allowance gives the necessary candidate-centered outer strip

$$\beta m_j \in [\ell_j - \epsilon_j, u_j + \epsilon_j].$$

Intersect these strips over the received rows, dividing with the correct sign whenever $m_j \neq 0$ and retaining the zero-multiplier feasibility tests. Here every intersection uses $j = 1, \dots, 33$. Denote the resulting force-only parameter set by $B_\theta = B_{\theta,33}$.

For any of these choices, the direct-data set

$$B_D = B_{D,33} = \bigcap_{j=1}^{33}\{\beta : |\beta m_j - y_j| \le \eta_j\}$$

is contained in the corresponding B_θ. The same componentwise construction applies separately to the 31 received original-data jet coefficients. Write $B_{D,64}$ for the direct-data set after adding these 31 constraints to the 33 force constraints, and $B_{\mathrm{comp},64}$ for the corresponding componentwise candidate outer set. Write $B_{\mathrm{agg},33}$ for the aggregate-loss outer set using only the 33 force rows. Then

$$B_{D,64} \subseteq B_{\mathrm{comp},64} \subseteq B_{\mathrm{agg},33}.$$

The first inclusion applies the componentwise triangle inequality to all 64 rows. The second uses $\delta_j \le \eta_j\sqrt{33\overline{L}_F}$ on the 33 force rows; adding jet constraints can only shrink the componentwise outer set. These 64 rows are not independent observations.

Proof. The first bound is the triangle inequality. For the second, $|g_j - y_j|/\eta_j \le \sqrt{33L_F} \le \sqrt{33\overline{L}_F}$. The rational-budget case is immediate. Since $g_j \in [\ell_j, u_j]$, each error allowance implies the displayed outer strip. Moreover, for any $v \in [\ell_j, u_j]$ and $|\beta m_j - y_j| \le \eta_j$, $|v - \beta m_j| \le \delta_j + \eta_j \le \epsilon_j$. In particular $[y_j - \eta_j, y_j + \eta_j]$ is contained in $[\ell_j - \epsilon_j, u_j + \epsilon_j]$. Dividing by a signed m_j and intersecting proves $B_D \subseteq B_\theta$. No cancellation of a same-sign fit residual is used to tighten the set.

Proposition S38. Received nonzero-loss full64 instance. *For the unchanged saved checkpoint, exact interval evaluation gives*

$$\frac{3}{100} < L_F \le \overline{L}_F \le \frac{12}{275}, \qquad \overline{L}_F \simeq 0.03783786916.$$

The lower inequality is checked from the interval distances of the actual network outputs to the labels; positivity of an upper bound alone is not used as proof of nonzero loss. Take the rational sufficient budget $\gamma = 6/5$, so that $\gamma^2/33 = 12/275$ and $\epsilon_j = 11\eta_j/5$. Without intersecting with the earlier direct-data parameter interval, the candidate-centered construction receives a nonempty exact rational interval B_θ^{rat}. Its endpoints are displayed outward as

$$B_\theta^{\mathrm{rat}} \subset [0.05999999283191808,\ 0.06000000327884982].$$

The exact endpoints, rather than these decimals, define the interval. For every β in that exact interval, the full 64-coordinate linear frequencies obey

$$|k \cdot \omega(\beta)| \geq \frac{2646043}{2147483648} > 0, \qquad 0 < |k|_1 \leq 4.$$

Together with the parameter-uniform certificate of nonzero modal tip overlaps, this proves the nondegenerate quartic twist and the qualitative local elliptic KAM conclusion for the declared full-coordinate family.

Proof. The 33 output intervals give strictly positive lower bounds on all 33 absolute fit residuals. Summing their squared normalized lower bounds and comparing exactly gives $L_F > 3/100$. Summing their corresponding upper bounds gives the exact rational $\overline{L}_F \leq 12/275$. The parameter strips are intersected using only these evaluations, the saved labels and their deterministic allowances; no material-truth parameter is loaded for the inference.

The new interval is wider than the earlier direct-data interval, so its spectral coverage is recomputed. All 128 exact center-root endpoints from the full64 certificate are retained. Generalized min–max bounds transport every root to the new interval, with exact dyadic square-root enclosures. Exact integer enumeration of all signed frequency sums up to order four, including repeated indices, then gives the displayed strict lower bound. No new eigenvalue proposal or truncated trial space is used. The same computations give strict divisor bounds $2472231/1073741824$ for the componentwise force-and-jet route and $2756265/2147483648$ for the aggregate route using an exact outward enclosure of $\sqrt{33\overline{L}_F}$.

The positive mass and elastic stiffness, nonzero tip overlaps and twist formula hold for the same exact source matrices and all $\beta > -1$. Apply the local elliptic KAM argument of Theorem S35 separately at each accepted parameter. This uses the global scalar constitutive prior to pass from reduced observations to the full-coordinate target family. It does not extend the dimension of the neural graph itself.

Remark S1. Interpretation of the sufficient budget. The threshold $12/275$ is a posteriori sufficient for this checkpoint, these label allowances, and this source-defined scalar material family. It is neither a necessary loss threshold nor a training stopping rule specified before training. Since $B_D \subseteq B_\theta$, this construction does not add identification power to the direct deterministic data. Its contribution is an explicit nonzero-loss interface compatible with the actual saved graph and the full-coordinate local KAM proof. It does not establish generic nonlinear identification, unknown-plant transfer, a numerical torus amplitude, or the necessity or superiority of a neural or complex-valued representation.

S24 A quantitative material-jet neighborhood with fixed quadratic part

This section concerns only the three-coordinate target-family restriction; its constants are not full-coordinate neural bounds. The following result permits general analytic cubic and quartic material corrections to the admitted quadratic target family. It supplies an explicit nonzero sufficient neighborhood; it does not assert that the finite observation interface has identified the unknown material jet inside it. Let I be the received interval for β, and let

$$H_\beta^0(q,p) = \tfrac{1}{2}p^T A^{-1} p + \tfrac{1+\beta}{2} q^T K_3 q + \kappa(1 - \cos(d^T q)).$$

Assume the uniform positive, simple spectrum, order-at-most-four nonresonance and nonsingular twist conditions of the admitted-family certificate. Use its canonical coordinates $q = L_\beta X$ and physical actions $I_j = (X_j^2 + Y_j^2)/2$, and write its twist as T_β^0. For a real analytic perturbation consider

$$R(q) = E_s \left[\sum_{|\alpha|=3} c_\alpha (q/\ell)^\alpha + \sum_{|\alpha|=4} e_\alpha (q/\ell)^\alpha \right] + O(|q|^5), \qquad |c_\alpha| \leq \eta_3, \quad |e_\alpha| \leq \eta_4. \tag{S112}$$

These are ordinary monomial coefficients, not Taylor derivatives. In particular R has no constant, linear or quadratic terms, and the quadratic part and frequencies remain exactly those of H^0_β. No numerical bound on the higher-order analytic remainder is required for the qualitative local statement below; such a bound would be necessary to assign a numerical neighborhood or torus radius.

Put $a_j = (X_j + iY_j)/\sqrt{2}$ and $\bar a_j = (X_j - iY_j)/\sqrt{2}$, so that $\{a_j, \bar a_k\} = -i\delta_{jk}$ and $I_j = a_j\bar a_j$. Let U_α denote the complex-coordinate polynomial obtained from $E_s(q/\ell)^\alpha$. For $|\alpha| = 3$, let V_α be the unique cubic solution of

$$\{H_2, V_\alpha\} = -U_\alpha.$$

For a homogeneous quartic polynomial P, define $\mathcal{T}(P)$ by

$$\Pi_{\mathrm{res}} P = \tfrac{1}{2} I^T \mathcal{T}(P) I,$$

where the projection retains the balanced monomials $a^\nu \bar a^\nu$. All quartic resonances are balanced by the order-four nonresonance condition. Enumerate the ten cubic indices in a fixed order. Define the matrices

$$A_\gamma = \mathcal{T}(U_\gamma), \quad |\gamma| = 4, \qquad B_{\alpha\alpha} = \tfrac{1}{2}\mathcal{T}(\{U_\alpha, V_\alpha\}),$$

$$B_{\alpha\rho} = \tfrac{1}{2}\mathcal{T}(\{U_\alpha, V_\rho\} + \{U_\rho, V_\alpha\}), \qquad \alpha < \rho.$$

All matrices depend on β through the frequencies and chart. Let G_4, G_3 be nonnegative entrywise upper-bound matrices satisfying, uniformly throughout the whole interval I,

$$(G_4)_{ij} \ge \sum_{|\gamma|=4} |((T^0_\beta)^{-1} A_\gamma)_{ij}|, \qquad (G_3)_{ij} \ge \sum_{\alpha\le\rho} |((T^0_\beta)^{-1} B_{\alpha\rho})_{ij}|.$$

Set $b_k = \max_i \sum_j (G_k)_{ij}$ for $k = 3, 4$.

Lemma S39. A fixed-quadratic robust elliptic KAM criterion. *If*

$$b_4\eta_4 + b_3\eta_3^2 < 1, \tag{S113}$$

then, for every $\beta \in I$ and every real analytic perturbation (S112)*, the Hamiltonian $H^0_\beta + R$ has nonsingular quartic Birkhoff twist and satisfies the qualitative local elliptic KAM criterion. In particular sufficiently small neighborhoods of its elliptic equilibrium contain a positive-measure family of invariant Lagrangian three-tori. The size of a suitable neighborhood may depend on the higher-order analytic remainder.*

Proof. The base Hamiltonian has no cubic term. The cubic homological equation and the degree-four Lie-series formula therefore give

$$H_4^{\mathrm{eff}} = H_4^0 + R_4 + \tfrac{1}{2}\{R_3, \chi_3\}, \qquad \chi_3 = \sum_{|\alpha|=3} c_\alpha V_\alpha.$$

The sign follows from $\{H_2, \chi_3\} = -R_3$: in the pullback Lie series $\{R_3, \chi_3\} + \frac{1}{2}\{\{H_2, \chi_3\}, \chi_3\}$ the two cubic-generated contributions combine to $\frac{1}{2}\{R_3, \chi_3\}$. Consequently the twist perturbation is exactly

$$\Delta T = \sum_{|\gamma|=4} e_\gamma A_\gamma + \sum_{\alpha\le\rho} c_\alpha c_\rho B_{\alpha\rho}.$$

The triangle inequality and induced infinity norm imply

$$\|(T^0_\beta)^{-1}\Delta T\|_\infty \le b_4\eta_4 + b_3\eta_3^2 < 1.$$

The Neumann-series criterion makes $T^0_\beta + \Delta T = T^0_\beta[\mathrm{Id} + (T^0_\beta)^{-1}\Delta T]$ invertible. The equilibrium Hessian and finite-order divisors are unchanged. Real analyticity, the finite normal-form construction and the nonsingular twist give the same qualitative local elliptic KAM conclusion as for the admitted family. Terms of degree at least five do not alter this twist.

An independent coefficient-norm majorant. For a polynomial in $(a, \bar{a})$ write $\|P\|_1$ for its coefficient one-norm. Let

$$t_i = \frac{\sqrt{2}}{\ell} \sum_j |(L_\beta)_{ij}|, \qquad C_k = E_s \sum_{|\alpha|=k} \prod_i t_i^{\alpha_i}.$$

Then $\|R_3\|_1 \le C_3\eta_3$ and $\|R_4\|_1 \le C_4\eta_4$. If δ_3 bounds all relevant cubic divisors away from zero, then $\|\chi_3\|_1 \le C_3\eta_3/\delta_3$. For homogeneous polynomials of degrees m, n, $\|\{P, Q\}\|_1 \le mn\|P\|_1\|Q\|_1$; this follows term by term because the sum of the absolute Poisson multipliers is bounded by mn. Resonant projection does not increase the coefficient one-norm, and extraction of the symmetric action Hessian gives $\|\mathcal{T}(P)\|_F \le 2\|\Pi_{\mathrm{res}}P\|_1$. Thus, uniformly on I,

$$\|\Delta T\|_2 \le 2C_4\eta_4 + \frac{9C_3^2}{\delta_3}\eta_3^2.$$

This estimate independently verifies that the cubic correction is quadratic in the cubic jet and is not omitted. The executable result uses the sharper 15 linear and 55 quadratic matrix responses, with outward Arb arithmetic for the full interval of canonical charts.

Scope of a received coefficient tolerance. The numerical tolerance applies to material coefficients in the exact normalization $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$, with the quadratic family jet fixed. The current finite-observation coefficient radii are compared with this sufficient bound, without declaring the unknown target received. An upper error bound exceeding the tolerance does not prove absence of KAM tori. It shows that this particular uniform robustness certificate has not been instantiated by that bound. Uncertainty in the quadratic jet requires a separate spectral and coordinate perturbation analysis.

Received nonzero target-jet tolerance. For the entire received scalar interval, outward evaluation of all 15 quartic responses and 55 symmetric cubic products gives the safe rounded constants

$$b_4 < 1.123431 \times 10^{11}, \qquad b_3 < 2.663083 \times 10^{12}.$$

The normalization is $\ell = 10^{-3}$ and $E_s = 6 \times 10^{-12}$. An independent computation in real (X, Y) coordinates inverts the 56-dimensional cubic homological operator and extracts the action Hessian by circle moments, without using the complex-polynomial implementation. Both implementations receive the coefficient rectangle

$$\eta_3 = 2^{-22} \simeq 2.384186 \times 10^{-7}, \qquad \eta_4 = 2^{-38} \simeq 3.637979 \times 10^{-12}, \qquad b_4\eta_4 + b_3\eta_3^2 < 0.561 < 1.$$

The common radius $\eta_3 = \eta_4 = 2^{-38}$ gives the stronger inverse margin below 0.409. These are sufficient radii, with no assertion that they are optimal.

The current candidate-to-target quartic interface radius of approximately 2.066906×10^{-6} is about 5.68×10^5 times the received common radius. The largest prescribed magnetic quartic coefficient, approximately 5.363314×10^{-9}, is about 1.47×10^3 times that radius. These comparisons use the ordinary normalized coefficient convention; they do not identify the unknown material perturbation relative to the quadratic target family. In particular, the known magnetic term is already included exactly in the base Hamiltonian and is not an uncertain material coefficient. The present certificate fixes the quadratic jet exactly and therefore does not receive the current unknown-target learning-to-KAM transfer. Failure of a sufficient bound is not a proof that target KAM tori are absent.

Comparison with the received material coefficient boxes. The comparison reference is the admitted material potential βE_0, whose cubic and quartic material coefficients vanish for every admitted β. The prescribed magnetic cosine remains in the base Hamiltonian. Using the actual eleven-configuration coefficient centers m_α and radii r_α gives the exact box majorants

$$\widehat{\eta}_k = \max_{|\alpha|=k}(|m_\alpha| + r_\alpha), \qquad k = 3, 4.$$

Their numerical values are

$$\widehat{\eta}_3 \simeq 8.168713334 \times 10^{-7}, \qquad \widehat{\eta}_4 \simeq 2.067036334 \times 10^{-6}.$$

Thus the respective box-to-safe-radius ratios are approximately 3.426 and 5.682×10^5 for the rectangle $(\eta_3, \eta_4) = (2^{-22}, 2^{-38})$. The sufficient inverse-margin majorant evaluates to approximately 232219, which exceeds one. This is a direct material-to-family comparison, independently of the candidate network's coefficients. It does not establish that a specific compatible target lacks KAM tori.

These boxes were themselves received under an explicit zero fifth-order tail and equilibrium prior for the declared material family. They cannot be promoted to coefficient bounds for an arbitrary unknown analytic material without a separately justified tail bound. Moreover, the robustness theorem fixes the quadratic jet exactly; the finite-data quadratic uncertainty has not been received by that theorem. Consequently this comparison supplies no completed unknown-target learning-to-KAM transfer.

S25 Physical power, learning errors, and bounded magnetic safety

All derivatives in this section use physical time. The full observation space has $n = 64$ coordinates; its three-coordinate restriction is a different dynamical system. Let

$$M = M_0 + M_r \succ 0, \qquad K_0 \succ 0, \qquad D_0 = \alpha M_0 \succeq 0, \qquad \alpha > 0,$$

with constant symmetric matrices. In particular, D_0 is not αM. For a full-column-rank $V \in \mathbb{R}^{64\times 3}$, put

$$A = V^T M V, \qquad B = A^{-1} V^T M, \qquad q = Bx, \qquad E_0(x) = \tfrac{1}{2} x^T K_0 x, \qquad \theta = b^T x.$$

Thus $BV = I_3$. Fix $\mu > 0$ and a constant bias coefficient $\kappa > 0$. The transverse magnetic field $u = B_y$ is measured in tesla. Define

$$\begin{aligned} U_N(x) &= E_0(x) + \kappa(1 - \cos\theta) + F_N(Bx, E_0(x)), \\ D_N(x) &= a_N(Bx, E_0(x)) D_0, \\ H_N(x, v) &= \tfrac{1}{2} v^T M v + U_N(x), \\ g(x) &= \mu \cos\theta\, b, \qquad y(x, v) = g(x)^T v. \end{aligned} \tag{S114}$$

Here F_N is real-valued on real inputs. The complex network is decoded in real coordinates before differentiation.

The actual network parameterization. With positive saved feature scales Q_j and E_*, distinct from the exact decimal normalization E_s, write $s = (q_1/Q_1, q_2/Q_2, q_3/Q_3, E/E_*)$. Let $N(s)$ be the real energy readout and $(R(s), I(s))$ the two real readouts of the separate complex dissipation network. The candidate is

$$\begin{aligned} F_N(q, E) &= c_E E + E_* \left[N(s) - N(0) - \sum_{j=1}^{3} \gamma_j s_j \right], \qquad \gamma_j = \partial_j N(0), \\ a_N(q, E) &= [1 + c_C + R(s) - R(0)]^2 + [I(s) - I(0)]^2. \end{aligned} \tag{S115}$$

The fourth linear feature term is retained. Removing $\partial_4 N(0) E/E_*$ would define another potential.

Proposition S40. Full-field structure and real-domain bounds. *For differentiable F_N, the full candidate equation*

$$\dot{x} = v, \qquad M\dot{v} = -\nabla U_N(x) - D_N(x) v + g(x) u$$

has the identities

$$\nabla_x F_N(Bx, E_0(x)) = B^T F_q + F_E K_0 x, \tag{S116}$$

$$\dot{H}_N = -a_N v^T D_0 v + u y. \tag{S117}$$

In particular, $D_N \succeq 0$. Suppose the split-tanh network is written as two real hidden layers with matrices W_1, W_2 and final real output rows C_R, C_I. For a real feature box $|s_j| \le r_j$, define

$$\delta_R = |C_R|\,|W_2|\,|W_1|r, \qquad \delta_I = |C_I|\,|W_2|\,|W_1|r,$$

where the products use entrywise absolute values and the displayed outputs are scalars. Then throughout that real box

$$a_- \le a_N \le a_+, \qquad a_- = (\max\{0, |1+c_C| - \delta_R\})^2, \quad a_+ = (|1+c_C| + \delta_R)^2 + \delta_I^2. \tag{S118}$$

If, in addition, $D_0 \succeq \lambda_0 M$ for $\lambda_0 > 0$ and $a_- > 0$, then $D_N \succeq a_- \lambda_0 M$ on the box.

Proof. Since $dq = B\,dx$ and $dE_0 = x^T K_0\,dx$, the chain rule gives (S116). Multiplication of the candidate acceleration equation by v^T proves (S117). For the Rayleigh function $\mathcal{R}_N(x,v) = a_N(Bx, E_0(x))v^T D_0 v/2$, the derivative with respect to v, with x fixed, is $a_N D_0 v$. There is no spatial derivative of a_N in its dissipated power. Real tanh is globally 1-Lipschitz. Applying this fact to the two real-coordinate hidden layers bounds their output differences from the origin by the two stated matrix products. The reverse triangle inequality bounds the first squared factor from below; the ordinary triangle inequality bounds both factors from above. The Loewner bound follows on multiplication by $D_0 \succeq 0$.

The lower bound in (S118) must be recomputed for the selected trained weights and the relevant full-state feature range. A sum of two squares alone does not exclude a common zero. For a fixed field value u_c, the alternative storage $H_N - \mu u_c \sin\theta$ dissipates at rate $-v^T D_N v$. For a differentiable varying field, including $-\mu u \sin\theta$ in the storage gives supply $-\mu \dot{u} \sin\theta$. With the storage in (S114), the supply is uy and has no $\dot{u}$ term. These are two storage conventions for the same force.

A sufficient coercivity test. For finite split-tanh weights, the real readout satisfies $|N(s)| \le C_N$ on all real features, for an explicit finite constant C_N obtained from its final output row and bias. Set

$$\xi_c = E_* B^T \operatorname{diag}(Q_1^{-1}, Q_2^{-1}, Q_3^{-1})(\gamma_1, \gamma_2, \gamma_3)^T.$$

If $1 + c_E > 0$, then for every $0 < \eta < 1 + c_E$,

$$U_N(x) \ge (1 + c_E - \eta)E_0(x) - E_*(C_N + |N(0)|) - \frac{\xi_c^T K_0^{-1} \xi_c}{2\eta}. \tag{S119}$$

Indeed, the centered position term is $-\xi_c^T x$ and $|\xi_c^T x| \le \sqrt{2E_0 \xi_c^T K_0^{-1} \xi_c} \le \eta E_0 + \xi_c^T K_0^{-1} \xi_c/(2\eta)$. Thus H_N has compact sublevels under this sufficient test. The architecture by itself does not constrain $c_E > -1$.

The physical plant and its errors. On an open configuration region Ω, suppose the physical plant is

$$\dot{x} = v, \qquad M\dot{v} = -\nabla U_*(x) - D_*(x)v + g(x)u + w(x,v,t), \qquad D_*(x) = D_*(x)^T \succeq 0. \tag{S120}$$

It has the same declared mass and magnetic port as the candidate. The generalized force w covers additional physical work, rather than conservative or damping errors already retained in U_*, D_*. Define

$$H_* = \tfrac{1}{2} v^T M v + U_*(x), \quad \Delta U = U_N - U_*, \quad G = \nabla \Delta U, \quad d_* = v^T D_* v, \quad d_N = v^T D_N v.$$

Unlike a full-phase network with learned momentum dependence, this candidate uses exactly $\dot{x} = M^{-1}p$. At the same observed full state (x, v), its port output is therefore exactly y; no potential gradient error changes this equality. Replacing (x, v) by a reduced state or a stale observation requires a separate output-error bound. Along the physical plant the two storage derivatives are

$$\dot{H}_* = -d_* + uy + v^T w, \qquad \dot{H}_N = -d_* + uy + v^T(G + w). \tag{S121}$$

Theorem S41. Physical safety and potential-value transfer. *Let Ω be bounded with compact closure. Assume U_N, U_* are C^2 and D_* is locally Lipschitz on a neighborhood of that closure, and*

$$|U_N - U_*| \le \varepsilon_U \quad \text{on } \overline{\Omega}. \tag{S122}$$

Let $U > 0$, $k > 0$ and use the actual continuous feedback

$$u = -\operatorname{sat}_U(ky), \qquad \operatorname{sat}_U(r) = \max(-U, \min(U, r)), \qquad \Phi(y) = |y| \min(k|y|, U).$$

Suppose $w = 0$, $x(0) \in \Omega$, and for some energy cap c,

$$H_N(x(0), v(0)) + \varepsilon_U \le c, \qquad \inf_{x \in \partial\Omega} U_N(x) - \varepsilon_U > c. \tag{S123}$$

Then the physical solution exists for all $t \ge 0$, remains in Ω, and satisfies

$$H_*(t) \le H_*(0) \le c, \qquad H_N(t) \le c + \varepsilon_U. \tag{S124}$$

Consequently any physical admissible set containing $\{(x, v) : x \in \overline{\Omega},\ H_N(x, v) \le c + \varepsilon_U\}$ contains this trajectory. Neither a gradient-error bound nor an approximation of D_* by D_N is needed for this conclusion.

Proof. The actual field has magnitude at most U and is locally Lipschitz in the state. The first identity in (S121) gives $\dot{H}_* = -d_* - \Phi(y) \le 0$ as long as the solution remains in the declared region. The initial inequality and (S122) give $H_*(0) \le c$. At a first configuration exit, continuity and the strict boundary inequality would imply $H_* \ge U_* \ge U_N - \varepsilon_U > c$, a contradiction. On the compact configuration closure, U_* is bounded below. The energy cap therefore bounds v because $M \succ 0$. The solution stays in a compact subset of its smooth domain, so the standard continuation theorem excludes finite-time escape. Finally $H_N = H_* + \Delta U \le c + \varepsilon_U$ proves the set inclusion.

The theorem concerns the true storage and the exact measured physical port. The learned energy supplies an initial-energy estimate and a guard conversion. Its role here is not a claim that a neural controller outperforms a physical passivity controller. If a positive-work force, an incorrect mass, or an incorrect input map is present, the corresponding extra power cannot be omitted.

Proposition S42. Quantitative error and observation budgets. *Set $r = (v^T M v)^{1/2}$. Suppose, on a common declared state region,*

$$\|M^{-1/2}(D_N - D_*)M^{-1/2}\|_2 \le \delta_D, \quad \|M^{-1/2}G\|_2 \le \varepsilon_g, \quad \|M^{-1/2}w\|_2 \le \varepsilon_w.$$

Then

$$d_* \ge \underline{d} := \max\{0, d_N - \delta_D r^2\}. \tag{S125}$$

For exact port feedback the two powers obey

$$\dot{H}_* \le -\underline{d} - \Phi(y) + \varepsilon_w r, \tag{S126}$$

$$\dot{H}_N \le -d_N + \delta_D r^2 - \Phi(y) + (\varepsilon_g + \varepsilon_w) r. \tag{S127}$$

If the applied feedback instead uses $\widehat{y}$ with $|y - \widehat{y}| \le \varepsilon_y$, let $a = \min(k|\widehat{y}|, U)$. The term $-\Phi(y)$ in either bound may be replaced by

$$-a(|\widehat{y}| - \varepsilon_y). \tag{S128}$$

Proof. The matrix error gives $|d_N - d_*| \le \delta_D r^2$; retain also $d_* \ge 0$. Weighted Cauchy–Schwarz gives $|v^T G| \le \varepsilon_g r$ and $v^T w \le \varepsilon_w r$. Substitution into (S121) gives the assertions. For the observed output, $uy = u\widehat{y} + u(y - \widehat{y}) \le -a|\widehat{y}| + a\varepsilon_y$.

For example, a certified $D_N \succeq \lambda_N M$ and $\delta_D < \lambda_N$ give true damping at least $(\lambda_N - \delta_D) r^2$. A velocity-proportional disturbance bound $\|M^{-1/2}w\| \le \nu r$ with $\nu \le \lambda_N - \delta_D$ is absorbed by this damping. A constant $\varepsilon_w > 0$ gives a linear term

in r and is not absorbed uniformly by quadratic damping near $r = 0$. The same issue affects a nonzero constant gradient bound in (S127). Empirical weak-loss values alone provide none of the uniform constants in this proposition.

Theorem S43. Boundary invariance and bounded-port feasibility. *Let H denote either H_* or H_N, with H of class C^2, and suppose the sublevel $S_c = \{(x, v) : H(x, v) \le c\}$ is compact, lies in the declared smooth domain, and has a regular boundary: $\nabla_{x,v} H \neq 0$ on ∂S_c. For a specified locally Lipschitz physical force $w(x, v, t)$, uniformly locally Lipschitz in the state on finite time intervals, define*

$$P_* = -d_* + v^T w \quad (H = H_*), \qquad P_N = -d_* + v^T (G + w) \quad (H = H_N).$$

Assume the corresponding functions are continuous in time. The saturated feedback above renders S_c positively invariant if

$$P(x, v, t) \le \Phi(y(x, v)) \quad \textit{for all } (x, v) \in \partial S_c, \quad t \ge 0. \tag{S129}$$

At a fixed boundary state and time, the smallest available energy derivative among all fields $|u| \le U$ equals

$$\min_{|u| \le U} (P + uy) = P - U|y|. \tag{S130}$$

Thus $P - U|y| \le 0$ at every boundary point is necessary for an invariant cap under a bounded continuous feedback. This pointwise feasibility condition alone asserts neither a continuous feedback selection nor an implemented controller.

Proof. Along the saturated closed loop, $\dot{H} = P - \Phi(y)$. The boundary condition is the inward tangent condition for the regular sublevel. To make the invariance argument explicit, a compact regular level has a tubular neighborhood in which distance to the level is bounded by a constant times $|H - c|$. The locally Lipschitz closed-loop field and the locally Lipschitz level normal therefore bound the positive part $h = (H - c)_+$ by $\dot{h} \le L_T h$ in that neighborhood on each finite interval $[0, T]$. Gronwall's inequality with $h(0) = 0$ precludes exit. Initial states strictly inside the sublevel are handled by their first boundary contact. Compactness gives continuation. The scalar minimization is obtained with $u = -U \operatorname{sign} y$ when $y \neq 0$, and with any input when $y = 0$. If its minimum were positive at a boundary state, every available continuous closed loop would initially point strictly out of the sublevel there, contradicting invariance from that initial state.

In particular, at $y = 0$ no magnetic field changes the instantaneous energy derivative. This occurs when $\cos\theta = 0$ or $b^T v = 0$. Restricting $|\theta| < \pi/2$ removes only the former possibility. On a verified part Γ of a boundary where $|y| \ge y_0 > 0$, choosing $k \ge U/y_0$ gives $\Phi(y) = U|y|$. A budget $P \le U y_0 - m$ with $m > 0$ then provides a strict inward margin m on Γ. The remaining boundary still requires (S129); a local nonzero-port calculation does not cover the turning points or the velocity directions orthogonal to b.

A centered, realizable obstruction for a learned-energy cap. Take two coordinates with $M = K_0 = I_2$, $D_* = dI_2$ for $d > 0$, $b = (1, 0)^T$, $\mu > 0$, and

$$U_* = E_0 + \kappa(1 - \cos x_1), \qquad U_N = (1 + \epsilon)E_0 + \kappa(1 - \cos x_1), \qquad \epsilon > 0.$$

This discrepancy is precisely the affine energy skip $F_N = \epsilon E_0$ with zero neural branch, not an uncentered linear force. At $x = (0, 1)^T$, $v = (0, t)^T$ and $0 < t < \epsilon/d$,

$$y = 0, \qquad \dot{H}_N = \epsilon t - dt^2 > 0, \qquad \dot{H}_* = -dt^2 < 0. \tag{S131}$$

Every magnetic input has the same derivative at this state. The H_N sublevel through it has a regular boundary and cannot be invariant under any bounded continuous magnetic feedback, although the true passive energy decreases. The parameters ϵ, t can be arbitrarily small. Hence a small conservative-gradient error and positive damping do not, by themselves, certify invariance of learned energy sublevels on the physical plant.

Restriction and execution scope. On $x = Vq$, $v = VA^{-1}p$, the restricted storage is

$$H_N^{\mathrm{r}}(q, p) = \tfrac{1}{2} p^T A^{-1} p + U_N(Vq), \qquad y^{\mathrm{r}} = \mu \cos(b^T Vq) b^T V A^{-1} p.$$

It has its own reduced power identity. It does not identify the physical output with y^{r} on an arbitrary full trajectory, prove invariance of $\mathrm{range}(V)$, or transfer a restricted KAM torus to the full forced dissipative system. A sampled, delayed, or native numerical realization additionally needs full-prefix work, observation, actuation, and numerical-defect bounds for this same physical model. The continuous boundary theorem does not silently inherit a previous held-command record or its safety certificate.

S25.1 Dissipation and coercivity for the calibrated weights

Fresh calculation with the calibrated weights gives $c_C \simeq 0.07237264643$ and $c_E \simeq 0.06002256805$. In particular, $1 + c_E > 0$, so the sufficient global real coercivity condition in (S119) holds for this same network. Its nonnegative dissipation-factor parameterization and full-field power identity hold algebraically. Exact native LDL factorizations again give $D_0 > (6/25)M\,\mathrm{s}^{-1}$, with the exact sum $M = M_0 + M_r$. Exact saved-weight absolute products give the following strict downward relaxations on the full-state feature boxes $|s|_\infty \le r$:

r	a_N lower bound	D_N/M lower bound in s^{-1}
$1/4$	1.0346097	0.2483063
$1/2$	0.9253337	0.2220801
1	0.7250739	0.1740177
$3/2$	0.5492037	0.1318088
2	0.3977230	0.0954535

The last column is a strict Loewner matrix lower bound for every full velocity in the stated configuration domain. The same-checkpoint calculation proves candidate dissipation and coercivity; it does not by itself identify physical damping or certify sampled magnetic feedback. A conditional physical potential guard requires its own uniform potential allowance and implementation assumptions.

S25.2 A conditional physical guard for the calibrated potential

The same saved weights also admit a full-configuration potential-value guard under an explicit physical family. Suppose

$$U_*(x) = (1+\beta)E_0(x) + \kappa(1 - \cos(b^T x)),$$

with constant β in the exact scalar compatibility interval in Section S13.1. This is a constitutive family and parameter-interval assumption, not an identification of an arbitrary physical potential from the simulation records. Let E_{net} be the exact saved binary energy scale and set

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad c = 2.05E_{\mathrm{net}}.$$

Fresh exact-weight bounds on the energy-head gradient and Hessian, an outward origin derivative calculation, and the exact native compliance matrix give the strict relaxations

$$\sup_{\overline{\Omega}_4} |U_N - U_*| < 0.003838998456E_{\mathrm{net}}, \qquad \inf_{\partial\Omega_4} U_N > 4.236161085119E_{\mathrm{net}}.$$

These constants are recomputed for the calibrated weights and the new compatibility interval. They do not reuse the earlier single-configuration checkpoint's margins. With $\varepsilon_U = 0.003838998456E_{\mathrm{net}}$, the certified boundary margin is

$$4.236161085119 - 0.003838998456 - 2.05 = 2.182322086663 > 0.$$

The initial set

$$\mathcal{I} = \{(x,v) : x \in \Omega_4, \quad H_N(x,v) \le 2.046161001544E_{\mathrm{net}}\}$$

contains the origin and an open neighborhood, by continuity and the algebraic centering $H_N(0,0) = 0$. Corollary S45 separately receives all 22 numerical initial records into this fixed set.

Assume the exact same native mass and physical magnetic port, locally Lipschitz physical damping $D_*(x) \succeq 0$, and no unmodeled work. Under actual continuous feedback $u = -\mathrm{sat}_U(ky)$ with the exact full-state port $y = \mu\cos(b^T x)b^T v$, any

$U, k > 0$, Theorem S41 applies to every initial state in $\mathcal{I}$. It gives existence for all $t \geq 0$ and

$$x(t) \in \Omega_4, \qquad H_*(t) \leq H_*(0) \leq c, \qquad H_N(t) \leq c + \varepsilon_U.$$

The physical damping assumption in this theorem is distinct from the candidate damping bounds. The candidate bound on the full Ω_4 domain is proved separately in Proposition S44; it does not identify physical damping. This is a conditional continuous-feedback safety conclusion; no sampled controller, sensor error, actuation delay, controller rollout or physical-target KAM transfer is certified here.

S25.3 Strict candidate dissipation on the full safety configuration domain

All constants in this subsection belong to the unchanged 918-real-parameter checkpoint used for the calibrated energy guard. Saved binary64 weights and feature scales are exact constants of the mathematical source graph. The native matrices are the exact source operators, with $M = M_0 + M_r \succ 0$, $K_0 \succ 0$ and $D_0 \succ (6/25)M\,\mathrm{s}^{-1}$. This is a statement about the candidate equations, not an identification of the physical dissipative force.

Let $T = BK_0^{-1}B^T$, $E_0 = x^T K_0 x/2$ and

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad s = (q_1/Q_1, q_2/Q_2, q_3/Q_3, E_0/E_{\mathrm{net}}), \quad q = Bx.$$

The positive Q_j carry the units of q_j in the declared native-coordinate normalization, and E_{net} is an energy in joules; the four features are dimensionless. The same-source compliance calculation supplies strict rational bounds $p_j^2 Q_j^2 > 2E_{\mathrm{net}} T_{jj}$. Here

$$(p_1, p_2, p_3) = (4.595063567868,\ 1.819558891632,\ 0.927250710462).$$

The displayed terminating decimals are exact rational upper bounds, not rounded evaluations of square roots. Put

$$r = (2p_1, 2p_2, 2p_3, 4) = (9.190127135736, 3.639117783264, 1.854501420924, 4).$$

Proposition S44. Anisotropic-domain damping bound. *For every real $x \in \overline{\Omega}_4$, the selected candidate satisfies*

$$0.005110793448 < a_N(x) < 5.344160513025, \tag{S132}$$

$$D_N(x) \succ 0.00122659042752M\,\mathrm{s}^{-1}. \tag{S133}$$

The conclusion holds on the full 64-coordinate ellipsoid, without assuming $x \in \mathrm{range}(V)$. For the exact candidate equation, define $H_N = v^T M v/2 + U_N(x)$ and $y = \mu\cos(b^T x)b^T v$. While $x(t) \in \overline{\Omega}_4$,

$$\dot{H}_N \leq -0.00122659042752\,\mathrm{s}^{-1} v^T M v + uy. \tag{S134}$$

The inequality is strict when $v \neq 0$. In particular, zero total field or exact continuous feedback $u = -\mathrm{sat}_U(ky)$ makes this candidate storage nonincreasing on that domain.

Proof. Weighted Cauchy–Schwarz with $K_0 \succ 0$ gives

$$|B_j x|^2 \leq (x^T K_0 x)\, T_{jj}.$$

Thus, for $E_0(x) \leq 4E_{\mathrm{net}}$, $|q_j/Q_j| < 2p_j = r_j$, and $0 \leq E_0/E_{\mathrm{net}} \leq 4$. Consequently the entire configuration ellipsoid maps into the stated anisotropic feature box. No reduction of a full state is used in this argument.

For completeness, write each complex affine weight as $A_\ell + iB_\ell$. The four real features enter the complex network as $s + i0$. In real coordinates its first weight block is

$$W_1 = \begin{pmatrix} A_1 \\ B_1 \end{pmatrix} \in \mathbb{R}^{24\times 4}, \qquad W_2 = \begin{pmatrix} A_2 & -B_2 \\ B_2 & A_2 \end{pmatrix} \in \mathbb{R}^{24\times 24},$$

and its two output rows are $C_R = (A_3, -B_3)$ and $C_I = (B_3, A_3)$. Both hidden activations apply ordinary real tanh separately to these 24 real coordinates. Since $|\tanh'(t)| \le 1$ for every real t, the global componentwise Lipschitz estimate, including arbitrary fixed biases, is

$$\begin{pmatrix} |R(s) - R(0)| \\ |I(s) - I(0)| \end{pmatrix} \le L|s|, \qquad L = \left| \begin{pmatrix} C_R \\ C_I \end{pmatrix} \right| |W_2|\,|W_1|.$$

Absolute values and inequalities are componentwise. The estimate is valid for all real features, not merely near zero. It uses no holomorphic-complex-tanh derivative assertion: the implemented activation is the realification of split tanh.

Reconstructing these blocks from the exact saved complex weights gives, on $|s_j| \le r_j$,

$$\delta_R = L_R r \approx 1.0008828328328028, \qquad \delta_I = L_I r \approx 1.0226300556580044.$$

These two displays are explanatory floating-point approximations; the certificate uses their exact rational values. The saved scalar is $c_C = 2607499387914309/36028797018963968$. Hence

$$m = |1 + c_C| - \delta_R > 0, \qquad m^2 \le a_N(x) \le (|1 + c_C| + \delta_R)^2 + \delta_I^2.$$

Exact rational arithmetic proves

$$m^2 > 0.005110793448, \quad (|1 + c_C| + \delta_R)^2 + \delta_I^2 < 5.344160513025, \quad (6/25)m^2 > 0.00122659042752.$$

Combining the lower factor bound with the received strict native mass-metric inequality for D_0 proves (S132)–(S133).

Finally, differentiate H_N along $M\dot{v} = -\nabla U_N - D_N v + \mu \cos(b^T x) b u$. The conservative terms cancel, with the complete energy-feature chain rule retained, yielding $\dot{H}_N = -v^T D_N v + uy$. Substitution of (S133) proves (S134). Its damping term has units $(\mathrm{s}^{-1})(\mathrm{J}) = \mathrm{J\,s}^{-1}$, as does uy. For saturated continuous feedback $uy = -|y| \min(k|y|, U) \le 0$.

Candidate trajectory scope. The already received guard gives $\inf_{\partial\Omega_4} U_N > 4.236161085119 E_{\mathrm{net}}$ and the initial set has $H_N(0) \le 2.046161001544 E_{\mathrm{net}}$ with $x(0) \in \Omega_4$. For the exact candidate under zero total field or the specified continuous saturated feedback, global nonnegative dissipation implies $\dot{H}_N \le 0$. A first configuration exit would require $H_N \ge U_N > 4.236161085119 E_{\mathrm{net}}$, contradicting the initial bound. The same selected weights satisfy $1 + c_E > 0$ and the received real coercivity test, so bounded-energy trajectories are in a compact phase-space set and smooth-vector-field continuation gives existence for all positive time. Thus this candidate retains the strict mass-metric damping estimate throughout those trajectories. No exponential decay of the full storage, asymptotic convergence to a specified equilibrium, sampled-control passivity, or physical-target damping error is asserted. On a physical plant, the derivative of the learned storage has its separate potential/damping/work error terms; the candidate inequality cannot replace that analysis.

S25.4 Current-network reception of the 22 recorded numerical initial states

Corollary S45. Fixed-guard reception of all 22 recorded numerical initial states. *Keep the current calibrated complex split-tanh potential U_N, the same exact native full-coordinate matrices $M = M_0 + M_r, K_0$, and the conditional physical-family assumptions of Section S25.2. Let E_{net} be its exact saved binary64 energy scale and set $E_0(x) = x^T K_0 x/2$. Retain the already certified domain, cap and error allowance without adjustment:*

$$\Omega_4 = \{x : E_0(x) < 4E_{\mathrm{net}}\}, \qquad c = \frac{41}{20} E_{\mathrm{net}}, \qquad \varepsilon_U = \frac{479874807}{125000000000} E_{\mathrm{net}}, \qquad c - \varepsilon_U = \frac{255770125193}{125000000000} E_{\mathrm{net}}.$$

Consider the finest signed records selected in the frozen 11-configuration training protocol: the original shifted configuration, directions 2 and 3, and design nodes 3 through 10, with each of their two velocity-sign labels. For each record use its own unique saved sample with time exactly zero and form

$$x_0 = (w_1, \theta_1, \ldots, w_{32}, \theta_{32})^T, \qquad v_0 = (w_{t,1}, \theta_{t,1}, \ldots, w_{t,32}, \theta_{t,32})^T.$$

Every saved binary64 component is interpreted exactly. Neither vector is replaced by Vq, by an intended preparation, or by the opposite sign. Write $H_N(x,v) = v^T Mv/2 + U_N(x)$ and $m_0 = c - \varepsilon_U - H_N(x_0, v_0)$. Both signs in each row satisfy the following strict rational bounds; all terminating decimals are exact outward relaxations:

configuration	$E_0(x_0)/E_{\mathrm{net}} <$	$H_N(x_0,v_0)/E_{\mathrm{net}} <$	$m_0/E_{\mathrm{net}} >$
original shifted	0.011840151418	0.972247611362	1.073913390182
direction 2	0.075607037915	0.237377388001	1.808783613543
direction 3	0.665892667009	0.785898582383	1.260262419161
jet node 3	0.699184876801	0.818661343066	1.227499658478
jet node 4	0.672114306437	0.795381639556	1.250779361988
jet node 5	0.248565856915	0.290319882622	1.755841118922
jet node 6	0.688816911554	0.745602603503	1.300558398041
jet node 7	0.686085823522	0.817887305482	1.228273696062
jet node 8	0.757581829134	0.887146476049	1.159014525495
jet node 9	0.198955734425	0.276686835522	1.769474166022
jet node 10	0.705414332297	0.819033952757	1.227127048787

Thus all 22 saved numerical initial states satisfy the same frozen initial guard. Across these states,

$$0.237377387997 < \frac{H_N(x_0,v_0)}{E_{\mathrm{net}}} < 0.972247611362, \qquad 1.073913390182 < \frac{m_0}{E_{\mathrm{net}}} < 1.808783613547.$$

The unchanged boundary guard has exact margin $2182322086663E_{\mathrm{net}}/10^{12} > 0$.

Proof. The receiver verifies the current guard's actual zero exit code and hashed result before receiving its fixed constants. Each of the 22 observation files is bound to its own completed producer receipt and a launcher receipt with actual MATLAB exit code zero, including the different receipt structures of the original and later producers. The 257 saved times, 33 node identities, and all four 257×33 position/velocity field arrays are checked separately for every record. Exact rational coordinate gaps at most $10^{-15}\,\mathrm{m}$ associate each saved node with a unique native finite-coordinate field row; this is a node-identity check, not an interpolation-error estimate. The saved clamped-root values at the initial sample are exactly zero.

The receiver extracts all 64 free coordinates and evaluates E_0 and $v^T Mv/2$ by exact rational arithmetic, adding $M_0 + M_r$ before use. The same-source successful native positive-pivot proof is received; no new LDL factorization is performed. For the energy features only, it computes the exact mass projection $q = Bx_0$, where $B = (V^T MV)^{-1} V^T M$; the full state remains x_0. The centered current energy-head expression is evaluated with outward Arb arithmetic at 512 bits, including its actual complex realification, the exact skip coefficient, the origin readout and the first three origin-gradient subtractions. The magnetic term uses the actual saved tip angle θ_{32}. This yields strict rational lower and upper endpoints for every H_N/E_{net} and m_0/E_{net}. Independent exact rational rounding proves the displayed relaxations. All 22 tests resolve strictly and pass; rejection would have been reported as a valid negative decision rather than as arithmetic failure. Since $E_0 < 4E_{\mathrm{net}}$ and $H_N + \varepsilon_U < c$, each saved vector pair satisfies the theorem's initial inequalities when used as an exact mathematical initial state. The 11 positive/negative pairs have nonzero recorded position and velocity-sum differences, which are retained and evaluated separately.

This corollary receives numerical recorded initial states, not physical measurements. No certified error bound relates a saved binary64 state to a true physical initial state. The continuous safety conclusion still requires the declared physical potential family, the same exact full-state port, $w = 0$, and actual continuous saturated feedback. These forced observation records do not show that such feedback was executed, or certify any sampled, delayed or sensor-error realization. No new native solve, training, gradient update, weight selection, guard adjustment, control run, or physical KAM transfer is established.

S26 A conditional sampled magnetic power guard

Let $x, v \in \mathbb{R}^{64}$ have the native transverse finite-element ordering, let $b = e_{64}$, and set

$$E_0(x) = \tfrac{1}{2} x^T K_0 x, \qquad U_\beta(x) = (1+\beta)E_0(x) + \kappa(1 - \cos b^T x). \tag{S135}$$

Here M, K_0 are the received positive definite native matrices and $\beta \in [\beta_-, \beta_+] \subset (-1, \infty)$ is constant. Consider the explicitly assumed physical family

$$M\dot{v} = -\nabla U_\beta(x) - a(x) D_0 v + \mu u b \cos(b^T x), \qquad \dot{x} = v, \tag{S136}$$

where $D_0 \succeq 0$, a is locally Lipschitz, and $0 \le a(x) \le \bar{a}$. The input u denotes the actual total transverse magnetic field, after quantization, with $|u| \le U$. The preparation cosine field is replaced, not superposed. Define

$$H_\beta = \tfrac{1}{2} v^T M v + U_\beta(x), \qquad y = \mu \cos(b^T x) b^T v. \tag{S137}$$

Then $\dot{H}_\beta = -a(x) v^T D_0 v + uy \le uy$. Discarding the nonnegative dissipation requires no bound on its derivative.

A full-state output-rate majorant. Fix an energy cap $c > 0$, and write $z = M^{-1} b$. On $H_\beta \le c$, set

$$V_c = \sqrt{2c\, b^T M^{-1} b}, \qquad \Theta_c = \sqrt{\frac{2c\, b^T K_0^{-1} b}{1+\beta_-}}, \qquad S_c = \min(1, \Theta_c), \tag{S138}$$

$$A_K = (1+\beta_+)\sqrt{\frac{2c\, z^T K_0 z}{1+\beta_-}}, \qquad A_D = \bar{a}\sqrt{2c\,(D_0 z)^T M^{-1} (D_0 z)}, \tag{S139}$$

$$A_B = (b^T M^{-1} b)(\kappa S_c + \mu U). \tag{S140}$$

For positive μ, κ, a valid majorant is

$$L_y = \mu\{S_c V_c^2 + A_K + A_D + A_B\}. \tag{S141}$$

Indeed, the energy cap bounds $|b^T v|$ and $|b^T x|$ by V_c and Θ_c. Cauchy–Schwarz in the K_0 and M metrics bounds the elastic and dissipative terms in $b^T \dot{v}$ by A_K and A_D, respectively. The remaining magnetic term is bounded by A_B. Differentiating $y = \mu \cos(b^T x) b^T v$ gives $|\dot{y}| \le L_y$. The assumption $\bar{a} < \infty$ is needed for this particular majorant; positive semidefinite dissipation alone does not provide it.

Sampled work and a strict energy ledger. At the actual hold start t_k, suppose a valid observation supplies $|y(t_k) - \widehat{y}_k| \le \varepsilon_k$. Apply a constant actual input u_k for a prescribed duration $h_k > 0$. For every prefix $0 \le s \le h_k$ remaining below the energy cap,

$$H_\beta(t_k + s) - H_\beta(t_k) \le F_k(s) := s(u_k \widehat{y}_k + |u_k|\varepsilon_k) + \tfrac{1}{2} s^2 |u_k| L_y. \tag{S142}$$

Since F_k is convex and $F_k(0) = 0$, $F_k(s) \le \max\{0, F_k(h_k)\}$. Let $\Delta_k = F_k(h_k)$ and let $\lceil \cdot \rceil_g$ denote upward rounding to a fixed positive energy grid g. Assume a trusted initial ledger satisfies $H_\beta(t_0) \le L_0 < c$ and $L_0 \in g\mathbb{Z}_{\ge 0}$. Accept a nonzero proposal only when it satisfies the actual input limit, the current observation is valid, and

$$\Delta_k \le \alpha(c - L_k), \qquad L_{k+1} = \big\lceil \max\{0, L_k + \Delta_k\} \big\rceil_g < c, \qquad 0 < \alpha < 1. \tag{S143}$$

Otherwise apply exactly zero and retain the ledger. The accepted hold obeys $H_\beta(t_{k+1}) \le L_{k+1}$ and remains strictly below c throughout. To prove the latter without circularity, suppose there is a first time at which $H_\beta = c$. Before that time (S141) applies, and (S142) implies

$$H_\beta(t_k + s) \le L_k + \max\{0, \Delta_k\} < c,$$

contradicting the first exit. A rejected or zero proposal uses $\dot{H}_\beta \le 0$ directly. Induction proves the strict energy cap for all holds. A fixed $h_k = h > 0$, or another schedule without finite-time accumulation, excludes Zeno accumulation. The coercive energy then bounds the state for all finite times.

For an opposite-sign proposal, the stronger condition $|\widehat{y}_k| \geq \varepsilon_k + L_y h_k/2$ makes $\Delta_k \leq 0$ and makes every prefix work bound nonpositive. This integral statement does not assert instantaneous $u_k y(t) \leq 0$; the sufficient instantaneous condition instead uses $|\widehat{y}_k| \geq \varepsilon_k + L_y h_k$.

Connection to the learned potential guard. Suppose the current checkpoint satisfies $|U_N - U_\beta| \leq \varepsilon_U$ on $\Omega_4 = \{E_0 < 4E_{\mathrm{net}}\}$. Each separately received numerical initial state can initialize the ledger by rounding an outward upper enclosure of $H_N(t_0) + \varepsilon_U$ upward to the grid, provided the rounded value is strictly below c. For the present $c = 41E_{\mathrm{net}}/20$ and received β_-, $c/(1+\beta_-) < 4E_{\mathrm{net}}$, so the physical energy cap also stays inside Ω_4. The theorem concerns the declared physical family, not a derivative match between the learned and physical potentials.

Observation and execution scope. A rigorous midpoint computation at a saved numerical state supplies an arithmetic observation enclosure. It does not certify a physical sensor. For a tip observation with actual hold-start errors $|\theta - \widehat{\theta}| \leq r_\theta$ and $|\omega - \widehat{\omega}| \leq r_\omega$, a sufficient additional port error allowance is

$$\mu\{r_\omega + |\widehat{\omega}| \min(2, r_\theta)\},$$

plus the outward error of computing $\mu\cos(\widehat{\theta})\widehat{\omega}$. Timing, sensing and total-field actuation must satisfy their declared contracts to apply the result to an experiment. This initial-proposal reception alone does not execute a controller. The separate numerical execution is treated in Section S26.3; neither reception claims a physical sampled controller run, physical parameter identification, convergence, or control performance.

S26.1 A received sampled-power instance for the calibrated guard

Use the same current checkpoint, $c = 41E_{\mathrm{net}}/20$, $\varepsilon_U = 479874807E_{\mathrm{net}}/125000000000$, the unchanged scalar compatibility interval, and the full 64-coordinate native matrices. For this instance assume additionally $D_*(x) = a(x)D_0$ with $0 \leq a(x) \leq 6/5$. This finite operator bound is an extra physical hypothesis; it is not inferred from positive semidefiniteness or from the candidate's feature-box dissipation bound.

Exact rational matrix operations and outward square roots give

$$|b^T x| < 0.001909972751, \qquad |b^T v| < 20.504956934$$

on $H_* \leq c$. With the actual total-field limit $U = 10^{-4}$ T, (S141) has the strict bound

$$L_y < 49.953396682 \ \mathrm{J\,T^{-1}s^{-2}}.$$

Here M is the exact sum of the saved M_0 and M_r entries, and $D_0 = \alpha M_0$ with the exact nominal coefficient from the native operator record. The shifted trajectory damping coefficient and translational-plus-rotary mass are not substituted into D_0.

For each of the 22 separately received numerical initial states, compute an outward enclosure of $y = \mu\cos\theta_{32}\, v_{\theta,32}$ and take its rational midpoint $\widehat{y}_0$. Its arithmetic enclosure radius is strictly below the prescribed allowance $\varepsilon_y = 10^{-8}\,\mathrm{J\,T^{-1}s^{-1}}$. Set the energy grid to $g = 10^{-9}E_{\mathrm{net}}$, the reserve fraction to $\alpha = 1/2$, and initialize

$$L_0 = \left\lceil \overline{H}_N(x_0, v_0) + \varepsilon_U \right\rceil_g.$$

The strict common bounds are $0.241216386 < L_0/E_{\mathrm{net}} < 0.976086611 < c/E_{\mathrm{net}}$. For the nonzero opposite-sign proposal $u_0 = -U\,\mathrm{sign}(\widehat{y}_0)$, exact evaluation of (S143) gives

Hold duration	Accepted nonzero proposals	Upper next ledger/E_{net}
$1\,\mu$s	22/22	0.976499156
$10\,\mu$s	22/22	1.017677105
$100\,\mu$s	0/22	0.976086611

For the $10\,\mu$s holds the computed work majorants obey $0.041462830554 < \Delta_0/E_{\mathrm{net}} < 0.041617370734$. The rejected $100\,\mu$s proposals use the exact zero fallback and retain their initial ledger. These comparisons keep the same field limit, error allowance, cap and rounding grid throughout.

The arbitrary-horizon conclusion follows from the work-ledger theorem only if every subsequent hold also receives a valid current observation and applies the accepted actual total command with the stated timing. This initial-proposal calculation evaluates saved numerical states without executing a trajectory. A separate numerical feedback execution and its reference-IVP certificate appear in Section S26.3. The error allowance is not a physical sensor certificate. Acceptance establishes a conditional energy guard, not monotone decrease on accepted holds or improved control performance.

S26.2 Continuous reception of the three held-field segments

Fix the declared 64-coordinate mathematical equation

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa \sin(b^{\top}x) - \mu u(t)\cos(b^{\top}x)\} = 0, \quad M = M_0 + M_r, \quad K = \tfrac{53}{50}K_0, \quad D = \alpha_{\mathrm{shift}} M_0,$$

where the saved matrix entries and $\alpha_{\mathrm{shift}} = 5754906970707699/9007199254740992$ are interpreted as exact rationals. Here $b = e_{64}$, time is in seconds, u is in tesla, and $y = \mu\cos(b^{\top}x)b^{\top}\dot{x}$ has units $\mathrm{J}/(\mathrm{T\,s})$. The field u is the actual, constant binary64 total transverse field on each segment; it replaces the preparation cosine. This declared IVP starts at the fresh seed's own saved position and velocity. It asserts neither an exact physical initial state nor identity with a compiled native equation.

Let Q be the piecewise cubic Hermite curve through every retained native position and velocity endpoint. Each cell uses its own exact dyadic duration. All 101 predecessor handoffs, including the physical-time coordinate, are received bitwise. Thus Q and $\dot{Q}$ are continuous across both interior cells and field switches. The exact equation has continuous position and velocity at a field switch, although acceleration may jump. Define

$$e = x - Q, \quad r^2 = \dot{e}^{\top}M\dot{e} + e^{\top}Ke, \quad c_M = \sqrt{b^{\top}M^{-1}b}, \quad c_K = \sqrt{b^{\top}K^{-1}b}.$$

The initial radius is zero once, at the declared saved seed, and is not reset at a continuation boundary. For $|u| \le U = 10^{-4}\,\mathrm{T}$, the scalar magnetic force is globally Lipschitz in angle with constant $\kappa + \mu U$. If ρ bounds the Hermite residual in the M^{-1} norm, then

$$D^{+}r \le ar + \rho, \qquad a = \tfrac{1}{2}(\kappa + \mu U)c_M c_K.$$

Indeed, the damping term is nonpositive, the residual contributes at most $r\rho$, and $|\dot{e}^{\top}b|\,|b^{\top}e| \le c_M c_K r^2/2$. For a cell of duration h with initial radius r_ℓ,

$$\bar{r}(s) = e^{as}r_\ell + \rho\,\frac{e^{as} - 1}{a}, \qquad 0 \le s \le h$$

is a nondecreasing prefix majorant. An outward grid ceiling of $\bar{r}(h)$ supplies the next initial radius. The global force bound removes a tube-closure assumption.

On one cell put $\theta_Q = b^{\top}Q$ and $\Theta \ge \sup|\theta_Q|$. Form the degree-nine polynomial residual

$$R_{\mathrm{pol}} = M\ddot{Q} + D\dot{Q} + KQ + b\{\kappa(\theta_Q - \theta_Q^3/6) - \mu u(1 - \theta_Q^2/2)\}.$$

The omitted scalar force is bounded on the whole cell by

$$\epsilon_{\mathrm{nl}} = \kappa\Theta^5/120 + \mu|u|\Theta^4/24.$$

These real Taylor bounds do not require a small-angle assumption. If $B_0, \dots, B_9$ are the degree-nine Bernstein coefficient vectors of R_{pol}, convexity of the norm gives

$$\rho = \max_j \sqrt{B_j^{\top}M^{-1}B_j} + c_M\epsilon_{\mathrm{nl}}.$$

The exact rational matrices and inverse are received once, then enclosed in 512-bit Arb balls. All cell arithmetic, including the quadratic forms and stable $\mathrm{expm1}(ah)$ evaluation, uses outward intervals; the reported radius is rounded upward. No per-cell matrix inverse or sampled-residual substitution is used.

At a saved endpoint with tip velocity $v_{\theta,Q}$,

$$|y(x,\dot{x}) - y(Q,\dot{Q})| \le \mu\{c_M r + |v_{\theta,Q}| \min(2, c_K r)\}.$$

For a whole-cell comparison, replace r by its prefix endpoint majorant and $|v_{\theta,Q}|$ by the Bernstein bound on the Hermite velocity. For the controller's actual logged exact rational value $\widehat{y}$, the separate outward readout allowance $|\,\mu\cos(\theta_Q)v_{\theta,Q} - \widehat{y}\,|$ must be added. In particular an exact midpoint of an Arb enclosure need not be converted to binary64. Only a receipt binding the actual logged value to the actual hold-start observation can receive the fixed prospective $\varepsilon_y = 10^{-8}\,\mathrm{J/(T\,s)}$ contract. A saved finite state alone is insufficient. Neither a successful IVP tube nor that sensor inequality alone receives the other causal ledger, cap, timing, actuation, or initial guard hypotheses of the sampled safety theorem.

S26.3 Continuous reception of the recorded calibrated feedback

Proposition S46. Recorded three-hold energy ledger. *Consider the 64-coordinate mathematical initial-value problem*

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa\sin\theta - \mu u(t)\cos\theta\} = 0, \qquad \theta = b^{\top}x, \quad b = e_{64},$$

with the exact saved matrices $M = M_0 + M_r$, $K = (53/50)K_0$, *and* $D = \alpha_{\mathrm{shift}}M_0$, *where* $\alpha_{\mathrm{shift}} = 5754906970707699/9007199254740992$. *Its initial position and velocity are the actual saved numerical seed state at* $t_0 \simeq 20\,\mu\mathrm{s}$. *Define*

$$H_\star = \tfrac{1}{2}\dot{x}^{\top}M\dot{x} + \tfrac{1}{2}x^{\top}Kx + \kappa(1-\cos\theta), \qquad y = \mu\cos\theta\, b^{\top}\dot{x}.$$

The three recorded holds have their own exact saved binary64 endpoints $t_0 < t_1 < t_2 < t_3 \simeq 50\,\mu\mathrm{s}$. *On these holds the actual total input is, respectively,* $-u_q, +u_q, 0$, *with*

$$u_q = \frac{1844674407370955}{18446744073709551616}\,\mathrm{T} < 10^{-4}\,\mathrm{T}.$$

The held field replaces the preparation cosine. For this declared IVP, the received bounds imply

$$H_\star(t_j) \le L_j, \qquad H_\star(t) \le 0.910436431\,E_{\mathrm{net}} < c = \tfrac{41}{20}E_{\mathrm{net}} \quad (t_0 \le t \le t_3),$$

where

$$(L_0, L_1, L_2, L_3)/E_{\mathrm{net}} = (0.836050803,\ 0.870553256,\ 0.910436431,\ 0.910436431).$$

This is a statement about the recorded three-hold declared IVP. It does not assert identity with the compiled native equation, a physical initial-state or sensor certificate, native dense-output accuracy, or any future rollout.

Proof. The actual native execution contains one fresh preparation solve and three continuation solves. Each continuation retains all 101 saved solution and time-derivative entries bitwise from its predecessor. The mathematical IVP starts only at the saved seed; the preceding preparation trajectory is not certified by this proposition. The three holds contain 514 retained raw states each, including an extremely short terminal cell of approximately $4.066\,10^{-19}\,\mathrm{s}$. The original all-cell reception is preserved and does not receive the fixed sensor budget.

The separate selected-partition protocol predeclares strides two, four and eight. For each hold, the stride-eight curve retains the original raw indices $0, 8, \ldots, 504, 513$, with zero-based indexing. It has 64 Hermite cells per hold and keeps every actual controller endpoint and its original position and velocity. The final short cell is merged into a wider cell. No source state, input, controller time or error budget is changed. For this new piecewise C^1 curve Q, every selected cell is certified on its entire interval using the residual of the declared 64-coordinate equation. The reconstruction need not interpolate omitted raw samples; no enclosure of those samples or native dense output is inferred.

The comparison radius satisfies

$$r^2 = (\dot{x}-\dot{Q})^{\top}M(\dot{x}-\dot{Q}) + (x-Q)^{\top}K(x-Q), \qquad D^+r \le ar + \rho,$$

where $a = (\kappa + \mu 10^{-4})\sqrt{(b^\top M^{-1} b)(b^\top K^{-1} b)}/2$. It is initialized once at $r(t_0) = 0$ and propagated across all three holds. The polynomial residual uses cubic sine and quadratic cosine terms. Its whole-cell remainder is bounded by $\kappa\Theta^5/120 + \mu|u|\Theta^4/24$. Degree-nine Bernstein coefficient vectors, the M^{-1} norm and outward 512-bit Arb arithmetic give the cell residual bound. Thus neither residual samples nor resetting the radius at field switches is used.

At a hold start, with $c_M = \sqrt{b^\top M^{-1} b}$ and $c_K = \sqrt{b^\top K^{-1} b}$, the received port allowance is

$$\mu\{c_M r + |v_{\theta,Q}| \min(2, c_K r)\} + \left|\mu \cos(\theta_Q) v_{\theta,Q} - \widehat{y}\right|.$$

The second term independently receives the exact rational midpoint $\widehat{y}$ actually used by the bridge; it is not converted to binary64. The resulting upward bounds at the three hold starts are

$$\delta_{y,0} \le 10^{-50}, \qquad \delta_{y,1} < 6.873047\,10^{-9}, \qquad \delta_{y,2} < 1.639569\,10^{-8} \quad \mathrm{J/(T\,s)}.$$

Only the first two holds have nonzero input. Their allowances fit the unchanged prospective $\varepsilon_y = 10^{-8}$. The computed third bound exceeds that budget, and the all-three-sensor contract remains unreceived.

Direct exact-model energy evaluation gives $H_\star(t_0) < 0.832211815\,E_{\mathrm{net}} \le L_0 < c$. An independent energy-dual calculation on $H_\star \le c$ gives an output-rate bound below $49.953391663\,\mathrm{J/(T\,s^2)}$; it fits the already frozen uniform bound $L_y < 49.953396682\,\mathrm{J/(T\,s^2)}$. For each nonzero hold, its actual exact duration $h_j = t_{j+1} - t_j$ is used in

$$\Delta_j = h_j\{u_j \widehat{y}_j + |u_j|\varepsilon_y\} + \tfrac{1}{2} h_j^2 |u_j| L_y.$$

Independent rational arithmetic receives the reserve gate, upward $10^{-9}E_{\mathrm{net}}$ grid ceiling, strict cap gate and matching single transactional commit. The quadratic prefix majorant and first-exit argument then give $H_\star(t) < c$ throughout the first two holds and $H_\star(t_2) \le L_2$. These holds spend a positive energy reserve; they are not certified to have nonincreasing energy.

On the final recorded hold the actual total field is identically zero. Since $D = \alpha_{\mathrm{shift}} M_0 \succeq 0$,

$$\dot{H}_\star = -\dot{x}^\top D \dot{x} \le 0, \qquad H_\star(t) \le L_2 = L_3 < c \quad (t_2 \le t \le t_3).$$

This branch requires neither a port observation nor L_y. It completes the recorded-interval ledger statement without receiving the third sensor contract.

Comparison of reconstructions. The original all-cell reconstruction and the predeclared stride-two and stride-four reconstructions did not receive the required port budget at the second nonzero hold. Their computed upper bounds exceed 10^{-8}; upward rounded displays are $4.972088\,10^{-4}$, $2.132616\,10^{-8}$ and $1.094943\,10^{-8}$, respectively. These are upper bounds, not lower bounds on the actual error. The stride-eight reconstruction receives both required nonzero-hold contracts; its third, unused sensor contract remains unreceived. All original negative outcomes, raw data and actual commands are retained.

S27 Signed work for the received feedback history

S27.1 Signed work on a received constant-field history

Fix the previously declared, saved-seed, full64 reference IVP

$$M\ddot{x} + D\dot{x} + Kx + b\{\kappa \sin\theta - \mu u(t)\cos\theta\} = 0, \quad \theta = b^\top x, \quad b = e_{64},$$

where $M = M_0 + M_r \succ 0$, $K = (53/50)K_0 \succ 0$, $D = \alpha_{\mathrm{shift}} M_0 \succeq 0$, and $\mu > 0$. The matrices, scalar parameters, initial data and executed held total fields have the exact-source conventions of the received reference IVP. There are no impulses at a change of field. The physical-storage function used in this reference equation is

$$H_\star(x, v) = \tfrac{1}{2} v^\top M v + \tfrac{1}{2} x^\top K x + \kappa(1 - \cos\theta).$$

It does not contain u, so it is continuous at a field switch. Multiplying the equation by $v^\top$ gives

$$\dot{H}_\star = -v^\top D v + \mu u \cos\theta\, \dot{\theta}.$$

Consequently, on one constant-field hold $[t_k, t_{k+1}]$,

$$H_\star(t) - H_\star(t_k) = -\int_{t_k}^{t} v^\top D v\, ds + \mu u_k \{\sin\theta(t) - \sin\theta(t_k)\}. \tag{S144}$$

Dropping the nonnegative dissipation yields an upper bound, without requiring a numerical velocity quadrature or a bound on a sensor readout.

Suppose the already received continuous comparison with the piecewise cubic Hermite curve Q supplies on every retained cell I_i a bound

$$(\dot{x} - \dot{Q})^\top M(\dot{x} - \dot{Q}) + (x - Q)^\top K(x - Q) \le r_i^2.$$

The propagated radius is initialized to zero only at the saved seed and is never reset at a cell boundary or field switch. Write $c_K = \sqrt{b^\top K^{-1} b}$ and $c_M = \sqrt{b^\top M^{-1} b}$. If $[q_i^-, q_i^+]$ and $[v_i^-, v_i^+]$ enclose the ranges of $b^\top Q$ and $b^\top \dot{Q}$ on I_i, respectively, then

$$\theta(I_i) \subseteq [q_i^- - c_K r_i, q_i^+ + c_K r_i],$$
$$\dot{\theta}(I_i) \subseteq [v_i^- - c_M r_i, v_i^+ + c_M r_i].$$

This follows from Cauchy–Schwarz in the K and M inner products. Bernstein coefficients of the exact cubic and its quadratic derivative provide the required whole-cell ranges, rather than sampled extrema. At a retained endpoint use its saved angle and the corresponding propagated endpoint radius instead of the whole-cell range.

For a real interval A, let $S(A)$ be any outward interval enclosing $\sin A$. For intervals A, B define

$$\mathcal{W}_u(A, B) = \mu u\{S(A) - S(B)\}, \qquad W_u^+(A, B) = \sup \mathcal{W}_u(A, B).$$

The subtraction is interval subtraction and is allowed to overestimate correlation between its two arguments. In particular, the same uncertain state is not treated as two independent physical states. If A_i encloses $\theta(I_i)$, A_k encloses the hold-start angle, and $E_k \ge H_\star(t_k)$, then every $t \in I_i$ satisfies

$$H_\star(t) \le E_k + W_{u_k}^+(A_i, A_k). \tag{S145}$$

For the hold endpoint interval A_{k+1},

$$E_{k+1} = E_k + W_{u_k}^+(A_{k+1}, A_k) \quad \Longrightarrow \quad E_{k+1} \ge H_\star(t_{k+1}).$$

The same estimates may be based at the left endpoint of an individual cell. The minimum of several valid upper bounds is again a valid upper bound. This permits interval dependence to be reduced by comparing both the hold anchor and the previous cell endpoint.

There is a separate sign test. If $\cos\theta \ge 0$ and $u_k\dot{\theta} \le 0$ throughout a cell, then $\dot{H}_\star \le 0$ throughout that cell, so its energy is nonincreasing. For $u_k = 0$ this conclusion needs no angle or velocity sign test. A proved sign test permits replacement of a cell's prefix and endpoint bounds by the already proved left-endpoint energy bound. A failed sign test is retained as a failure of that sufficient condition; it is not a proof that the energy increases. If this sign test fails anywhere on a hold, a bound $H_\star(t) \le H_\star(t_k)$ throughout that hold must not be described as pairwise monotonicity on the hold.

All bounds used below can be rational. On $[-1, 1]$, $\cos z \ge 1 - z^2/2 > 0$, so sine is increasing. Taylor's theorem gives

$$P_7(z) = z - z^3/6 + z^5/120 - z^7/5040, \qquad |\sin z - P_7(z)| \le |z|^9/9!.$$

For $[a,b]\subset[-1,1]$ use

$$S([a,b])=[P_7(a)-|a|^9/9!,\ P_7(b)+|b|^9/9!].$$

An exact rational ceiling for c_K, c_M and exact rational Bernstein coefficients therefore suffice for signed magnetic-work bounds. For comparison, the weaker absolute-work estimate is

$$H_\star(t)\le H_\star(t_0)+\sum_{I_i\text{ before }t}\mu|u_i|h_i\{\sup_{I_i}|b^\top\dot Q|+c_M r_i\},$$

where the final partial cell may be enlarged to its entire duration.

Taking the maximum of the received prefix bounds and the initial bound gives a uniform energy cap on precisely the recorded finite time interval. This is an offline theorem about the same declared continuous reference IVP and the same executed fields. It changes neither the controller nor its original prospective guard, ledger, sensor budget or negative sensor receptions. It does not enclose a compiled native equation, COMSOL dense output, omitted native samples, a physical plant, an actuator or a sensor, and implies no future-time guarantee, tracking performance or complex-network advantage.

Offline signed-work reception of the executed reference history. For the same saved-seed reference IVP, the same three executed total fields, and the independently received stride-eight comparison, exact rational reconstruction of all 192 retained tip-angle cubics gives

$$H_\star(t)<0.832211815\,E_{\mathrm{net}},\qquad t_0\le t\le t_3,\qquad (t_0,t_3)\simeq(20,50)\,\mu\mathrm{s}.$$

The exact endpoints are the saved binary64 clocks $t_0=5902958103587057/295147905179352825856\,\mathrm{s}$ and $t_3=7378697629483821/147573952589676412928\,\mathrm{s}$. The original prospective ledger bound $0.910436431\,E_{\mathrm{net}}$ remains valid and unchanged; the smaller offline bound uses the signed identity $\mu u_k[\sin\theta(t)-\sin\theta(t_k)]$ on each constant-field hold. No new native solve, training update, ODE integration, or inferred damping work enters this calculation. The nonnegative dissipation is discarded.

The received upper bounds at the three hold endpoints are, with outward decimal ceilings,

$$H_\star(t_1)<0.829688104\,E_{\mathrm{net}},\qquad H_\star(t_2),H_\star(t_3)<0.827405859\,E_{\mathrm{net}}.$$

On every hold, the received energy is bounded by that hold's actual starting energy. A stronger pairwise monotonicity conclusion is received throughout the second hold, where the continuous angle tube has positive cosine and the tip-velocity tube has nonpositive sign, and throughout the third hold, whose total field is exactly zero. Fifteen cells of the first hold do not satisfy the nonpositive-port sufficient condition; therefore no monotonicity claim is made for the entire three-hold history.

The calculation receives existing continuous reference-IVP radii and rebuilds the exact tip Hermite ranges and signed work, rather than rederiving all full64 residuals. It preserves the original failed third sensor-budget reception. Its scope remains this particular reference IVP and recorded time interval; it supplies no compiled-equation, native dense-output, physical sensing or actuation, future-rollout, tracking, or complex-network superiority certificate.

S28 Long-time numerical prediction diagnostics

S28.1 Post hoc decomposition in all nominal mechanical modes

We examine the fixed original 918-parameter checkpoint on the single positive-initial-velocity coarse-step case over $0\le t\le 0.6\,\mathrm{s}$. This analysis describes the already observed prediction error. It neither selects a model nor replaces the full-state evaluation by a selected mode. Let L_K and L_M denote the saved Cholesky factors, and define the rounded nominal matrices $K_0=L_KL_K^{\mathsf T}$ and $M=L_ML_M^{\mathsf T}$. The mass-normalized nominal modes satisfy

$$K_0S=MS\Omega^2,\qquad S^{\mathsf T}MS=I_{64},\qquad \Omega=\mathrm{diag}(\omega_1,\dots,\omega_{64}).\tag{S146}$$

These are the modes of the saved nominal quadratic model, rather than instantaneous nonlinear modes or an exact eigensystem of the physical device.

Table S2. Post hoc nominal-mode decomposition for the fixed original 918-parameter checkpoint and one positive coarse-step case. All shares refer to squared mechanical-energy coordinates; the peak share uses the maximum of the unchanged full-state metric.

$\omega_j h_c$	Count	Initial energy	Peak error	Integrated error
$[0, 0.1]$	16	0.89088316	0.11128504	0.14585627
$(0.1, 1]$	33	0.08326342	0.62204564	0.65501561
$(1, 2]$	15	0.02585342	0.26666932	0.19912812
$(2, \infty)$	0	0	0	0

For the saved positive coordinate scale s, the full-state observation is $z^{\mathrm{obs}} = (L_K^{\mathsf{T}} x/s, L_M^{\mathsf{T}} \dot{x}/s)$. Write the prediction error as $\delta z = (\delta z_q, \delta z_p)$ and retain the same initial-state denominator as the primary metric, $D = \|z^{\mathrm{obs}}(0)\|_2^2$. The nominal modal transforms and normalized squared errors are

$$T_q = \Omega^{-1} S^{\mathsf{T}} L_K, \qquad T_p = S^{\mathsf{T}} L_M, \tag{S147}$$

$$e_j^2(t) = \frac{(T_q \delta z_q(t))_j^2 + (T_p \delta z_p(t))_j^2}{D}, \qquad E^2(t) = \frac{\|\delta z(t)\|_2^2}{D}. \tag{S148}$$

Both transforms are orthogonal in exact arithmetic, hence $\sum_{j=1}^{64} e_j^2(t) = E^2(t)$. Equivalently, the position modal coordinates are $\Omega S^{\mathsf{T}} M x/s$ and the velocity modal coordinates are $S^{\mathsf{T}} M \dot{x}/s$. Their squared magnitudes partition the nominal quadratic mechanical energy. The independently recomputed full-error curve agreed with the saved primary curve within 3.4×10^{-16}; the maximum absolute discrepancy in the pointwise squared-error decomposition was 2.8×10^{-14}. These are floating-point consistency checks.

Table S2 reports all 64 modes, grouped after observation by $\omega_j h_c$, where $h_c = 12.5\,\mu$s is the native coarse-step setting. For a band B, the initial energy share is $\sum_{j \in B} [(T_q z_q^{\mathrm{obs}}(0))_j^2 + (T_p z_p^{\mathrm{obs}}(0))_j^2]/D$. The peak-error share is $\sum_{j \in B} e_j^2(t_*)/E^2(t_*)$, evaluated at the common full-state maximum $t_* = 0.3131125\,$s. It does not sum the separate maxima of individual modes. The integrated share is $\sum_{j \in B} \int_0^{0.6} e_j^2(t)\,dt / \int_0^{0.6} E^2(t)\,dt$, evaluated by the composite trapezoidal rule on all 96,001 common output times, with spacing $6.25\,\mu$s.

The modes with $\omega_j h_c > 0.1$ carry about 10.91% of the initial nominal mechanical energy but 88.87% of the squared error at the full-state maximum and 85.41% of its time integral. This is a descriptive association with frequency. The band boundaries are not certified time-integration error thresholds, and this decomposition does not identify COMSOL step-size error as the cause of the mismatch. The positive coarse/fine comparison is reported separately in Section S28.2; it does not certify a converged reference. The full-state maximum remains approximately 0.654780; a much smaller first-mode error cannot replace this primary result. A single fixed checkpoint and one coarse-step case also provide no paired-case conclusion or evidence of a general advantage of complex-valued networks.

S28.2 Fine-step observation scores from retained numerical predictions

The positive coarse- and fine-step COMSOL records have bitwise-identical output times, prescribed magnetic inputs, and observed initial positions and velocities in all 64 free coordinates. Inspection of the frozen prediction code establishes that the predictor right-hand side depends only on time, its current state, the prescribed input, and the saved model parameters. Subsequent observation rows enter the error evaluation rather than the predictor dynamics. The archived predictions retain their saved numerical initial vectors. Thus the 26 existing IVP integrations, comprising 13 frozen predictors under both numerical settings, were reused to score the fine-step observations. No new integration, parameter fitting, or checkpoint selection was performed. Recomputing the original coarse error curves reproduced the saved curves within 3.4×10^{-16}. The direct observation-coordinate reconstruction and the archived momentum-first initialization differ by at most 1.84×10^{-15} per component in the scaled initial chart. The two formulas coincide in exact arithmetic for an exact factorization $M = L_M L_M^T$; this is not a bitwise floating-point claim. Reuse here scores the fixed archived numerical trajectories; it does not supply a rigorous roundoff-to-exact-IVP or physical initial-state transfer certificate.

All fine-step scores retain the full 128-dimensional energy coordinates and the actual initial-state denominator. Over the 96,001 common output times, the primary maximum errors of all 13 predictors range from 0.58083122 to 0.58113085. The original 918-parameter checkpoint has maximum, RMS, and endpoint errors of 0.58092591, 0.50106948, and 0.48303912, respectively. Here RMS is the square root of the unweighted mean of the squared normalized errors over all output times.

Table S3. All five positive fine-step maximum-error contrasts. Entries are multiplied by 10^{-5}; each row retains all three fixed training seeds. SD uses divisor two and describes training-seed variability on this one record. The original 918-parameter checkpoint is outside these contrasts.

Contrast	Seed 43	Seed 44	Seed 45	Mean	Sample SD
R−GR	11.79209	3.54427	20.44787	11.92808	8.45262
C−GC	10.31492	11.67295	3.65521	8.54769	4.29108
R−C	-8.54026	-8.92145	16.76004	-0.23389	14.71841
GR−GC	-10.01743	-0.79277	-0.03262	-3.61427	5.55831
(GR−GC)−(R−C)	-1.47717	8.12869	-16.79266	-3.38038	12.56921

The maximum primary/reference predictor discrepancies range from 4.2470×10^{-5} to 4.2541×10^{-5}, and all 13 fine-step numerical-sensitivity flags are false under the original criterion. These discrepancies remain numerical diagnostics rather than certified integration-error bounds or a certificate of resolved model rankings. The original checkpoint's fixed first-mode maximum error is 5.6290×10^{-6} with the same full-state initial denominator; the full-state result remains the primary metric.

The five fixed comparisons retain all three training seeds, with the original checkpoint reported separately. For the primary maximum error, the $R - C$ differences change sign across seeds: their mean is -2.3389×10^{-6} and sample standard deviation is 1.4718×10^{-4}. The fine-step record therefore supplies no general advantage of complex-valued predictors. These are scores against a numerically generated fine-step record. The coarse/fine native discrepancy itself reaches 0.64373173 under the common initial-state normalization; neither record is established as converged physical truth by this two-setting comparison. The negative-initial-velocity cases remain outside this result, and the fine-step tip and weak-momentum diagnostics have not been recomputed.

Native numerical step sensitivity. The fine record uses an internal step $6.25\,\mu$s and the same 0.6 s horizon, $96\,001$ output times and source model as the coarse record. The initial velocity discrepancy from the prescribed field is 9.012×10^{-9} in the native component maximum, inside the original guard; that guard is not a rigorous solver-error certificate. Under the common full-state initial normalization, the coarse/fine native difference has maximum 0.643731731, RMS 0.544956606 and endpoint 0.539518613. The maximum occurs at $t = 0.3400375$ s. The same nominal-mode partition assigns about 92.13% of the squared peak discrepancy and 90.03% of its time integral to modes with $\omega_j h_c > 0.1$. Both native records therefore require further resolution evidence before either serves as a converged reference for long-time accuracy. This comparison does not establish a convergence order, a physical error bound, or a causal partition of model and native-solver errors.

References

Abbott JJ, Diller E and Petruska AJ (2020) Magnetic methods in robotics. *Annual Review of Control, Robotics, and Autonomous Systems* 3: 57–90. doi: 10.1146/annurev-control-081219-082713

Chen Y, Matsubara T and Yaguchi T (2022) KAM theory meets statistical learning theory: Hamiltonian neural networks with non-zero training loss. *Proceedings of the AAAI Conference on Artificial Intelligence* 36(6): 6322–6332. doi: 10.1609/aaai.v36i6.20582

Desai SA, Mattheakis M, Sondak D, Protopapas P and Roberts SJ (2021) Port-Hamiltonian neural networks for learning explicit time-dependent dynamical systems. *Physical Review E* 104: 034312. doi: 10.1103/PhysRevE.104.034312

Greydanus S, Dzamba M and Yosinski J (2019) Hamiltonian neural networks. In: *Advances in Neural Information Processing Systems*, vol. 32. https://arxiv.org/abs/1906.01563.

Hairer E (2010) Geometric numerical integration. Lecture 2: Symplectic integrators. Lecture notes, TU München, January–February 2010. https://www.unige.ch/~hairer/poly_geoint/week2.pdf.

Haro À and Luque A (2019) A-posteriori KAM theory with optimal estimates for partially integrable systems. *Journal of Differential Equations* 266(2–3): 1605–1674. doi: 10.1016/j.jde.2018.08.003 Preprint version used: https://arxiv.org/abs/1711.05515v1.

Kim Y, Parada GA, Liu S and Zhao X (2019) Ferromagnetic soft continuum robots. *Science Robotics* 4(33): eaax7329. doi: 10.1126/scirobotics.aax7329

Pöschel J (2001) A lecture on the classical KAM theorem. In: *Proceedings of Symposia in Pure Mathematics*, vol. 69, pp. 707–732. doi: 10.1090/pspum/069/1858551 Preprint version used: https://arxiv.org/abs/0908.2234v1 (Version 1.3, 2009).

Trabelsi C, Bilaniuk O, Zhang Y et al. (2018) Deep complex networks. In: *International Conference on Learning Representations*. https://arxiv.org/abs/1705.09792.

Zhang SQ, Gao W and Zhou ZH (2022) Towards understanding theoretical advantages of complex-reaction networks. *Neural Networks* 151: 80–93. doi: 10.1016/j.neunet.2022.03.024

Zhong YD, Dey B and Chakraborty A (2020) Dissipative SymODEN: Encoding Hamiltonian dynamics with dissipation and control into deep learning. In: *ICLR Workshop on Integration of Deep Neural Models and Differential Equations*. https://arxiv.org/abs/2002.08860.